PAC-Bayesian Learning

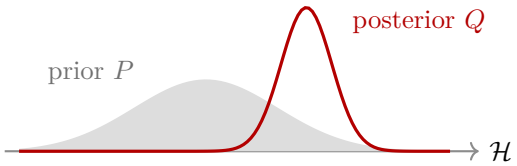
A Theoretical Framework for Ensemble Methods

Master MALIA – Ensemble Methods

Guillaume Metzler
Institut de Communication (ICOM)
Université de Lyon, Université Lumière Lyon 2
Laboratoire ERIC UR 3083, Lyon, France
guillaume.metzler@univ-lyon2.fr

Abstract

Most ensemble methods — bagging, random forests, boosting, stacking, and even kernel machines with random features — output a *weighted majority vote*, that is, a probability distribution over a set of base classifiers. These notes present PAC-Bayesian theory, a framework whose object of study is precisely such a distribution. We derive the classical PAC-Bayesian generalization bounds from a single general theorem, show how the risk of a majority vote can be controlled through the quality *and the diversity* of its voters, and develop the *self-bounding* point of view: learning algorithms that minimize a generalization bound and therefore output a predictor together with its own risk certificate. Each section combines theory (with proofs), intuition, worked examples and exercises, and is accompanied by Python notebooks, a tutorial sheet, a practical session and a project, all available on the course webpage.



$$R_{\mathcal{D}}(B_Q) \;\leq\; \mathcal{B}\big(\widehat{\mathrm{stat}}_S(Q),\, \mathrm{KL}(Q\|P),\, m,\, \delta\big)$$

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Foreword

These notes extend the *Ensemble Methods* course. In the first part of the course we studied how to combine several models — by bagging, random forests, boosting, gradient boosting or stacking — and we saw that most of these algorithms output a *(weighted) majority vote* of base classifiers. We also saw that classical generalization bounds (VC dimension, Rademacher complexity) explain why a single model generalizes, but that they tell us very little about why *combining* models helps.

The aim of this part is to present **PAC-Bayesian theory**, a framework in which the object of study is not a single hypothesis but a *distribution* over hypotheses. A weighted majority vote is precisely such a distribution. PAC-Bayesian theory therefore provides a natural language to (i) derive tight, often *non-vacuous*, generalization guarantees for ensembles, (ii) understand the role of the *diversity* of the voters, and (iii) design *self-bounding* learning algorithms, which minimize a generalization bound instead of an empirical risk and come with a risk certificate for free.

How to read these notes. Each section opens with a short box *About this section*, which states the question the section answers and announces its content. It closes with a summary *To remember*, a few short questions *Check your understanding* (answered in Appendix A.5) and exercises. A single experiment, a random forest on the `digits` dataset, runs through all the sections in teal boxes *Running example*: following it is the quickest way to see how the pieces of the theory fit together. Detailed solutions to all the exercises are gathered in a separate document, *Solutions to the exercises*, available on the course webpage. Definitions are in blue boxes and theorems in red boxes, always with their proofs or proof sketches; boxes *In a nutshell* give the intuition behind a result, historical notes and pitfalls complete the picture, and violet boxes *Recent work* present results of the last few years related to each section. Most notions are illustrated by running the algorithms on the datasets of the course (`breast cancer`, `digits`, two moons and simulated data). As in the slides of the course, long computations are written one step per line, and a grey arrow ↓ like this explains how each line is obtained from the previous one; coloured braces name the terms of the main bounds.

The code of the figures. Every figure can be reproduced. The end of each caption indicates its source: either a function of the Python script `make_figures.py` (which relies on the module `pbtools.py` of the course), or a TikZ file of the folder `tikz/`, which can be included in another document with `\input` or compiled alone with `build_tikz.sh`. The script, the TikZ files and the style files of these notes are gathered in an archive available on the course webpage, together with a `README` listing all the figures; most experiments are also revisited in the notebooks. The dependencies between the sections are shown in Fig. 1: a first reading can follow the path 1–2–3–5–6.

Accompanying material. Five Python notebooks (student and solution versions), a tutorial sheet (TD), a practical session (TP) and a project are available on the course webpage, in the section *Apprentissage PAC-Bayes*. The notebooks follow the sections: notebook 1 (Sections 1–2), notebook 2 (Sections 3–4), notebook 3 (Section 5), notebook 4 (Sections 6–7, forests), notebook 5 (Section 7, linear classifiers and kernels).

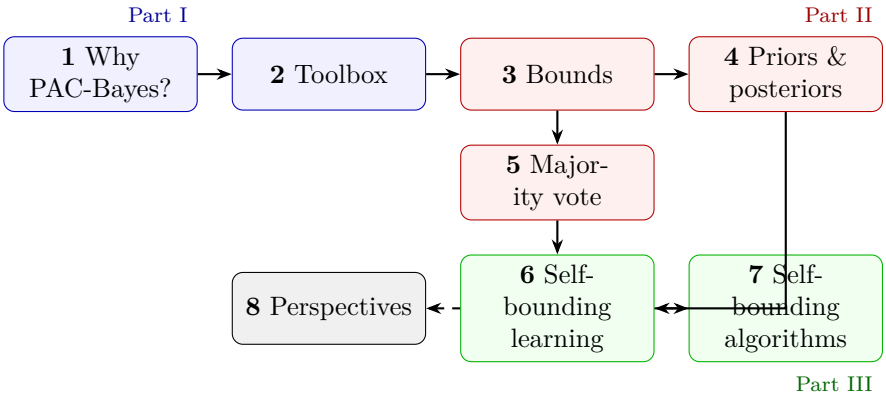


Figure 1: Dependencies between the sections. Blue: foundations; red: PAC-Bayesian bounds; green: self-bounding learning. [code: `tikz/roadmap.tex`]

Prerequisites. The first part of the course (risk, empirical risk, bias–variance decomposition, loss functions, Rademacher bound, bagging, random forests, AdaBoost, gradient boosting) and basic probability (expectation, Markov's inequality, independence).

Suggested schedule. Sections 1–4 in a first 3-hour lecture, Sections 5–7 in a second one; Section 8 is a reading section.

Main references. For a first reading, we recommend Alquier (2024), Guedj (2019) and Langford (2005). The majority-vote and self-bounding points of view follow mainly Germain et al. (2015), Masegosa et al. (2020) and Viallard et al. (2021). Catoni (2007) and Hellström et al. (2025) are advanced references. On ensemble methods themselves, Dietterich (2000), Zhou (2012) and Schapire and Freund (2012) complement the first part of the course, and Shalev-Shwartz and Ben-David (2014) and Mohri et al. (2018) cover the statistical learning theory it relies on.

Notation

$\mathcal{X}$, $\mathcal{Y} = \{-1, +1\}$	input space, (binary) output space
$\mathcal{D}$	unknown data distribution on $\mathcal{X} \times \mathcal{Y}$
$S = \{(x_i, y_i)\}_{i=1}^m \sim \mathcal{D}^m$	learning sample of m i.i.d. examples
$\mathcal{H}$	set of *voters* (base classifiers) $h : \mathcal{X} \to \{-1, +1\}$ (or $[-1, 1]$)
ℓ	loss function, $\ell(h, (x, y)) \in [0, 1]$; by default the 0-1 loss $\mathbb{1}[h(x) \neq y]$
$R_{\mathcal{D}}(h)$, $\widehat{R}_S(h)$	true risk and empirical risk of h
$\mathcal{M}(\mathcal{H})$	set of probability distributions on $\mathcal{H}$
$P \in \mathcal{M}(\mathcal{H})$	*prior* distribution (chosen before seeing S)
$Q \in \mathcal{M}(\mathcal{H})$	*posterior* distribution (chosen after seeing S); also denoted ρ
$\mathrm{KL}(Q\|P)$	Kullback–Leibler divergence $\mathbb{E}_{h\sim Q}\left[\ln \frac{Q(h)}{P(h)}\right]$
$\mathrm{kl}(q\|p)$	binary KL divergence $q \ln \frac{q}{p} + (1-q) \ln \frac{1-q}{1-p}$
$\overline{\mathrm{kl}}^{-1}(q, \varepsilon)$, $\underline{\mathrm{kl}}^{-1}(q, \varepsilon)$	upper / lower inverses of the binary kl
G_Q, $R_{\mathcal{D}}(G_Q)$	Gibbs (stochastic) classifier and its risk $\mathbb{E}_{h\sim Q}[R_{\mathcal{D}}(h)]$
B_Q	Q-weighted majority vote $B_Q(x) = \mathrm{sign}\left(\mathbb{E}_{h\sim Q}[h(x)]\right)$
$W_Q(x, y)$, $M_Q(x, y)$	weight of the erring voters, margin $y\, \mathbb{E}_{h\sim Q}[h(x)] = 1 - 2W_Q$
$e_{\mathcal{D}}(Q)$, $d_{\mathcal{D}}(Q)$	joint error (tandem loss) and disagreement of the voters
$\delta \in (0, 1]$	confidence: bounds hold with probability $\geq 1 - \delta$ over S

We write ln for the natural logarithm and "w.p. $\geq 1 - \delta$" for "with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$".

Part I

Foundations

A random forest whose trees err on 19% of the test digits, but whose vote errs on only 4.4%: this experiment opens the course and raises its two questions. Why is a vote so much better than its voters, and can we guarantee its accuracy without a test set? This first part explains why ensemble methods call for a dedicated theory, and gathers the mathematical tools used in the rest of the notes: concentration inequalities, the Kullback–Leibler divergence and the change of measure inequality. The Occam bound, presented at the end of the part, already contains the main idea of PAC-Bayes: the price of choosing a hypothesis with the data can be measured, and paid.

1 Why PAC-Bayes for ensemble methods?

About this section

Guiding question. *Why is a vote of mediocre classifiers so good, and can we guarantee its accuracy without a test set?*

We start from where the Ensemble Methods *course stopped: averaging models reduces their variance, and the Rademacher bound controls the generalization of one model. We then move to classification, observe that all the ensemble methods of the course — bagging, random forests, boosting, stacking, and even kernel machines — produce the same object, a probability distribution over base classifiers used to vote, and explain why the bounds seen so far say little about such votes. We finally introduce the three ingredients of PAC-Bayesian theory: a prior, a posterior, and a bound that holds for every posterior. A short history and a map of the notes close the section.*

1.1 Where we left the Ensemble Methods course

Before introducing anything new, let us recall the two results of the first part of the course that motivate these notes: why averaging models helps, and how we measured the generalization of a model. The notation is that of the course: a sample $S = \{(x_i, y_i)\}_{i=1}^m$ of m examples, models $h_1, \dots, h_T$, and their combination H_T.

Averaging reduces the variance. In regression, we observed that polynomials of high degree learned on different samples are very different from each other, whereas their average is stable (left panel of Fig. 2). The explanation is a short computation. Write $h_t(x) = f(x) + \varepsilon_t(x)$, where f is the target function and $\varepsilon_t(x)$ the error of the t-th model, and assume that the errors are centred, with the same variance σ^2 and the same correlation ρ between two different models. The squared error of the average $H_T = \frac{1}{T}\sum_t h_t$ is then

$$\begin{aligned}
\mathbb{E}\big[(H_T(x) - f(x))^2\big] &= \mathbb{E}\left[\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_t(x)\right)^2\right] \\
&\quad\downarrow \text{ we develop the square} \\
&= \frac{1}{T^2}\sum_{t=1}^{T}\mathbb{E}\big[\varepsilon_t(x)^2\big] + \frac{1}{T^2}\sum_{t\neq t'}\mathbb{E}\big[\varepsilon_t(x)\varepsilon_{t'}(x)\big] \\
&\quad\downarrow \text{ } T \text{ variances } \sigma^2 \text{ and } T(T-1) \text{ covariances } \rho\sigma^2 \\
&= \frac{T\sigma^2 + T(T-1)\rho\sigma^2}{T^2} = \underbrace{\rho\,\sigma^2}_{\text{shared error}} + \underbrace{\frac{(1-\rho)\,\sigma^2}{T}}_{\text{averaged out}}.
\end{aligned} \tag{1.1}$$

When the errors are uncorrelated ($\rho = 0$), averaging T models divides the error of a single model by T: this is the computation of the course, and the reason why the bias–variance decomposition (Geman et al., 1992; Hastie et al., 2009) is the classical language of ensemble methods (Breiman, 1996b; Zhou, 2012). When they are correlated, only the part $(1-\rho)\sigma^2$ of the error that the models do *not* share is averaged out, and the error of the average cannot go below $\rho\sigma^2$, however many models we add (right panel of Fig. 2). Two lessons are worth keeping. First, the gain of an ensemble comes from the *diversity*

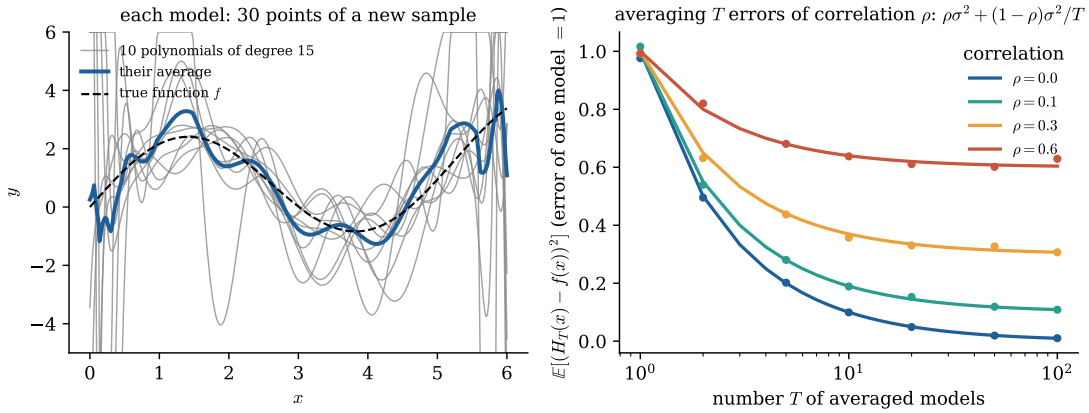


Figure 2: Averaging reduces the variance (regression). Left: ten polynomials of degree 15, each fitted on a new sample of 30 points, and their average; the experiment of the Ensemble Methods course. Right: error of the average of T models whose errors have variance $\sigma^2 = 1$ and correlation ρ; curves: formula (1.1), dots: simulation. [code: `make_figures.py: fig_averaging_variance`]

of its members: identical models ($\rho = 1$) gain nothing. This is why bagging and random forests inject randomness in the construction of the trees. Second, for the squared loss the average is never worse than its members on average, since by Jensen's inequality $\left(\frac{1}{T}\sum_t \varepsilon_t\right)^2 \le \frac{1}{T}\sum_t \varepsilon_t^2$. The gap in this inequality, $\frac{1}{T}\sum_t(\varepsilon_t - \bar{\varepsilon})^2$, is the *ambiguity* of the ensemble (Krogh and Vedelsby, 1995); Wood et al. (2023) have recently shown that such exact bias–variance–*diversity* decompositions hold for other losses (cross-entropy, Poisson), and that diversity is a quantity to be traded off, not blindly maximized.

What changes in classification. In classification, the models are classifiers $h_t : \mathcal{X} \to \{-1, +1\}$ and they are combined by a vote, $H_T(x) = \text{sign}\left(\frac{1}{T}\sum_t h_t(x)\right)$. The computation above does not apply any more, because the 0-1 loss is not the squared loss, and the picture changes in two ways. The good news is that a vote can be *much* better than its voters: it only errs on a point when at least half of the voters err on it simultaneously. If the voters err independently with probability 0.3, the vote of 101 of them almost never errs (Fig. 3, left). The bad news is that this requires diverse voters even more than in regression: as soon as their errors are correlated, they pile up on the same points and the vote keeps a large part of the error of a single voter. The right panel of Fig. 3 shows why: the *average* fraction of erring voters is always 0.3, but only its *spread* decides how often it exceeds $\frac{1}{2}$. A vote can even be worse than its average voter — by a factor at most 2, as we will prove in Section 5.

How we measured generalization. The course also studied the gap between the empirical risk $R_S(h)$ of a model and its true risk $R(h)$. We decomposed the excess risk into an approximation error, due to the choice of the class $\mathcal{H}$, and an estimation error, and we bounded the latter with the *Rademacher complexity* of the class (Koltchinskii and Panchenko, 2000; Bartlett and Mendelson, 2003): with probability at least $1-\delta$, for all $h \in \mathcal{H}$, $R(h) \le R_S(h) + 2\mathfrak{R}_m(\mathcal{H}) + \sqrt{\ln(1/\delta)/(2m)}$ (Theorem 3.1 of the course). For boosting, we proved that the *training* error of AdaBoost decreases exponentially fast, $R_S(H_T) \le e^{-2\gamma^2 T}$ when every weak learner has an edge γ over random guessing (Freund and Schapire, 1997; Schapire and Freund, 2012). The course ended on a remark: these

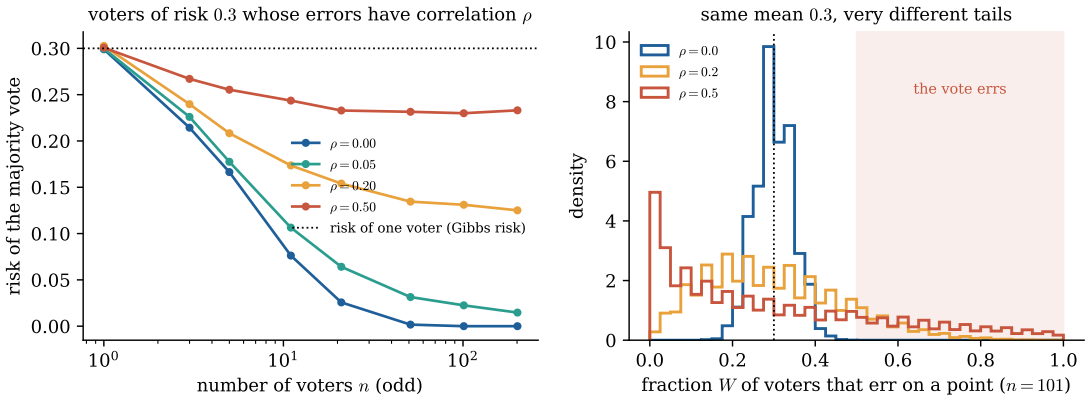


Figure 3: Votes of correlated voters (classification). Each voter errs with probability 0.3, and the errors of two voters are positively correlated (latent Gaussian model with correlation parameter ρ). Left: risk of the majority vote as the number of voters grows. Right: distribution of the fraction of erring voters on a random point, for 101 voters; the vote errs when this fraction exceeds $\frac{1}{2}$ (shaded area). [code: `make_figures.py: fig_vote_correlation`]

tools explain why a *single* model generalizes, and why boosting fits the training data, but not why a combination of models generalizes better than its members. It mentioned PAC-Bayesian theory as the natural tool for ensembles. These notes start exactly there; Fig. 4 lists the notions of the course that we are going to extend, and where.

1.2 A puzzling experiment

Let us start with an experiment that will accompany us throughout these notes. We train a random forest of 100 trees on the `digits` dataset of `scikit-learn`, where the task is to decide whether a handwritten digit is smaller than 5. We use 1257 examples for training and keep 540 for testing. Taken one by one, the trees are mediocre: their test errors range from 14% to 25%, with an average of 19%. Yet the forest, which simply lets the 100 trees vote, errs on only 4.4% of the test digits. The vote is more than four times better than an average tree, and three times better than the best one (Fig. 5): the classification counterpart of the averaging effect recalled above.

This experiment raises two questions, which organize the whole course. The first one is *explanatory*: why is a vote of mediocre classifiers so much better than its members, and what property of the ensemble makes this possible? The second one is *practical*: could we have *guaranteed* that the forest is accurate without a test set, that is, from the training data alone? PAC-Bayesian theory answers both. To see why it is the right tool, we first need to notice that the forest is not an isolated case.

> **Running example** We will come back to this forest in every section. In Section 3 we certify the risk of an average tree, in Section 5 we certify the vote itself and see why its accuracy is linked to the diversity of the trees, and in Sections 6 and 7 we learn new weights for the trees by minimizing the certificates.

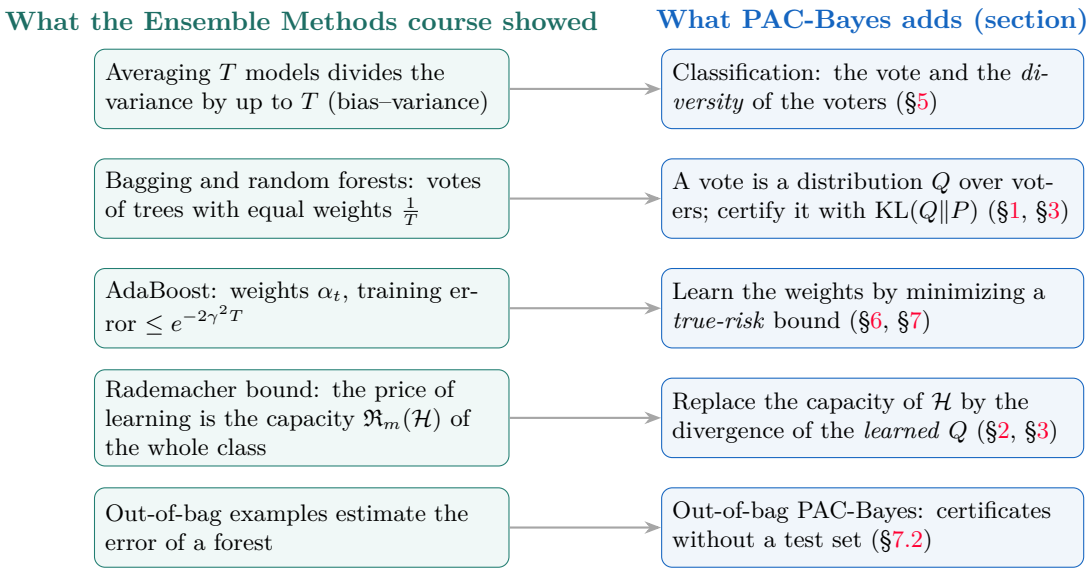


Figure 4: From the Ensemble Methods course to these notes: each notion of the course (left) is extended by PAC-Bayesian theory in the section indicated on the right. [code: `tikz/course_bridge.tex`]

1.3 Ensemble methods produce majority votes

In the first part of the course we built several *ensemble* predictors from a family of base classifiers $h_1, \ldots, h_n$, which we will call *voters* from now on. The methods rely on very different ideas, but they all end up with the same kind of predictor.

In bagging (Breiman, 1996a), each tree h_t is learned on a bootstrap (Efron and Tibshirani, 1993) sample S_t of the learning sample S, and random forests (Breiman, 2001) add some randomness in the construction of each tree by drawing, at each split, a random subset of the features. The final prediction is obtained by letting the trees vote,

$$H(x) = \operatorname{sign}\Big(\frac{1}{n}\sum_{t=1}^{n} h_t(x)\Big),$$

a majority vote in which every tree has the same weight $\frac{1}{n}$.

Boosting looks different at first sight. AdaBoost (Freund and Schapire, 1997) builds the voters sequentially: at round t, the weak learner h_t is trained on a reweighted sample and receives the weight $\alpha_t = \frac{1}{2}\ln\frac{1-\varepsilon_t}{\varepsilon_t}$, which is non-negative as soon as its weighted error ε_t is smaller than $\frac{1}{2}$. But the final classifier can be written in two equivalent ways,

$$H(x) = \operatorname{sign}\Big(\sum_{t=1}^{n} \alpha_t h_t(x)\Big) = \operatorname{sign}\Big(\sum_{t=1}^{n} Q(h_t)\, h_t(x)\Big), \qquad \text{with} \qquad Q(h_t) = \frac{\alpha_t}{\sum_s \alpha_s},$$

because dividing the score by the positive constant $\sum_s \alpha_s$ does not change its sign. The normalized weights $Q(h_1), \ldots, Q(h_n)$ are non-negative and sum to one. In other words, AdaBoost outputs a *probability distribution* over the weak learners, and predicts with the corresponding weighted majority vote. Figure 6 shows such a distribution on a simulated problem. Gradient boosting (Friedman, 2001) follows the same pattern with

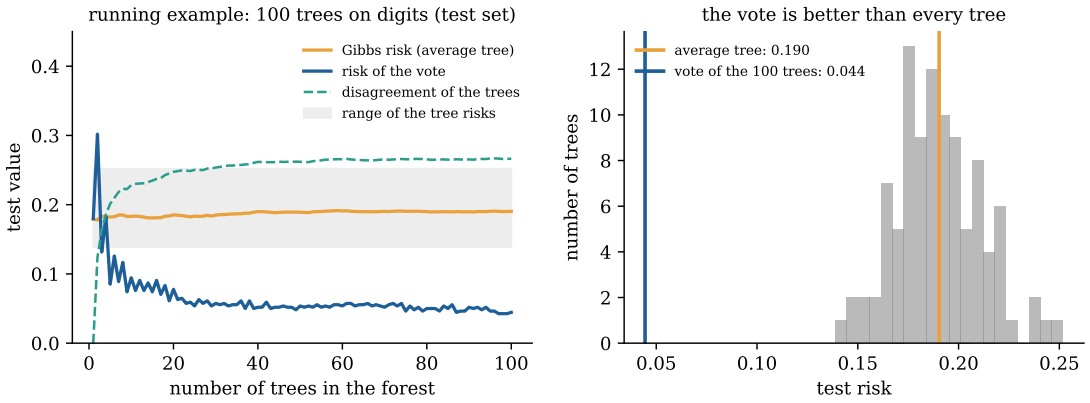


Figure 5: The running example. Left: test risk of the vote of the first n trees, average test risk of these trees (the *Gibbs risk*, see below) and their disagreement, as n grows. Right: the test risks of the 100 trees; the vote is better than every one of them. [code: `make_figures.py: fig_forest_growth`]

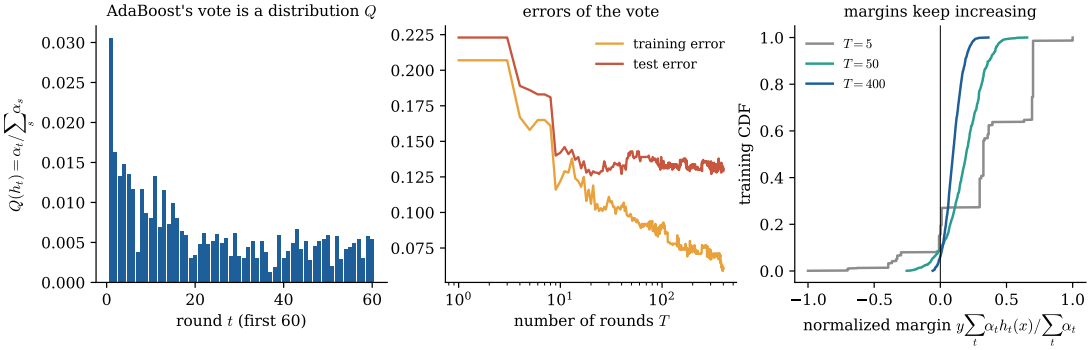


Figure 6: AdaBoost with decision stumps on a simulated problem (1000 training examples, 20 features). Left: the normalized weights $Q(h_t)$ of the first 60 stumps form a probability distribution over the weak learners. Middle: the training error of the vote goes to zero while the test error stabilizes. Right: even after the training error is small, the distribution of the normalized margins keeps moving to the right; this observation motivated the margin theory of boosting (Schapire et al., 1998), to which we come back in Section 5.9. [code: `make_figures.py: fig_adaboost`]

real-valued regressors, and stacking (Wolpert, 1992) with a linear meta-learner learns the weights of a vote on a held-out sample.

Even kernel methods fit in this picture. With random Fourier features (Rahimi and Recht, 2007), a kernel machine is approximated by $\operatorname{sign}\left(\sum_k \alpha_k \cos(\omega_k^\top x + b_k)\right)$, where the frequencies ω_k are drawn from a distribution determined by the kernel. This is a vote over an infinite family of voters indexed by ω (Letarte et al., 2019b), and it is precisely the model studied in the project of the first part of the course.

In all these cases, the learned predictor has the form

$$B_Q(x) = \operatorname{sign}\left(\mathop{\mathbb{E}}_{h\sim Q}[h(x)]\right), \qquad Q \in \mathcal{M}(\mathcal{H}), \tag{1.2}$$

where $\mathcal{H}$ is a (finite or infinite) set of voters and Q a probability distribution on $\mathcal{H}$ (Fig. 7). Table 1 translates the notation of the Ensemble Methods course into the

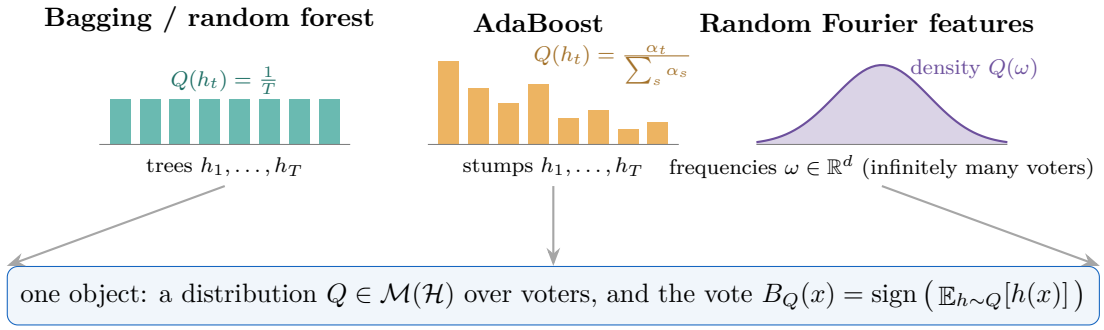


Figure 7: Three ensemble methods of the course, one object. Bagging and random forests put the same weight on every tree, AdaBoost normalizes the weights α_t of its weak learners, and a kernel machine with random features draws its voters from a continuous distribution. In each case the predictor is the Q-weighted majority vote B_Q. [code: `tikz/ensembles_as_q.tex`]

Table 1: From the notation of the Ensemble Methods course to the notation of these notes.

Ensemble Methods course	**These notes**	**Comment**
models $h_1, \ldots, h_T$	voters $h \in \mathcal{H}$	$\mathcal{H}$ may be finite (n voters) or infinite
weights $\frac{1}{T}$ or $\frac{\alpha_t}{\sum_s \alpha_s}$	posterior $Q \in \mathcal{M}(\mathcal{H})$	learned from the sample S
—	prior $P \in \mathcal{M}(\mathcal{H})$	fixed *before* seeing S
combined model H_T	majority vote B_Q	$B_Q(x) = \text{sign}\,\mathbb{E}_{h\sim Q} h(x)$
average error of the models	Gibbs risk $R_{\mathcal{D}}(G_Q)$	risk of a voter drawn from Q
$R(h)$, $R_S(h)$	$R_{\mathcal{D}}(h)$, $\widehat{R}_S(h)$	true and empirical risks
capacity $\mathfrak{R}_m(\mathcal{H})$	divergence $\text{KL}(Q\|P)$	price of learning
correlation of the errors	disagreement $d_{\mathcal{D}}(Q)$, joint error $e_{\mathcal{D}}(Q)$	diversity (Section 5)

notation of these notes, which is the standard one of the PAC-Bayesian literature. The learning problem for ensembles can therefore be rephrased in a single sentence: *how should we choose a good distribution Q over the voters?* This is exactly the question addressed by PAC-Bayesian theory. Figure 8 shows, on a two-dimensional problem, the phenomenon of our puzzling experiment: the individual trees of a random forest are irregular and rather inaccurate, whereas their vote is smooth and accurate.

1.4 What classical bounds say about majority votes

Can the generalization bounds of the first part of the course answer our two questions? Let us write the Rademacher bound recalled in Section 1.1 with the notation of these notes (Bartlett and Mendelson, 2002; Mohri et al., 2018): for a loss with values in $[0, 1]$, with probability at least $1 - \delta$ over the draw of $S \sim \mathcal{D}^m$, simultaneously for all $h \in \mathcal{H}$,

$$R_{\mathcal{D}}(h) \leq \widehat{R}_S(h) + 2\,\mathfrak{R}_m(\mathcal{H}) + \sqrt{\frac{\ln(1/\delta)}{2m}}. \tag{1.3}$$

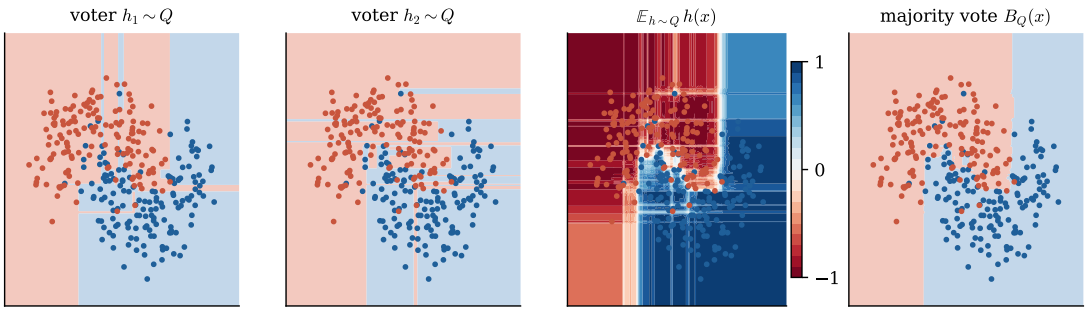


Figure 8: Two voters h_1, h_2 drawn from the posterior Q of a random forest (uniform over 200 randomized trees), the average vote $\mathbb{E}_{h\sim Q} h(x)$ and the majority vote B_Q on the `moons` dataset. Individual voters are very irregular; the vote is smoother and more accurate. Explaining, and certifying, this improvement is the purpose of this part of the course. [code: `make_figures.py: fig_gibbs_vs_mv`]

Such a bound is called *uniform*: it holds for every hypothesis of the class, whatever the way it has been chosen, and the price to pay is a measure of the complexity of the whole class. There are two reasons why this is not satisfactory for ensembles.

The first one is good news in disguise: taking averages does not make a class more complex. The empirical Rademacher complexity of the convex hull $\mathrm{conv}(\mathcal{H})$, that is of the set of all weighted averages of voters, is exactly the same as that of $\mathcal{H}$, because a linear function reaches its supremum over a convex set at one of its extreme points (Exercise 1.2). Schapire et al. (1998) and Koltchinskii and Panchenko (2002) exploited this remark to derive *margin bounds* for voting classifiers that do not depend on the number of voters. This was the first explanation of the surprising behaviour of AdaBoost in Fig. 6, which does not overfit when more rounds are added. Unfortunately, for realistic sample sizes these bounds are almost always larger than one, and hence useless as guarantees.

The second reason is more fundamental: a uniform bound ignores *what has been learned*. Whether the forest puts all its weight on a few very good trees or spreads it over many diverse trees, the bound (1.3) pays the same price. Intuitively, a distribution Q that remains "close" to something we could have chosen *before* seeing the data should be cheaper than a distribution tuned very precisely on the sample. PAC-Bayesian theory turns this intuition into a theorem: the complexity of the class is replaced by a *divergence* between the learned distribution Q, called the *posterior*, and a reference distribution P fixed before seeing the data, called the *prior*.

1.5 The PAC-Bayesian principle

Figure 9 summarizes the setting. Before seeing the data, we fix a prior P over the voters, for instance the uniform distribution over the trees of a forest. A learning algorithm then looks at the sample and produces a posterior Q, which may depend on the data in any way we like. Finally, a bound compares the two, and certifies the predictor associated with Q.

Which predictor does a PAC-Bayesian bound certify? Its central object is not the vote but a randomized cousin of it, the *Gibbs classifier* G_Q: to predict the label of a point x, it draws a voter h at random according to Q and outputs $h(x)$. Since a new

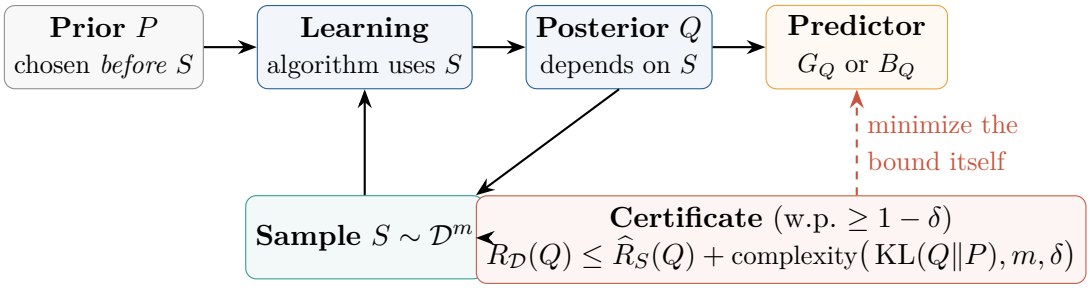


Figure 9: The PAC-Bayesian setting. The prior P encodes what we believe before seeing the data; the posterior Q may depend arbitrarily on S. The generalization bound holds *simultaneously for all posteriors*, so it can be used both as a certificate and as an objective to minimize (*self-bounding* learning, Section 6). [code: `tikz/principle.tex`]

voter is drawn for each prediction, its risk is simply the average risk of the voters,

$$R_{\mathcal{D}}(G_Q) = \mathop{\mathbb{E}}_{h\sim Q}\big[R_{\mathcal{D}}(h)\big]\,, \qquad \widehat{R}_S(G_Q) = \mathop{\mathbb{E}}_{h\sim Q}\Big[\widehat{R}_S(h)\Big]\,.$$

In our running example, the Gibbs classifier is "pick a tree at random and use it", and its test error is the 19% of an average tree. A typical PAC-Bayesian theorem (McAllester, 1999b, in the form of Maurer, 2004) then reads as follows: for any prior P that does not depend on S, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$,

$$\forall Q \in \mathcal{M}(\mathcal{H}), \qquad \underbrace{R_{\mathcal{D}}(G_Q)}_{\text{unknown}} \leq \underbrace{\widehat{R}_S(G_Q)}_{\text{computed on } S} + \sqrt{\frac{\underbrace{\mathrm{KL}(Q\|P)}_{\text{distance to the prior}} + \ln \underbrace{\frac{2\sqrt{m}}{\delta}}_{\text{confidence}}}{2m}}. \tag{1.4}$$

Let us read this statement slowly, because it contains the whole theory in a nutshell. On the left is the unknown quantity we care about, the true risk of the Gibbs classifier. On the right, the first term is its empirical risk, which we can compute. The second term is a price, which grows with the Kullback–Leibler divergence $\mathrm{KL}(Q\|P)$ between the posterior and the prior, and decreases with the sample size m. Three features make this statement very different from (1.3). First, the price depends on the *learned* distribution Q and on the prior, and not on the size of $\mathcal{H}$, which may even be infinite and uncountable. Second, the bound holds *for all posteriors simultaneously*: we are free to choose Q after having seen the data, and in particular to choose the posterior that minimizes the right-hand side. Third, every quantity on the right can be computed from the sample, so the bound is a genuine *risk certificate*: a number which, with high probability, is larger than the true risk of the predictor, and which can be reported next to it.

One difficulty remains: in practice we predict with the majority vote B_Q, not with the Gibbs classifier, and in our experiment the two differ enormously (4.4% against 19%). Section 5 is devoted to the relations between the two predictors. We will see that the gap between them is governed by the *disagreement* between the voters, that is by their diversity, which answers the explanatory question raised at the beginning of this section.

In a nutshell PAC-Bayes = PAC + Bayes. From the PAC framework (Probably Approximately Correct, Valiant, 1984) it inherits the form of the guarantee: a bound on the true risk which holds with high probability over the sample, whatever the data distribution $\mathcal{D}$. From Bayesian statistics it borrows the vocabulary of priors and posteriors. But, unlike in Bayesian inference, the posterior is not required to be given by Bayes' rule, and the data are not assumed to be generated by a probabilistic model: the guarantee is purely frequentist.

1.6 A short history

The first PAC-Bayesian analyses are due to Shawe-Taylor and Williamson (1997) and to McAllester (1998, 1999b,a), who combined the PAC framework with Bayesian priors and thereby generalized the *Occam's razor* bound of Blumer et al. (1987), which we present in Section 2.4. The bound was then sharpened by using the binary Kullback–Leibler divergence (Seeger, 2002; Langford, 2005), and Maurer (2004) gave it the simple constant that we use today. Independently, Catoni (2007) (see also Audibert, 2004; Zhang, 2006) developed a very complete theory, emphasizing the role of the Gibbs posterior and the analogy with statistical physics. At the same time, Langford and Shawe-Taylor (2002) and McAllester (2003b) used PAC-Bayes to explain why large margins make support vector machines generalize.

The link with majority votes appeared early (McAllester, 1999a), but it became central with the work of Lacasse et al. (2006), who introduced the *C-bound*, and of Germain et al. (2009, 2015), who turned PAC-Bayesian bounds into learning algorithms such as MinCq (Roy et al., 2011). The field gained a new momentum in 2017, when Dziugaite and Roy (2017) obtained the first non-vacuous bounds for deep neural networks. Since then, a series of works has produced tight bounds for random forests (Lorenzen et al., 2019), second-order bounds for majority votes (Masegosa et al., 2020; Wu et al., 2021), and algorithms which minimize them directly (Viallard et al., 2021; Zantedeschi et al., 2021). Recent surveys are those of Guedj (2019), Alquier (2024) and Hellström et al. (2025). Figure 10 summarizes these milestones.

1.7 Organization of the notes

The notes follow the two questions of our puzzling experiment, in four parts. *Part I* gathers the foundations. After this introduction, Section 2 presents the mathematical tools — concentration inequalities, the Kullback–Leibler divergence and the change of measure inequality — and ends with the Occam bound, the ancestor of all PAC-Bayesian bounds. *Part II* builds the certificates. Section 3 proves a general PAC-Bayesian theorem and derives from it the classical bounds of McAllester, Seeger, Maurer and Catoni; Section 4 discusses how to choose the posterior and the prior; Section 5 finally goes from the Gibbs classifier to the majority vote, where the diversity of the voters comes into play and where our first question receives its answer. *Part III* turns the certificates into learning objectives: Section 6 explains why a bound can be minimized without losing its validity, and Section 7 presents, and runs, the resulting algorithms for forests, boosting-like methods, linear classifiers, kernels and neural networks. *Part IV* (Section 8) opens on current research.

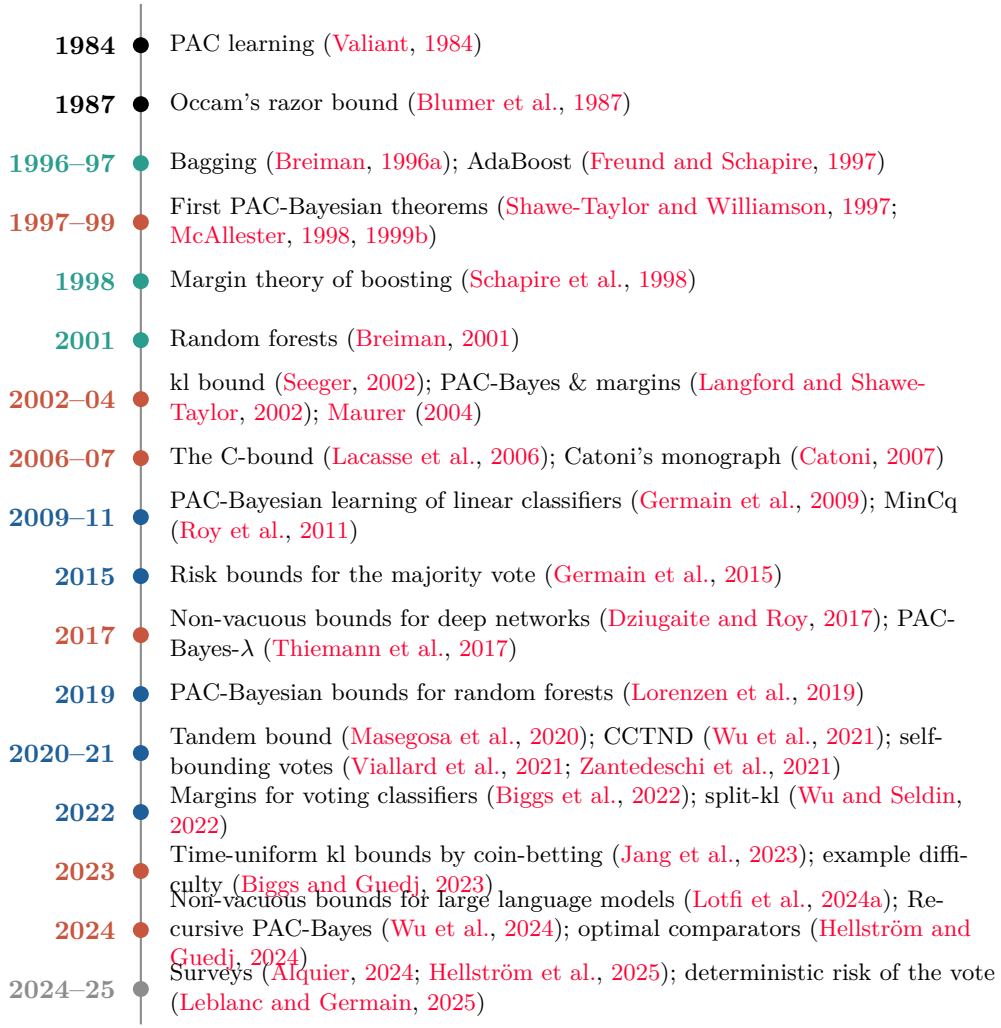


Figure 10: A few milestones. Black: learning theory; teal: ensemble methods; red: PAC-Bayesian bounds; blue: PAC-Bayes for majority votes; grey: surveys. [code: `tikz/timeline.tex`]

To remember from Section 1

Ensemble methods learn a distribution Q over a set of voters and predict with the corresponding majority vote B_Q, which can be much more accurate than its voters. PAC-Bayesian bounds replace the complexity of the whole class by the divergence $\mathrm{KL}(Q\|P)$ between this distribution and a prior fixed before seeing the data, and they hold for all posteriors simultaneously. They certify the Gibbs classifier (a voter drawn at random) directly, the majority vote through the relations of Section 5, and they can themselves be used as learning objectives (Section 6).

Check your understanding

Short questions to answer without computation; answers in Appendix A.5.

Q1.1. Why is the output of AdaBoost a probability distribution over the weak learners?

Q1.2. In the running example, what are the test errors of the Gibbs classifier and of the majority vote?

Q1.3. What may the posterior depend on, and what may the prior not depend on?

Q1.4. Why is the Rademacher complexity of the convex hull of $\mathcal{H}$ not larger than that of $\mathcal{H}$?

Exercises

Exercise 1.1. Write the predictors of bagging, of AdaBoost and of a kernel machine with random Fourier features in the form (1.2). In each case, what is the set of voters $\mathcal{H}$, and what is the distribution Q?

Exercise 1.2. Show that the empirical Rademacher complexity of $\mathrm{conv}(\mathcal{H})$ equals the one of $\mathcal{H}$. *Hint:* for a fixed vector of signs σ, the function $f \mapsto \frac{1}{m}\sum_i \sigma_i f(x_i)$ is linear in f.

Exercise 1.3. In Fig. 6, the training error reaches zero after a few dozen rounds. Using the right panel, explain why a bound based on the empirical margin distribution can keep decreasing afterwards.

Exercise 1.4. **(Comprehension)** True or false? Justify in one or two sentences. (a) Since the vote of the running example is better than an average tree, the risk of a majority vote is always smaller than the risk of its Gibbs classifier, $R_{\mathcal{D}}(B_Q) \le R_{\mathcal{D}}(G_Q)$. (b) The Gibbs risk $R_{\mathcal{D}}(G_Q)$ always lies between the smallest and the largest risk of the voters in the support of Q.

Exercise 1.5. **(Comprehension)** A student proposes the following trick: "since the bound (1.4) holds for any prior, I learn my posterior Q and then take $P = Q$, so that $\mathrm{KL}(Q\|P) = 0$ and the certificate is simply $\widehat{R}_S(G_Q) + \sqrt{\ln(2\sqrt{m}/\delta)/(2m)}$." Explain what is wrong.

Exercise 1.6. **(Application)** AdaBoost produces three weak learners with weighted errors $\varepsilon_1 = 0.1$, $\varepsilon_2 = 0.2$ and $\varepsilon_3 = 0.3$. Compute their weights α_t and the normalized distribution Q. On a point x where $h_1(x) = +1$ and $h_2(x) = h_3(x) = -1$, what does the vote predict? Is the best weak learner always obeyed?

Exercise 1.7. **(Application)** Consider the bound (1.4) with $\delta = 0.05$ and $m = 1000$. Compute its complexity term for the uniform posterior $Q = P$, and for a Dirac posterior on one of $n = 100$ voters with the uniform prior (for which $\mathrm{KL}(Q\|P) = \ln n$). How many examples are needed for the complexity term of the uniform posterior to be below 0.02?

Exercise 1.8. **(Proof)** Let $\mathcal{H} = \{h_1, \dots, h_n\}$ be finite and $Q \in \mathcal{M}(\mathcal{H})$. Show that $R_{\mathcal{D}}(G_Q) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[W_Q(x,y)]$ with $W_Q(x,y) = \sum_t Q(h_t)\mathbb{1}[h_t(x) \neq y]$, and deduce that $\min_{t:Q(h_t)>0} R_{\mathcal{D}}(h_t) \le R_{\mathcal{D}}(G_Q) \le \max_{t:Q(h_t)>0} R_{\mathcal{D}}(h_t)$.

Exercise 1.9. **(Proof)** For voters with values in $\{-1,+1\}$, show that $\mathbb{1}[h(x) \neq y] = \frac{1 - y\,h(x)}{2}$, deduce that the margin $M_Q(x,y) = y\,\mathbb{E}_{h\sim Q}[h(x)]$ equals $1 - 2W_Q(x,y)$, and show that if the vote B_Q errs on (x,y), then $W_Q(x,y) \ge \frac{1}{2}$ (whatever the convention chosen for $\mathrm{sign}(0)$).

Exercise 1.10. **(Proof)** Let $\varepsilon_1, \dots, \varepsilon_T$ be centred random variables with the same variance σ^2 and the same correlation ρ between two different ones. (a) Prove formula (1.1) for $\mathbb{E}\big[(\frac{1}{T}\sum_t \varepsilon_t)^2\big]$. (b) Deduce that $\rho \ge -\frac{1}{T-1}$. (c) Show that $\big(\frac{1}{T}\sum_t \varepsilon_t\big)^2 \le \frac{1}{T}\sum_t \varepsilon_t^2$ pointwise, and interpret this inequality for ensembles.

Exercise 1.11. **(Application)** In the setting of Fig. 2 ($\sigma^2 = 1$), how many models must be averaged to divide the error of one model by 10 when $\rho = 0$? When $\rho = 0.3$, what is the smallest achievable error, and how many models are needed to be within 10% of it? Relate the answer to the random selection of features in random forests, which lowers the correlation between the trees.

2 Mathematical toolbox

About this section

Guiding question. *How can an error measured on a sample be turned into a guarantee on the true error, first for one classifier, then for many?*

We fix the formal setting (Gibbs classifier and majority vote), and we gather the probabilistic tools that are used over and over in PAC-Bayesian proofs: concentration inequalities for a single voter, stated with the binary Kullback–Leibler divergence, the Kullback–Leibler divergence between distributions over voters, and the change of measure inequality, which is the key step of every PAC-Bayesian bound. The section ends with the Occam bound, which already contains the main idea of PAC-Bayes.

The section climbs three steps. We first learn how to certify *one* classifier fixed in advance (concentration inequalities). We then introduce the tool that lets us move from a distribution fixed in advance to a distribution chosen with the data (the change of measure). Finally, combining the two in the simplest case gives the Occam bound, which certifies a classifier *chosen* among many. The next section will take the last step, from one chosen classifier to a chosen distribution over classifiers.

2.1 Setting

Before stating any bound, we must fix the objects that the bounds talk about: the data, the voters, their risks, and the two ways of combining voters. We consider binary classification. Examples (x, y) are drawn independently from an unknown distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$ with $\mathcal{Y} = \{-1, +1\}$, and we observe a learning sample $S = \{(x_i, y_i)\}_{i=1}^m \sim \mathcal{D}^m$. We are given a set $\mathcal{H}$ of voters and a loss function $\ell : \mathcal{H} \times (\mathcal{X} \times \mathcal{Y}) \to [0, 1]$, which will be the 0-1 loss $\ell(h, (x, y)) = \mathbb{1}[h(x) \neq y]$ unless stated otherwise. The true risk and the empirical risk of a voter are

$$R_{\mathcal{D}}(h) = \mathop{\mathbb{E}}_{(x,y)\sim\mathcal{D}} [\ell(h, (x, y))] \qquad \text{and} \qquad \widehat{R}_S(h) = \frac{1}{m} \sum_{i=1}^{m} \ell(h, (x_i, y_i)).$$

For voters with values in $\{-1, +1\}$ we will often use the identity $\mathbb{1}[h(x) \neq y] = \frac{1 - y\,h(x)}{2}$, which turns the 0-1 loss into a linear function of the prediction.

Definition 2.1: Gibbs classifier and majority vote

Let $Q \in \mathcal{M}(\mathcal{H})$ be a distribution over $\mathcal{H}$. The **Gibbs classifier** G_Q predicts the label of x by drawing $h \sim Q$ and returning $h(x)$. Its true and empirical risks are

$$R_{\mathcal{D}}(G_Q) = \mathop{\mathbb{E}}_{h\sim Q} [R_{\mathcal{D}}(h)], \qquad \widehat{R}_S(G_Q) = \mathop{\mathbb{E}}_{h\sim Q} \left[\widehat{R}_S(h)\right].$$

The **(Q-weighted) majority vote** is the deterministic classifier $B_Q(x) = \operatorname{sign}\left(\mathbb{E}_{h\sim Q}[h(x)]\right)$, with the convention $\operatorname{sign}(0) = -1$, so that a tie is counted as an error in the analysis. Its risk is $R_{\mathcal{D}}(B_Q) = \mathbb{P}_{(x,y)\sim\mathcal{D}}(B_Q(x) \neq y)$.

The two predictors behave very differently. The Gibbs risk is *linear* in Q, which makes it easy to analyse, and it always lies between the risk of the best and the risk

of the worst voter. The risk of the majority vote is not linear in Q, and it can be much smaller than the Gibbs risk: in the running example of Section 1, the Gibbs risk of the forest is 19% while its vote only errs on 4.4% of the test points. A convenient way to relate the two is to exchange the expectations over the voters and over the data (Fubini's theorem):

$$R_{\mathcal{D}}(G_Q) = \mathop{\mathbb{E}}_{(x,y)\sim\mathcal{D}}[W_Q(x,y)]\,, \qquad \text{where} \qquad W_Q(x,y) = \mathop{\mathbb{E}}_{h\sim Q}[\mathbb{1}[h(x)\neq y]]$$

is the proportion, weighted by Q, of voters that err on (x,y). The random variable W_Q will be the main character of Section 5. For now, the important point is that the Gibbs risk is an average of the risks of single voters, so our first task is to control the risk of one voter.

2.2 Concentration inequalities

All generalization bounds rely on the same basic fact: for a *fixed* voter h, chosen before seeing the data, the empirical risk $\widehat{R}_S(h)$ is an average of m independent random variables with values in $[0,1]$ and mean $R_{\mathcal{D}}(h)$, hence it concentrates around $R_{\mathcal{D}}(h)$. The tools below quantify this concentration. We start with the simplest of all.

Lemma 2.2: Markov's inequality

Let $Z \geq 0$ be a random variable and $a > 0$. Then $\mathbb{P}(Z \geq a) \leq \frac{\mathbb{E}Z}{a}$. Equivalently, for any $\delta \in (0,1]$, with probability at least $1-\delta$, $Z \leq \frac{\mathbb{E}Z}{\delta}$.

Proof. Since $Z \geq 0$, we have $a\mathbb{1}[Z \geq a] \leq Z$ pointwise (Fig. 11): the left-hand side is 0 when $Z < a$, and $a \leq Z$ otherwise. Taking expectations,

$$\begin{aligned} a\,\mathbb{P}(Z \geq a) &= \mathbb{E}\big[a\mathbb{1}[Z \geq a]\big] \\ &\quad\downarrow \text{ monotonicity of the expectation and } a\mathbb{1}[Z \geq a] \leq Z \\ &\leq \mathbb{E}Z. \end{aligned}$$

For the second form, apply the first one with $a = \mathbb{E}Z/\delta$. □

Markov's inequality looks weak, but it becomes remarkably powerful when it is applied to a well-chosen transformation of the variable of interest. Applied to the exponential $e^{\lambda(p-\hat{p})}$, it gives Hoeffding's inequality.

Lemma 2.3: Hoeffding's inequality (Hoeffding, 1963)

Let $Z_1, \ldots, Z_m$ be i.i.d. random variables in $[0,1]$ with mean p, and $\hat{p} = \frac{1}{m}\sum_i Z_i$. For any $\lambda \in \mathbb{R}$, $\mathbb{E}\, e^{\lambda(p-\hat{p})} \leq e^{\lambda^2/(8m)}$, and, for $\varepsilon > 0$, $\mathbb{P}(p - \hat{p} \geq \varepsilon) \leq e^{-2m\varepsilon^2}$.

Proof. We admit *Hoeffding's lemma*: a random variable U with zero mean and values in an interval of length c satisfies $\mathbb{E}e^{\lambda U} \leq e^{\lambda^2 c^2/8}$ for every $\lambda \in \mathbb{R}$ (see Boucheron et al.,

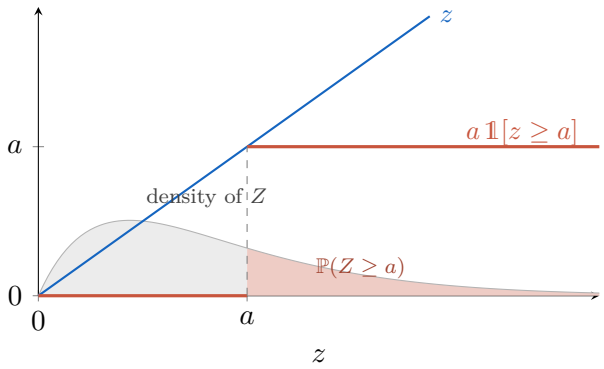


Figure 11: Why Markov's inequality holds: for a non-negative variable, the step $z \mapsto a\mathbb{1}[z \geq a]$ (red) lies below the identity $z \mapsto z$ (blue), so that $a\,\mathbb{P}(Z \geq a) \leq \mathbb{E}Z$. The shaded red area is the tail probability. [code: `tikz/markov.tex`]

2013). The variables $U_i = \frac{p - Z_i}{m}$ are independent, centred, and take their values in an interval of length $\frac{1}{m}$, so that

$$\begin{aligned}
\mathbb{E}\, e^{\lambda(p-\hat{p})} &= \mathbb{E} \prod_{i=1}^{m} e^{\lambda U_i} \\
&\downarrow \text{ independence of } Z_1, \dots, Z_m \\
&= \prod_{i=1}^{m} \mathbb{E}\, e^{\lambda U_i} \\
&\downarrow \text{ Hoeffding's lemma with } c = \frac{1}{m} \\
&\leq \prod_{i=1}^{m} e^{\lambda^2/(8m^2)} = e^{\lambda^2/(8m)}.
\end{aligned}$$

For the tail, we apply Markov's inequality to the exponential (this is *Chernoff's method*): for every $\lambda > 0$,

$$\begin{aligned}
\mathbb{P}(p - \hat{p} \geq \varepsilon) &= \mathbb{P}\big(e^{\lambda(p-\hat{p})} \geq e^{\lambda\varepsilon}\big) \\
&\downarrow \text{ Markov's inequality} \\
&\leq e^{-\lambda\varepsilon}\, \mathbb{E}\, e^{\lambda(p-\hat{p})} \\
&\downarrow \text{ first part} \\
&\leq e^{-\lambda\varepsilon + \lambda^2/(8m)} \\
&\downarrow \text{ the exponent is minimal for } \lambda = 4m\varepsilon \\
&= e^{-2m\varepsilon^2}.
\end{aligned}$$

□

Inverting Hoeffding's inequality, the true risk of a fixed voter satisfies $R_{\mathcal{D}}(h) \leq \widehat{R}_S(h) + \sqrt{\ln(1/\delta)/(2m)}$ with probability at least $1 - \delta$. In words, the empirical risk underestimates the true risk by more than $\sqrt{\ln(1/\delta)/(2m)}$ with probability at most δ. This is already a certificate, but its width does not depend on the value of the risk, which is wasteful when the risk is small: a Bernoulli variable of mean 0.01 fluctuates much less than a Bernoulli variable of mean 0.5.

The sharpest statement for Bernoulli variables uses the **binary Kullback–**

Leibler divergence

$$\mathrm{kl}(q\|p) = q\ln\frac{q}{p} + (1-q)\ln\frac{1-q}{1-p}, \qquad q,p\in[0,1],$$

which is the Kullback–Leibler divergence between two Bernoulli distributions of parameters q and p.

The binary kl has a very concrete meaning: it measures how *surprising* it is to observe a frequency q of errors when the true error rate is p. Indeed, the probability that m independent coin flips of bias p produce a frequency close to q behaves like $e^{-m\,\mathrm{kl}(q\|p)}$. This is Sanov's theorem for coins, and the next lemma makes one side of it precise. A small kl means that the observation is plausible under p, while a kl of ε means that it has a probability of order $e^{-m\varepsilon}$. Turning this around, the true risks p compatible with an observed empirical risk $\hat{q}$ are those for which $\mathrm{kl}(\hat{q}\|p)$ is small.

Lemma 2.4: Chernoff's bound and Maurer's lemma

Let $Z_1,\dots,Z_m$ be i.i.d. random variables in $[0,1]$ with mean p and average $\hat{p}$. Chernoff's bound states that, for $\varepsilon\in(0,p)$, $\mathbb{P}\big(\hat{p}\le p-\varepsilon\big)\le e^{-m\,\mathrm{kl}(p-\varepsilon\|p)}$. Maurer's lemma (Maurer, 2004) states that, for $m\ge 8$,

$$\mathbb{E}\Big[e^{m\,\mathrm{kl}(\hat{p}\|p)}\Big] \le \xi(m) := \sum_{k=0}^{m}\binom{m}{k}\Big(\frac{k}{m}\Big)^k\Big(1-\frac{k}{m}\Big)^{m-k}, \tag{2.1}$$

with $\sqrt{m}\le\xi(m)\le 2\sqrt{m}$; the first inequality in (2.1) is an equality when the Z_i are Bernoulli variables.

The two statements say the same thing in two different forms. Chernoff's bound controls a *tail*: a large deviation of the empirical mean is exponentially unlikely, at the rate given by the kl. Maurer's lemma controls an *exponential moment*: on average, $e^{m\,\mathrm{kl}(\hat{p}\|p)}$ is only of order $\sqrt{m}$, although it could a priori be as large as $e^{m\ln 2}$. The first form is the natural one for a single classifier; the second is the one that PAC-Bayesian proofs need, because an expectation can be averaged over voters, whereas a tail probability cannot.

Proof sketch. The key step in the second statement is that, for any convex function f on $[0,1]^m$, $\mathbb{E}f(Z_1,\dots,Z_m)\le\mathbb{E}f(B_1,\dots,B_m)$ where the B_i are i.i.d. Bernoulli(p) variables: a variable in $[0,1]$ is "less spread" than a Bernoulli variable with the same mean. Since $q\mapsto e^{m\,\mathrm{kl}(q\|p)}$ is convex, it suffices to treat the case $m\hat{p}\sim\mathrm{Bin}(m,p)$, for which

$$\mathbb{E}e^{m\,\mathrm{kl}(\hat{p}\|p)} = \sum_k\binom{m}{k}p^k(1-p)^{m-k}\Big(\frac{k/m}{p}\Big)^k\Big(\frac{1-k/m}{1-p}\Big)^{m-k} = \xi(m),$$

a quantity which, remarkably, does not depend on p; Stirling's formula gives the bounds on $\xi(m)$. □

Inverting the kl. Both statements bound the kl between the empirical and the true risk, whereas what we want is an explicit upper bound on the true risk. We therefore need to invert the binary kl. For a fixed q, the function $p\mapsto\mathrm{kl}(q\|p)$ is convex, vanishes

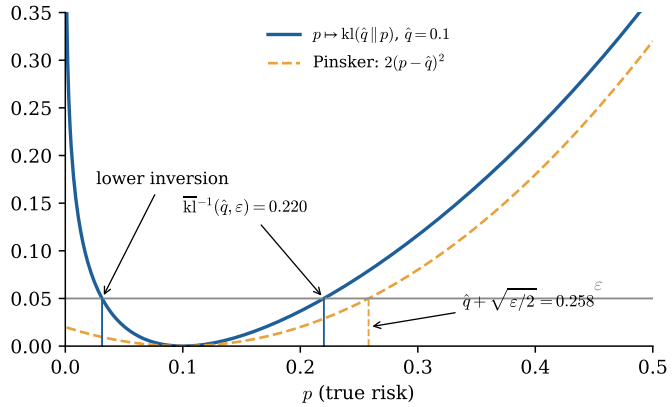


Figure 12: The function $p \mapsto \mathrm{kl}(\hat{q}\|p)$ for $\hat{q} = 0.1$. For $\varepsilon = 0.05$, the values of p compatible with $\mathrm{kl}(\hat{q}\|p) \leq \varepsilon$ form an interval whose upper end $\overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon) = 0.220$ is smaller than the Pinsker relaxation $\hat{q} + \sqrt{\varepsilon/2} = 0.258$. [code: `make_figures.py: fig_kl_inverse`]

Algorithm 1: Upper inversion of the binary kl by bisection

Input: empirical value $\hat{q} \in [0,1]$, budget $\varepsilon \geq 0$, tolerance τ

$a \leftarrow \hat{q}$, $b \leftarrow 1$;
while $b - a > \tau$ **do**
  $c \leftarrow (a+b)/2$;
  if $\mathrm{kl}(\hat{q}\|c) > \varepsilon$ **then** $b \leftarrow c$;
  else $a \leftarrow c$;

Output: b ; // $\geq \overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon)$, within τ

at $p = q$ and increases on $[q, 1]$ (see Fig. 12). The inequality $\mathrm{kl}(\hat{q}\|p) \leq \varepsilon$ therefore defines an interval $[\underline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon), \overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon)]$ of admissible values of p, with

$$\overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon) = \sup\{p \in [\hat{q}, 1] : \mathrm{kl}(\hat{q}\|p) \leq \varepsilon\}, \qquad \underline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon) = \inf\{p \in [0, \hat{q}] : \mathrm{kl}(\hat{q}\|p) \leq \varepsilon\}.$$

These quantities have no closed form, but since the function is monotone on each side of $\hat{q}$ they are computed to any precision by a few iterations of bisection (Algorithm 1). In the rest of the notes, $\overline{\mathrm{kl}}^{-1}$ will be our standard way of turning an empirical quantity into a certified upper bound.

Example 2.1 (Inverting the kl by hand). Let $\hat{q} = 0.1$ and $\varepsilon = 0.05$. We know that $\overline{\mathrm{kl}}^{-1}(0.1, 0.05) \geq 0.1$. Since $\mathrm{kl}(0.1\|0.2) \approx 0.037 < 0.05$ and $\mathrm{kl}(0.1\|0.25) \approx 0.072 > 0.05$, the answer lies in $[0.2, 0.25]$; the midpoint gives $\mathrm{kl}(0.1\|0.225) \approx 0.053 > 0.05$, so the answer is in $[0.2, 0.225]$, and so on. After twenty steps the interval has length 10^{-6} and we find $\overline{\mathrm{kl}}^{-1}(0.1, 0.05) \approx 0.220$.

Bisection gives numbers, but not formulas. To understand how kl-based bounds behave, and to derive bounds with a closed form, we also need explicit (if looser) upper bounds on $\overline{\mathrm{kl}}^{-1}$; the next two inequalities provide them.

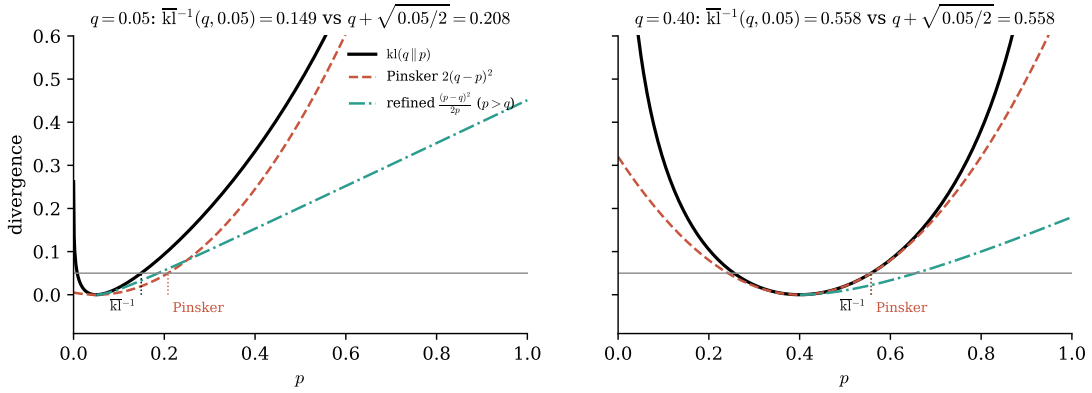


Figure 13: The binary kl $p \mapsto \mathrm{kl}(q\|p)$ (black) and its quadratic lower bounds, for $q = 0.05$ (left) and $q = 0.4$ (right). The admissible values of p are those below the horizontal line $\varepsilon = 0.05$. Near $q = 0$ the kl is much steeper than Pinsker's parabola, so its inverse $\overline{\mathrm{kl}}^{-1}$ is much smaller than $q + \sqrt{\varepsilon/2}$; near $q = \frac{1}{2}$ the two almost coincide. [code: `make_figures.py: fig_kl_pinsker`]

Lemma 2.5: Pinsker-type inequalities

For $q, p \in [0, 1]$, $\mathrm{kl}(q\|p) \geq 2(q-p)^2$ (Pinsker's inequality), hence $\overline{\mathrm{kl}}^{-1}(q, \varepsilon) \leq q + \sqrt{\varepsilon/2}$. Moreover, if $q < p$, then $\mathrm{kl}(q\|p) \geq \frac{(p-q)^2}{2p}$ (refined Pinsker inequality), hence $\overline{\mathrm{kl}}^{-1}(q, \varepsilon) \leq q + \sqrt{2q\varepsilon} + 2\varepsilon$.

Figure 13 compares the binary kl with its two quadratic lower bounds. Pinsker's inequality recovers a Hoeffding-type width $\sqrt{\varepsilon/2}$. The refined inequality explains why it is so much better for small risks: when q is close to zero, the width of the confidence interval is of order ε instead of $\sqrt{\varepsilon}$, i.e. of order $1/m$ instead of $1/\sqrt{m}$ when $\varepsilon \propto 1/m$. This "fast rate" is visible on the left panel of Fig. 14: for an empirical risk of 1% and $m = 1000$ examples, the kl interval is four times narrower than the Hoeffding interval. The right panel checks by simulation that both intervals are valid: their failure frequency always stays below δ, and usually well below, as these inequalities are worst-case statements.

Running example Suppose that we had fixed the forest before seeing the 540 test digits, and that we use them only to certify it: this is the classical *test-set bound*. The vote errs on 4.4% of them; with $\delta = 0.05$, Hoeffding's inequality certifies a true risk below $0.044 + \sqrt{\ln 20/1080} = 0.097$, and the kl inversion below 0.069. For an average tree (19% of test errors), the two certificates are 0.243 and 0.234: the kl only makes a difference for small risks. These numbers will serve as a reference. Our goal is to obtain comparable guarantees *without* setting aside a test sample, from the training data alone.

Second moments. The inequalities above only use the fact that the variables are bounded. Majority votes will require a finer tool, which also uses the *variance* of a random variable, and not only its mean.

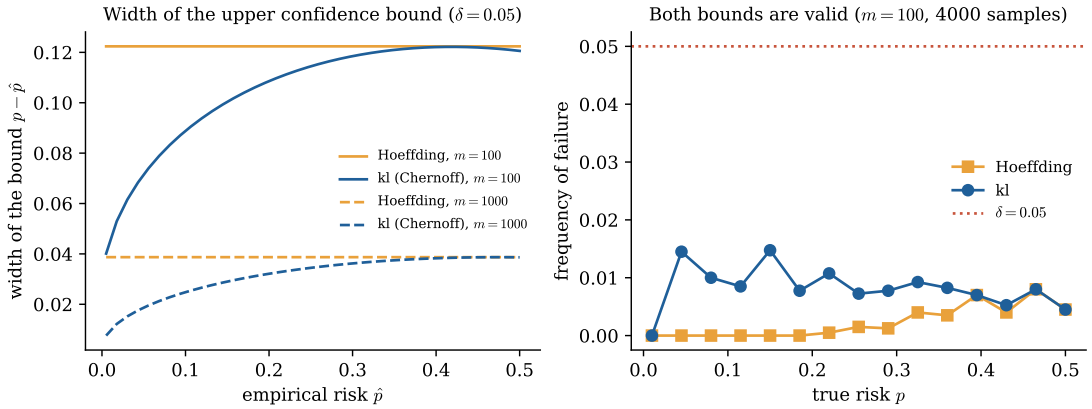


Figure 14: Left: width $p - \hat{p}$ of the upper confidence bound on the risk of a fixed voter given by Hoeffding's inequality (constant) and by the kl inversion, as a function of the empirical risk, for $m = 100$ and $m = 1000$. Right: frequency, over 4000 simulated samples of size 100, of the event "the true risk is above the bound"; both bounds are valid. [code: `make_figures.py: fig_concentration`]

Lemma 2.6: Chebyshev–Cantelli inequality

Let Z be a random variable with mean μ and variance v. For every $a > 0$, $\mathbb{P}(Z - \mu \geq a) \leq \frac{v}{v+a^2}$ and $\mathbb{P}(Z - \mu \leq -a) \leq \frac{v}{v+a^2}$.

Proof. For any $t \geq 0$, $\{Z - \mu \geq a\} \subseteq \{(Z - \mu + t)^2 \geq (a+t)^2\}$, so by Markov's inequality $\mathbb{P}(Z - \mu \geq a) \leq \frac{\mathbb{E}(Z-\mu+t)^2}{(a+t)^2} = \frac{v+t^2}{(a+t)^2}$. The right-hand side is minimal for $t = v/a$, where it equals $\frac{v}{v+a^2}$. Applying the result to $-Z$ gives the lower tail. □

In words, a variable whose variance is small compared with a^2 rarely deviates from its mean by more than a, whatever its distribution. Note also the structure of the proof, which we will meet again in Section 5: we control a tail probability by the expectation of a well-chosen non-negative function, and then optimize over a free parameter of this function.

2.3 The Kullback–Leibler divergence and the change of measure

Everything so far concerns a voter fixed *before* seeing the data. Learning is precisely the opposite: the algorithm looks at the sample and chooses. To account for this choice, PAC-Bayes compares the distribution Q that has been chosen with a distribution P that was fixed in advance. We need two things: a way to measure how far Q is from P, and a way to transfer statements that are true under P into statements under Q. The Kullback–Leibler divergence and the change of measure inequality provide exactly these two things.

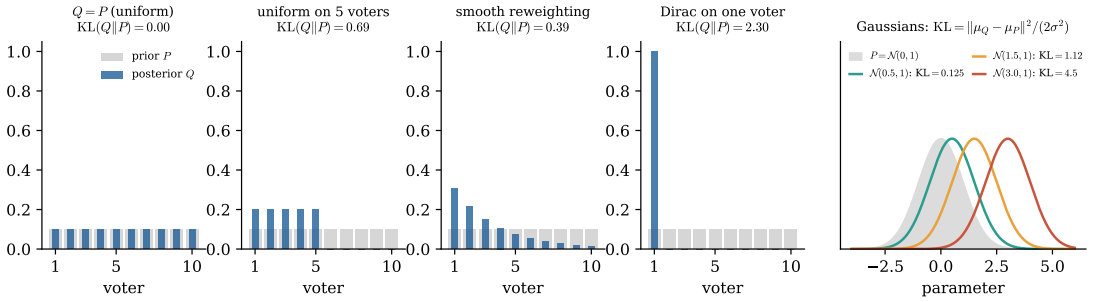


Figure 15: Orders of magnitude of $\mathrm{KL}(Q\|P)$. Four posteriors (blue) on $n = 10$ voters with the uniform prior (grey): the uniform posterior costs nothing, the uniform distribution on half of the voters costs $\ln 2$, and the Dirac on one voter costs the maximal value $\ln 10$. Right: for Gaussians with the same variance, the KL grows like the squared distance between the means.
[code: `make_figures.py: fig_kl_examples`]

Definition 2.7: Kullback–Leibler divergence

For $Q, P \in \mathcal{M}(\mathcal{H})$ such that Q is absolutely continuous w.r.t. P,

$$\mathrm{KL}(Q\|P) = \mathop{\mathbb{E}}_{h\sim Q}\left[\ln \frac{dQ}{dP}(h)\right] \quad \Big(= \sum_{h\in\mathcal{H}} Q(h) \ln \frac{Q(h)}{P(h)} \ \text{ if } \mathcal{H} \text{ is finite}\Big),$$

and $\mathrm{KL}(Q\|P) = +\infty$ otherwise.

The KL divergence, also called relative entropy (Cover and Thomas, 2006), is always non-negative and it vanishes only when $Q = P$; this is Gibbs' inequality, a consequence of Jensen's inequality applied to the concave logarithm. It is not symmetric, and it is not a distance, but it behaves as a measure of how far the posterior has moved away from the prior. A few examples help to get a feeling for its values.

On a finite set of n voters with the uniform prior, it takes the simple form $\mathrm{KL}(Q\|P) = \ln n - H(Q)$, where $H(Q) = -\sum_h Q(h) \ln Q(h)$ is the Shannon entropy of Q. The KL is then zero for the uniform posterior and maximal, equal to $\ln n$, for a Dirac on a single voter; if Q is uniform on k of the n voters, $\mathrm{KL}(Q\|P) = \ln \frac{n}{k}$: concentrating on a tenth of the voters costs $\ln 10$ nats. For product distributions the KL divergence adds up, $\mathrm{KL}(Q_1 \otimes Q_2\|P_1 \otimes P_2) = \mathrm{KL}(Q_1\|P_1) + \mathrm{KL}(Q_2\|P_2)$; in particular $\mathrm{KL}(Q^2\|P^2) = 2\,\mathrm{KL}(Q\|P)$ where $Q^2 = Q \otimes Q$ is the distribution of a pair of independent voters, an identity which will be used for majority votes. Finally, for Gaussian distributions with the same covariance, $\mathrm{KL}\left(\mathcal{N}(\mu_1, \sigma^2 I_d)\,\|\,\mathcal{N}(\mu_0, \sigma^2 I_d)\right) = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$. These properties are proved in the tutorial sheet, and Fig. 15 gives an idea of the values taken by the KL divergence on these examples: a few nats already correspond to a posterior that has moved far away from the prior.

We now have a way to measure how far Q is from P; it remains to transfer statements from one to the other. The following inequality, the heart of every PAC-Bayesian proof, does exactly that. It allows us to replace an expectation under the data-dependent posterior Q by an expectation under the data-independent prior P, at the price of $\mathrm{KL}(Q\|P)$.

Lemma 2.8: Change of measure (Donsker and Varadhan, 1975; Csiszár, 1975)

For any measurable function $\phi : \mathcal{H} \to \mathbb{R}$ and any $P \in \mathcal{M}(\mathcal{H})$,

$$\forall Q \in \mathcal{M}(\mathcal{H}), \qquad \mathop{\mathbb{E}}_{h\sim Q}[\phi(h)] \;\le\; \mathrm{KL}(Q\|P) + \ln \mathop{\mathbb{E}}_{h\sim P}\Big[e^{\phi(h)}\Big]. \tag{2.2}$$

Moreover, when $\mathbb{E}_P e^{\phi} < \infty$, equality holds for the *Gibbs distribution* $Q_\phi(h) = \frac{e^{\phi(h)}P(h)}{\mathbb{E}_P e^{\phi}}$, so that

$$\ln \mathop{\mathbb{E}}_{h\sim P}\Big[e^{\phi(h)}\Big] = \sup_{Q\in\mathcal{M}(\mathcal{H})} \Big\{ \mathop{\mathbb{E}}_{h\sim Q}[\phi(h)] - \mathrm{KL}(Q\|P)\Big\}. \tag{2.3}$$

Proof. Assume $\mathrm{KL}(Q\|P) < \infty$ (otherwise there is nothing to prove) and let Q_ϕ be the Gibbs distribution. We compute the divergence between Q and Q_ϕ (written for a finite $\mathcal{H}$ to keep the notation light):

$$\begin{aligned} 0 \le \mathrm{KL}(Q\|Q_\phi) &= \sum_h Q(h) \ln \frac{Q(h)}{Q_\phi(h)} \\ &\downarrow \text{ definition of } Q_\phi\text{: } \ln Q_\phi(h) = \phi(h) + \ln P(h) - \ln \mathbb{E}_P e^{\phi} \\ &= \sum_h Q(h)\Big[\ln \frac{Q(h)}{P(h)} - \phi(h) + \ln \mathbb{E}_P e^{\phi}\Big] \\ &\downarrow \text{ linearity, and } \sum_h Q(h) = 1 \\ &= \mathrm{KL}(Q\|P) - \mathbb{E}_Q \phi + \ln \mathbb{E}_P e^{\phi}. \end{aligned}$$

Rearranging gives (2.2), with equality if and only if $\mathrm{KL}(Q\|Q_\phi) = 0$, that is $Q = Q_\phi$. □

In plain words, the average of ϕ under *any* posterior is controlled by an exponential average under the prior, which does not depend on Q, plus a penalty that grows as Q moves away from P. Since the inequality holds for all Q at once, the posterior may be chosen after the fact. The proof shows even more than the inequality: the gap in (2.2) is exactly $\mathrm{KL}(Q\|Q_\phi)$, the divergence between Q and the Gibbs distribution. Figure 16 illustrates this on a small example with 25 voters: along the segment joining an arbitrary posterior Q to Q_ϕ, the quantity $\mathbb{E}_Q\phi - \mathrm{KL}(Q\|P)$ increases until it reaches the value $\ln \mathbb{E}_P e^{\phi}$ at Q_ϕ.

In a nutshell Equation (2.3) says that $\ln \mathbb{E}_P e^{\phi}$ is the maximal value of a "fit minus complexity" criterion $\mathbb{E}_Q\phi - \mathrm{KL}(Q\|P)$, and that the best posterior is obtained by *exponentially reweighting* the prior. This variational principle is the origin of the *Gibbs posterior* (Section 4.1) and of the link between PAC-Bayes and Bayesian inference.

Recent work: beyond the Kullback–Leibler divergence The change of measure is the only place where the KL divergence enters PAC-Bayes, and recent works replace it. Ohnishi and Honorio (2021) derive change of measure inequalities for f-divergences such as the χ^2 divergence, which only require the second

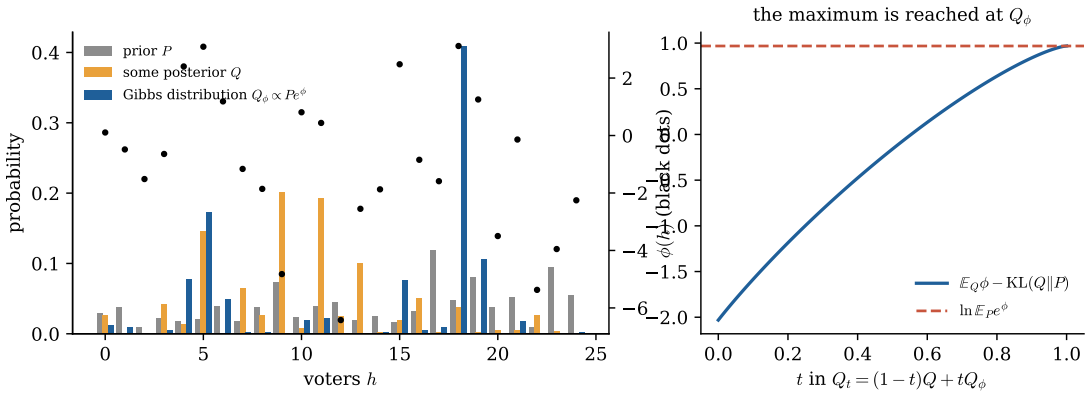


Figure 16: The change of measure on 25 voters. Left: a prior P, a function ϕ (black dots), an arbitrary posterior Q and the Gibbs distribution $Q_\phi \propto Pe^\phi$, which puts its mass on the voters where ϕ is large while remaining close to P. Right: value of $\mathbb{E}_{Q_t}\phi - \mathrm{KL}(Q_t\|P)$ along $Q_t = (1-t)Q + tQ_\phi$; it never exceeds $\ln \mathbb{E}_P e^\phi$ and reaches it at Q_ϕ. [code: `make_figures.py: fig_change_of_measure`]

moment of ϕ under the prior instead of its exponential moment, and thus apply to heavy-tailed losses. Lugosi and Neu (2022) show that any *strongly convex* function of the posterior can play the role of the KL, and Kuzborskij et al. (2024) build, from coin-betting arguments, a divergence that is never much larger than $\sqrt{\mathrm{KL}(Q\|P)\cdot \mathrm{TV}(Q,P)} + \mathrm{TV}(Q,P)$, where TV is the total variation distance; since $\mathrm{TV} \le 1$, their bounds improve on the KL ones precisely when the posterior is far from the prior. Section 8 returns to these alternatives.

2.4 Warm-up: the Occam bound

Before combining the two tools in full generality, let us see how far the concentration inequalities alone can take us when the learner selects one classifier among many. Consider a *countable* class $\mathcal{H}$ and a prior P on $\mathcal{H}$, which can be thought of as a "description length": $\ln \frac{1}{P(h)}$ is the number of nats needed to describe h with a code adapted to P.

Theorem 2.9: Occam's razor bound (Blumer et al., 1987; Langford, 2005)

For any countable $\mathcal{H}$, any prior P with $P(h) > 0$ for all h, and any $\delta \in (0,1]$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$,

$$\forall h \in \mathcal{H}, \qquad R_{\mathcal{D}}(h) \le \overline{\mathrm{kl}}^{-1}\Big(\widehat{R}_S(h), \frac{\ln \frac{1}{P(h)} + \ln \frac{1}{\delta}}{m}\Big),$$

$$\text{hence} \qquad R_{\mathcal{D}}(h) \le \widehat{R}_S(h) + \sqrt{\frac{\ln \frac{1}{P(h)} + \ln \frac{1}{\delta}}{2m}}.$$

Proof. For a fixed h, $m\widehat{R}_S(h)$ is a sum of m i.i.d. Bernoulli variables of mean $R_{\mathcal{D}}(h)$.

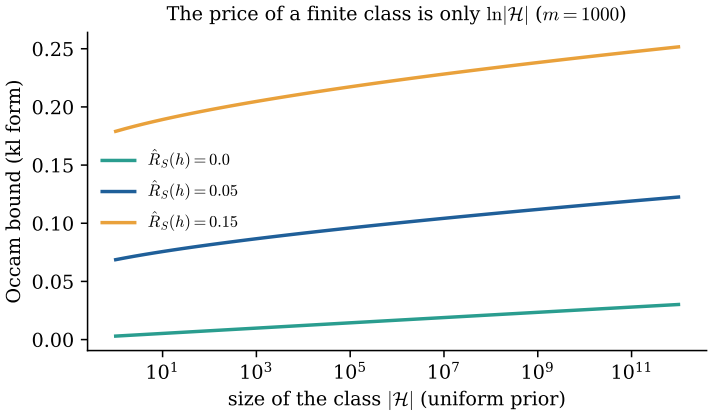


Figure 17: The Occam bound (kl form, $\delta = 0.05$, $m = 1000$) as a function of the size of the class, for three values of the empirical risk. The complexity only grows like $\ln|\mathcal{H}|$. [code: `make_figures.py: fig_occam`]

By Chernoff's bound (lower tail of $\widehat{R}_S(h)$), the event

$$\{\widehat{R}_S(h) < R_{\mathcal{D}}(h) \text{ and } \mathrm{kl}(\widehat{R}_S(h)\|R_{\mathcal{D}}(h)) > \varepsilon_h\} = \{R_{\mathcal{D}}(h) > \overline{\mathrm{kl}}^{-1}(\widehat{R}_S(h), \varepsilon_h)\}$$

has probability at most $e^{-m\varepsilon_h} = \delta P(h)$ when $\varepsilon_h = \frac{1}{m}\ln\frac{1}{\delta P(h)}$. A union bound over $h \in \mathcal{H}$ gives a total failure probability at most $\sum_h \delta P(h) = \delta$. The second statement follows from Pinsker's inequality. □

The theorem certifies every hypothesis of the class at once, so it remains valid for the one the learner finally picks. Its proof shows how the prior works: it *distributes the confidence budget* δ over the hypotheses, each h receiving the share $\delta P(h)$. A hypothesis that was *a priori* likely is cheap, and an unlikely one is expensive. With the uniform prior on a finite class, $\ln\frac{1}{P(h)} = \ln|\mathcal{H}|$ and we recover the classical bound for finite classes seen in the first part of the course. As shown in Fig. 17, the price of a large finite class is surprisingly small: with $m = 1000$ examples and an empirical risk of 5%, going from 10 to 10^{12} candidate classifiers only moves the bound from about 0.08 to 0.12.

The Occam bound already has the PAC-Bayesian flavour, but it has two limits. It only certifies a *single* hypothesis — in the language of the next sections, a Dirac posterior, for which $\mathrm{KL}(\delta_h\|P) = \ln\frac{1}{P(h)}$ — and it is useless for uncountable classes, where $P(h) = 0$ for every h. PAC-Bayes removes both limits by considering *averages* of hypotheses: averaging over a posterior which puts a large mass around a good hypothesis costs much less than pinpointing this hypothesis exactly. Making this idea precise, with the change of measure in place of the union bound, is the purpose of the next section.

Recent work: Occam bounds for large language models The Occam bound has found a spectacular application to modern models, whose number of parameters is much larger than the number of examples. The idea is to choose as prior a *code*: $P(h) \propto 2^{-|h|}$, where $|h|$ is the length in bits of a compressed description of h, so that $\ln\frac{1}{P(h)}$ is the size of the compressed model. Lotfi et al. (2024a) compress large language models (low-rank adaptation in a random subspace, then quantization) and certify their next-token prediction with a bound of

the form

$$R(h) \leq \widehat{R}_S(h) + \Delta\sqrt{\frac{\ln \frac{1}{P(h)} + \ln \frac{1}{\delta}}{2m}},$$

where Δ is the range of the (smoothed, hence bounded) log-loss; they obtain the first non-vacuous bounds for models of the size of GPT-2, and Lotfi et al. (2024b) extend them to LLaMA-2 models with up to 70 billion parameters by counting tokens, rather than documents, as data points (with a martingale argument, since tokens are not independent). In the same spirit, Akinwande et al. (2024) explain why prompt engineering for zero-shot image classifiers does not overfit: a prompt is a short sequence of words, and a language model provides a natural prior P on prompts. They report, for instance, a bound of 31.9% for a test error of 25.6% on ImageNet with handcrafted prompts.

To remember from Section 2

For a fixed voter, the empirical risk concentrates around the true risk; the binary kl gives the sharpest statement, and its inversion $\overline{\mathrm{kl}}^{-1}$, computed by bisection, turns an empirical risk into a certified upper bound, with a fast rate when the risk is small. The change of measure inequality $\mathbb{E}_Q \phi \leq \mathrm{KL}(Q\|P) + \ln \mathbb{E}_P e^{\phi}$ transfers a statement about a data-independent prior to any posterior, at the price $\mathrm{KL}(Q\|P)$, and it is an equality for the Gibbs distribution. The Occam bound is the special case of Dirac posteriors on a countable class; its price is $\ln \frac{1}{P(h)}$.

Check your understanding

Short questions to answer without computation; answers in Appendix A.5.

Q2.1. Why is the kl confidence interval narrower than Hoeffding's when the empirical risk is small?

Q2.2. What is $\mathrm{KL}(Q\|P)$ when P is uniform on n voters and Q is uniform on k of them?

Q2.3. For which posterior is the change of measure inequality an equality?

Q2.4. In the Occam bound, what plays the role of the complexity of the chosen hypothesis?

Exercises

Exercise 2.1. Compute $\mathrm{kl}(0.1\|0.2)$ and $\mathrm{kl}(0.2\|0.1)$. Is the binary kl symmetric?

Exercise 2.2. Show that the Gibbs distribution Q_ϕ of Lemma 2.8 is the only posterior for which the change of measure is an equality.

Exercise 2.3. Using the Occam bound in Pinsker form, how many examples are needed to certify a risk below 0.15 for a classifier with empirical risk 0.1 chosen among 10^6 candidates, with $\delta = 0.05$?

Exercise 2.4. Using Pinsker's inequality, show that the kl interval of Fig. 14 is never wider than the Hoeffding interval. Using notebook 1, determine for which empirical risks it is at least twice narrower when $m = 1000$.

Exercise 2.5. **(Comprehension)** True or false? Justify briefly. (a) If P and Q are two distributions on a finite set of voters, $\mathrm{KL}(Q\|P)$ is finite. (b) If P is uniform on n voters, then $\mathrm{KL}(Q\|P) \leq \ln n$ for every posterior Q. (c) In the Occam bound with the uniform prior, doubling the number of candidate classifiers doubles the complexity term $\ln \frac{1}{P(h)}$.

Exercise 2.6. **(Application)** Let P be uniform on $n = 4$ voters. Compute $\mathrm{KL}(Q\|P)$ for $Q = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0)$, for $Q' = (0.7, 0.1, 0.1, 0.1)$ and for the Dirac δ_{h_1}, both from the definition and with the formula $\ln n - H(Q)$. Compute also $\mathrm{KL}(P\|Q')$ and $\mathrm{KL}(P\|Q)$, and comment.

Exercise 2.7. **(Application)** Check the two test-set certificates of the running example of this section: a vote with 4.4% of errors on 540 test points, certified by Hoeffding's inequality (0.097) and by the kl inversion (0.069) with $\delta = 0.05$. Recompute both certificates with $\delta = 0.01$, and comment on the cost of the extra confidence.

Exercise 2.8. **(Proof)** Prove Markov's inequality (Lemma 2.2) in detail, then deduce Chebyshev's inequality: for a random variable Z with mean μ and variance v, $\mathbb{P}(|Z - \mu| \geq a) \leq \frac{v}{a^2}$ for every $a > 0$. Compare it with the one-sided Cantelli inequality (Lemma 2.6) when $a^2 = v$.

Exercise 2.9. **(Proof)** Admitting Hoeffding's lemma $\mathbb{E}\, e^{\lambda(p-\hat{p})} \leq e^{\lambda^2/(8m)}$ (Lemma 2.3), prove the tail bound $\mathbb{P}(p - \hat{p} \geq \varepsilon) \leq e^{-2m\varepsilon^2}$ by optimizing over $\lambda > 0$, and deduce that $p \leq \hat{p} + \sqrt{\ln(1/\delta)/(2m)}$ with probability at least $1 - \delta$.

Exercise 2.10. **(Proof)** Let Q_1, P_1 be distributions on a finite set $\mathcal{H}_1$ and Q_2, P_2 distributions on a finite set $\mathcal{H}_2$, with P_1, P_2 positive. Show that $\mathrm{KL}(Q_1 \otimes Q_2 \| P_1 \otimes P_2) = \mathrm{KL}(Q_1\|P_1) + \mathrm{KL}(Q_2\|P_2)$, and deduce that $\mathrm{KL}(Q^2\|P^2) = 2\,\mathrm{KL}(Q\|P)$, the identity used for the tandem bound. Show also that $\mathrm{KL}(\delta_h\|P) = \ln \frac{1}{P(h)}$ for a Dirac posterior.

Part II

PAC-Bayesian generalization bounds

This part builds the certificates. We first bound the risk of the Gibbs classifier, a voter drawn at random, from a general theorem to the classical bounds of the literature. We then discuss how to choose the prior and the posterior, and how to compute a bound in practice. We finally turn to the risk of the majority vote itself: this is where the diversity of the voters appears, and where the first question of the course receives its answer.

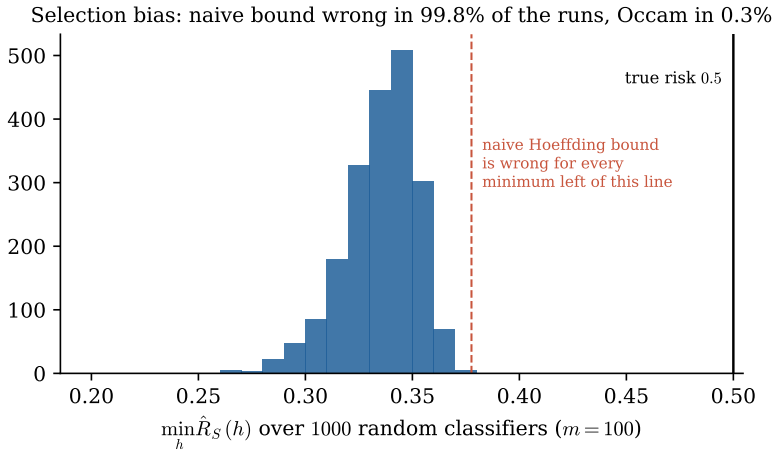


Figure 18: Selection bias. Distribution, over 2000 repetitions, of the smallest empirical risk among 1000 random classifiers of true risk 0.5, measured on $m = 100$ examples. A confidence bound computed for the selected classifier as if it had been fixed in advance is violated in almost every run. [code: `make_figures.py: fig_uniformity`]

3 PAC-Bayesian generalization bounds

About this section

Guiding question. *How much can the true risk of a randomly drawn voter exceed its empirical risk, when the distribution of the voters has been chosen with the data?*

This section bounds the true risk of the Gibbs classifier G_Q. We first explain why a useful bound must hold simultaneously for all posteriors, then state and prove a general PAC-Bayesian theorem whose proof has only three steps (Jensen, change of measure, Markov). The classical bounds of McAllester, Seeger–Maurer and Catoni, and the PAC-Bayes-λ bound, are all obtained as corollaries, and we compare them numerically to learn how to read the different terms of a bound.

3.1 Why the bound must hold for all posteriors

Before stating any PAC-Bayesian bound, we must understand what kind of guarantee we need. A learning algorithm looks at the data before choosing its output. If we compute a confidence bound for the chosen predictor as if it had been fixed in advance, we forget this selection step, and the bound can be badly wrong. The experiment reported in Fig. 18 shows how badly. We draw 1000 classifiers that answer completely at random, so that each of them has a true risk of exactly 0.5, and we evaluate them on a sample of $m = 100$ examples. By chance alone, the best of them has an empirical risk of about 0.34 on average. If we now apply Hoeffding's inequality to this best classifier as if it were fixed, we obtain an "upper bound" which is below 0.5 in 99.8% of the runs, although it should fail in at most 5% of them. The Occam bound of Section 2.4, which pays $\ln 1000$ for the selection, fails in only 0.3% of the runs.

This is why all the bounds of this section are *uniform*: they hold, with probability at least $1 - \delta$, for all posteriors Q at the same time. The posterior can then be chosen by any procedure that looks at the data — including the minimization of the bound itself — without invalidating the guarantee. The price of uniformity is the complexity

term $\mathrm{KL}(Q\|P)$, which plays the role that $\ln|\mathcal{H}|$ played in the Occam bound. The next subsection shows that this price is enough, with a single theorem.

3.2 A general PAC-Bayesian theorem

Rather than proving each classical bound separately, we prove one theorem that contains them all, in the form given by Germain et al. (2009) and Bégin et al. (2016). It is parametrized by a convex function Δ, which measures the discrepancy between an empirical risk and a true risk.

Theorem 3.1: General PAC-Bayesian theorem (Germain et al., 2009; Bégin et al., 2016)

Let $\Delta : [0,1]\times[0,1] \to \mathbb{R}$ be a *jointly convex* function. For any distribution $\mathcal{D}$ on $\mathcal{X}\times\mathcal{Y}$, any set of voters $\mathcal{H}$, any loss ℓ with values in $[0,1]$, any prior $P \in \mathcal{M}(\mathcal{H})$ and any $\delta \in (0,1]$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$,

$$\forall Q \in \mathcal{M}(\mathcal{H}), \quad \Delta\big(\widehat{R}_S(G_Q), R_{\mathcal{D}}(G_Q)\big) \le \frac{1}{m}\left[\mathrm{KL}(Q\|P) + \ln\frac{\mathcal{I}_\Delta(m)}{\delta}\right], \tag{3.1}$$

where $\mathcal{I}_\Delta(m) = \mathbb{E}_{S'\sim\mathcal{D}^m}\Big[\mathbb{E}_{h\sim P}\Big[e^{m\Delta\left(\widehat{R}_{S'}(h), R_{\mathcal{D}}(h)\right)}\Big]\Big]$.

In words, with high probability, and for every posterior at once, the gap between the empirical and the true Gibbs risk, measured by Δ, is at most a complexity per example: the KL divergence plus a confidence term, divided by m. Only $\mathcal{I}_\Delta(m)$ is left to compute, and it involves the prior alone.

Before the proof, let us describe its strategy. We want to control a quantity that involves the posterior Q, which depends on the data, and we only know how to control quantities that involve voters fixed in advance. The proof therefore moves the problem in three steps. First, it replaces the discrepancy of the *average* voter by the average discrepancy of the voters (Jensen). Second, it replaces the average under Q by an average under the prior P, at the price $\mathrm{KL}(Q\|P)$ (change of measure). Third, it controls the resulting quantity, which no longer involves Q, on average over the samples (Markov). The function Δ only matters in the last step, and each choice of Δ will give one of the classical bounds. Figure 19 summarizes this strategy; keep it in mind while reading the proof.

Proof. Fix a sample S and a posterior Q. The first two steps are deterministic inequalities:

$$\begin{aligned} m\,\Delta\big(\widehat{R}_S(G_Q), R_{\mathcal{D}}(G_Q)\big) &= m\,\Delta\Big(\mathop{\mathbb{E}}_{h\sim Q}\big[\widehat{R}_S(h)\big], \mathop{\mathbb{E}}_{h\sim Q}\big[R_{\mathcal{D}}(h)\big]\Big) \\ &\quad\downarrow \text{ Step 1: Jensen's inequality, } \Delta \text{ being jointly convex} \\ &\le \mathop{\mathbb{E}}_{h\sim Q}\Big[m\,\Delta\big(\widehat{R}_S(h), R_{\mathcal{D}}(h)\big)\Big] \\ &\quad\downarrow \text{ Step 2: change of measure (Lemma 2.8) with } \phi = m\Delta(\widehat{R}_S, R_{\mathcal{D}}) \\ &\le \mathrm{KL}(Q\|P) + \ln \underbrace{\mathop{\mathbb{E}}_{h\sim P}\Big[e^{m\Delta\left(\widehat{R}_S(h), R_{\mathcal{D}}(h)\right)}\Big]}_{=:\, Z(S),\ \text{free of } Q}. \end{aligned}$$

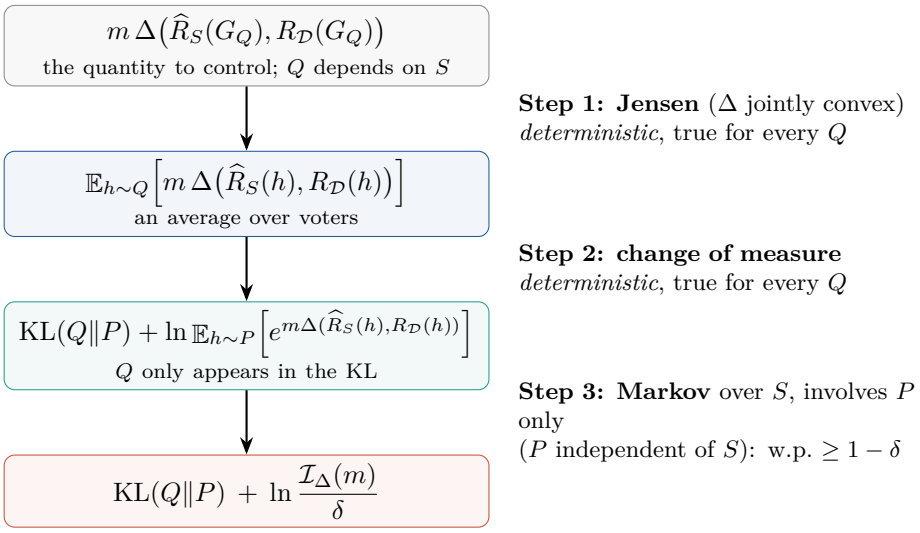


Figure 19: The three steps of the proof of Theorem 3.1. Only the last one is probabilistic, and it does not involve the posterior: this is why the bound holds simultaneously for all Q. [code: `tikz/proof_steps.tex`]

The posterior has now disappeared from the last term, which only involves the prior: $Z(S)$ is a non-negative random variable that depends on the sample only. The third step bounds it with Markov's inequality: with probability at least $1-\delta$ over S,

$$Z(S) \leq \frac{1}{\delta} \mathop{\mathbb{E}}_{S'\sim\mathcal{D}^m}[Z(S')]$$
$$\downarrow \text{ Fubini: } P \text{ does not depend on } S', \text{ the expectations can be exchanged}$$
$$= \frac{1}{\delta} \mathop{\mathbb{E}}_{h\sim P}\left[\mathop{\mathbb{E}}_{S'\sim\mathcal{D}^m}\left[e^{m\Delta(\widehat{R}_{S'}(h), R_{\mathcal{D}}(h))}\right]\right] = \frac{\mathcal{I}_\Delta(m)}{\delta}.$$

On this event of probability at least $1-\delta$, inequality (3.1) holds for *every* Q, because the first two steps are deterministic and valid for all posteriors simultaneously. □

The structure of this proof deserves attention, because it explains most of what follows. The only probabilistic statement is made in the third step, and it only concerns the prior: this is where the prior must be independent of the sample, since we need to exchange $\mathbb{E}_{S'}$ and $\mathbb{E}_{h\sim P}$ with P fixed. The posterior, on the other hand, only appears in the deterministic steps, and it may depend on S in an arbitrary way. This asymmetry is what makes PAC-Bayesian bounds usable as learning objectives: we are free to minimize them over Q.

To obtain a usable bound, it remains to choose Δ so that $\mathcal{I}_\Delta(m)$ can be computed or bounded. For a *fixed* voter h, $m\widehat{R}_S(h)$ is a sum of m independent variables in $[0,1]$ with mean $R_{\mathcal{D}}(h)$, so the concentration results of Section 2 apply. Each of them yields one of the classical bounds, and the next subsection goes through them in turn: the kl form (the tightest), its relaxation by Pinsker's inequality (the most readable), Catoni's bound, and two linear bounds (the easiest to optimize).

3.3 The classical bounds

3.3.1 Seeger's bound (kl form)

The most natural choice is the binary kl itself, since Maurer's lemma tells us exactly how large its exponential moment is.

Corollary 3.2: Seeger (2002); Langford (2005), with the constant of Maurer (2004)

For $m \geq 8$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, for all $Q \in \mathcal{M}(\mathcal{H})$,

$$\mathrm{kl}\left(\widehat{R}_S(G_Q) \,\|\, R_{\mathcal{D}}(G_Q)\right) \leq \frac{\mathrm{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}}{m}, \tag{3.2}$$

that is, $R_{\mathcal{D}}(G_Q) \leq \overline{\mathrm{kl}}^{-1}\Big(\widehat{R}_S(G_Q),\ \frac{\mathrm{KL}(Q\|P)+\ln \frac{2\sqrt{m}}{\delta}}{m}\Big)$.

Proof. Take $\Delta = \mathrm{kl}$, which is jointly convex. By Maurer's lemma (Lemma 2.4), for every fixed h, $\mathbb{E}_{S'} e^{m\,\mathrm{kl}(\widehat{R}_{S'}(h)\|R_{\mathcal{D}}(h))} \leq \xi(m) \leq 2\sqrt{m}$, hence $\mathcal{I}_{\mathrm{kl}}(m) \leq 2\sqrt{m}$. □

In plain words, the true Gibbs risk is at most the largest value that is still "plausible", in the sense of the kl, given the observed Gibbs risk and a budget equal to the complexity per example. Seeger's bound has no free parameter and no closed form, but it is computed in a few microseconds by the bisection of Algorithm 1. As the numerical comparison below will show, it is among the tightest bounds available, and it is the one we use by default to compute risk certificates. It inherits the fast rate of the kl: when the empirical risk is zero, it gives $R_{\mathcal{D}}(G_Q) \leq 1-e^{-\varepsilon} \leq \varepsilon$ with $\varepsilon = \frac{1}{m}[\mathrm{KL}(Q\|P)+\ln \frac{2\sqrt{m}}{\delta}]$, a rate in $1/m$. Its only weakness is that it is implicit, which makes it hard to read; the next bound trades some tightness for readability.

Recent work: a certificate valid for all sample sizes Seeger's bound holds for a sample size m fixed in advance. If examples arrive one at a time and we want to stop as soon as the certificate is good enough, the bound must hold for all m simultaneously; a union bound over m with confidence $\delta/(m(m+1))$ costs about $2\ln m$ extra nats. Jang et al. (2023) obtain such *time-uniform* bounds by a coin-betting argument: the wealth of a gambler who bets on the deviations $\widehat{R}_S(h) - R_{\mathcal{D}}(h)$ is a non-negative supermartingale, and Ville's maximal inequality controls it at all times. One of their corollaries is exactly Seeger's bound, valid for all m at once and with the constant $2\sqrt{m}$ replaced by $\sqrt{\pi}\,\Gamma(m+1)/\Gamma(m+\frac{1}{2}) \approx \sqrt{\pi m}$: with probability at least $1-\delta$, for all $m \geq 1$ and all Q,

$$\mathrm{kl}\left(\widehat{R}_S(G_Q) \,\|\, R_{\mathcal{D}}(G_Q)\right) \leq \frac{1}{m}\Big[\mathrm{KL}(Q\|P) + \ln \frac{\sqrt{\pi}\,\Gamma(m+1)}{\Gamma(m+\frac{1}{2})} + \ln \frac{1}{\delta}\Big].$$

Figure 20 shows that time uniformity is therefore free: the new constant is even slightly smaller than Maurer's. Related anytime-valid bounds based on supermartingales are due to Chugg et al. (2023) and, for heavy-tailed losses, to Haddouche and Guedj (2023).

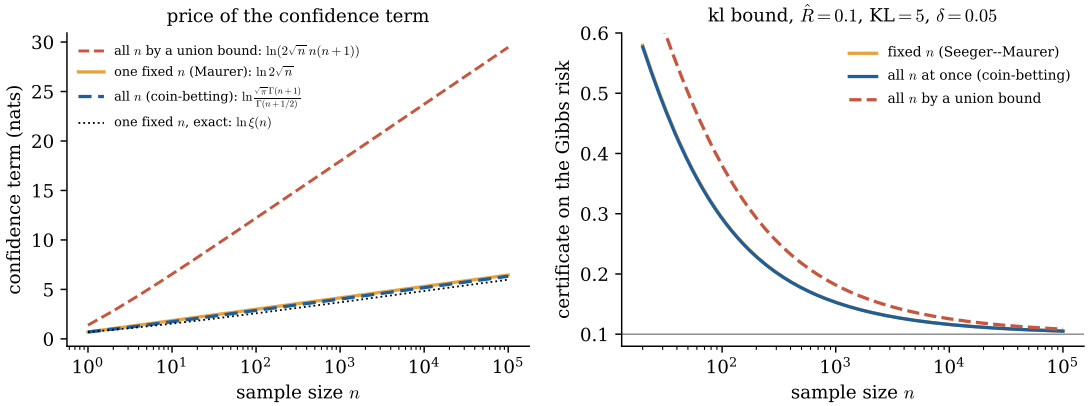


Figure 20: Time-uniform certificates. Left: confidence term of the kl bound for a fixed sample size (Maurer's $\ln 2\sqrt{n}$ and its exact value $\ln \xi(n)$), for all sample sizes at once by coin-betting (Jang et al., 2023), and by a union bound over n. Right: the resulting certificates for an empirical Gibbs risk of 0.1 and $\mathrm{KL}(Q\|P) = 5$; the fixed-n and coin-betting curves are indistinguishable. [code: `make_figures.py: fig_anytime`]

3.3.2 McAllester's bound

Relaxing the kl with Pinsker's inequality gives the historical PAC-Bayesian bound, whose form is the easiest to interpret.

Corollary 3.3: McAllester (1999b, 2003a), in the form of Maurer (2004)

For $m \geq 8$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, for all $Q \in \mathcal{M}(\mathcal{H})$,

$$R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \sqrt{\frac{\mathrm{KL}(Q\|P) + \ln\frac{2\sqrt{m}}{\delta}}{2m}}. \tag{3.3}$$

Proof. Apply Pinsker's inequality $\mathrm{kl}(q\|p) \geq 2(q-p)^2$ to (3.2). □

McAllester's bound reads "true risk ≤ empirical risk + complexity",

$$R_{\mathcal{D}}(G_Q) \;\leq\; \underbrace{\widehat{R}_S(G_Q)}_{\text{fit to the data}} \;+\; \underbrace{\sqrt{\frac{\mathrm{KL}(Q\|P) + \ln\frac{2\sqrt{m}}{\delta}}{2m}}}_{\text{price of choosing } Q \text{ with the data}},$$

where the complexity is a square root of the KL per example. It has the same structure as the VC or Rademacher bounds of the first part of the course, the capacity of the class being replaced by the divergence of the chosen posterior from the prior. Its drawback is the $1/\sqrt{m}$ rate, which it keeps even when the empirical risk is zero; Seeger's bound is always at least as tight.

> **Historical note** The original bounds of McAllester (1998, 1999b) had slightly worse constants, with a numerator of the form $\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta} + O(\ln m)$ and a denominator $2m-1$. The kl form was introduced by Seeger (2002), see also Langford (2005), and the clean constant $\ln\frac{2\sqrt{m}}{\delta}$, valid for any loss in $[0,1]$, is due

to Maurer (2004).

3.3.3 Catoni's bound

Can we avoid the $\ln 2\sqrt{m}$ term altogether? Catoni (2007) uses a different comparison function, for which the exponential moment is not bounded by $2\sqrt{m}$ but by 1, at the cost of a parameter that must be fixed in advance.

Corollary 3.4: Catoni (2007), see also Germain et al. (2009)

For any fixed $C > 0$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, for all Q,

$$R_{\mathcal{D}}(G_Q) \le \frac{1}{1-e^{-C}} \left[1 - \exp\left(- C\, \widehat{R}_S(G_Q) - \frac{\mathrm{KL}(Q\|P) + \ln \frac{1}{\delta}}{m} \right) \right]. \tag{3.4}$$

Proof. Take $\Delta(q,p) = \mathcal{F}_C(p) - C\,q$ with $\mathcal{F}_C(p) = -\ln\left(1 - p(1-e^{-C})\right)$, which is convex. For a fixed h with risk $p = R_{\mathcal{D}}(h)$, the convexity of $u \mapsto e^{-Cu}$ on $[0,1]$ gives $e^{-Cu} \le 1 - u(1-e^{-C})$, hence, by independence of the examples, $\mathbb{E}_{S'} e^{-Cm\widehat{R}_{S'}(h)} \le \left(1 - p(1-e^{-C})\right)^m = e^{-m\mathcal{F}_C(p)}$. Therefore $\mathbb{E}_{S'} e^{m\Delta(\widehat{R}_{S'}(h),p)} \le 1$ and $\mathcal{I}_\Delta(m) \le 1$. Rearranging $\mathcal{F}_C(R_{\mathcal{D}}(G_Q)) \le C\widehat{R}_S(G_Q) + \frac{1}{m}[\mathrm{KL} + \ln\frac{1}{\delta}]$ gives (3.4). □

The parameter C weighs the empirical risk against the complexity. It cannot be optimized after seeing the data for free, but the price is small. If we apply the bound with confidence δ/k to each value of a grid $C_1, \dots, C_k$ fixed in advance, a union bound shows that it holds for all of them simultaneously; the best value can then be chosen at a cost of only $\ln k$ in the numerator. With such a grid, Catoni's bound is numerically very close to Seeger's, and even marginally smaller when the KL is small, since it does not pay the $\ln 2\sqrt{m}$ term. Using $1 - e^{-x} \le x$, it also implies the *linear* bound $R_{\mathcal{D}}(G_Q) \le \frac{1}{1-e^{-C}}\left[C\,\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)+\ln(1/\delta)}{m}\right]$. Casado et al. (2024) recently showed how to optimize such a parameter *exactly*, without a grid, even for unbounded losses: their *PAC-Bayes-Chernoff* bound, $\mathbb{E}_Q R_{\mathcal{D}} \le \mathbb{E}_Q \widehat{R}_S + \inf_\lambda \left\{ \frac{\mathrm{KL}(Q\|P)+\ln(m/\delta)}{\lambda(m-1)} + \frac{1}{\lambda}\mathbb{E}_Q \Lambda_h(\lambda) \right\}$, where Λ_h is the cumulant generating function of the loss of h, costs only a $\ln m$ term for the uniformity in λ, as in Cramér–Chernoff's classical bound.

3.3.4 Linear bounds and the PAC-Bayes-λ bound

The last two bounds of this section are *linear* in $\widehat{R}_S(G_Q)$ and in $\mathrm{KL}(Q\|P)$. This is a weakness as far as tightness is concerned, but a great strength for learning: minimizing a linear combination of the empirical risk and of the KL over Q has an explicit solution, the Gibbs posterior of Section 4.1.

Corollary 3.5: Hoeffding-based bound (Catoni, 2007; Alquier, 2024)

For any fixed $\lambda > 0$, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, for all Q,

$$R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P) + \ln\frac{1}{\delta}}{\lambda m} + \frac{\lambda}{8}. \tag{3.5}$$

Proof. Take $\Delta(q,p) = \lambda(p-q)$, which is linear, hence convex. By Hoeffding's lemma (Lemma 2.3) applied with parameter λm, $\mathbb{E}_{S'} e^{\lambda m(R_{\mathcal{D}}(h) - \widehat{R}_{S'}(h))} \leq e^{\lambda^2 m/8}$, so that $\frac{1}{m}\ln \mathcal{I}_\Delta(m) \leq \lambda^2/8$. Divide by λ. □

The bound reads "empirical risk + complexity/λ + $\lambda/8$": a large λ makes the KL cheap but increases the last term, and a small λ does the opposite. The Hoeffding-based bound is the favourite of the statistical literature on PAC-Bayes (Alquier, 2024), because it leads to clean oracle inequalities. Its parameter λ must again be fixed in advance. The next bound, obtained by relaxing Seeger's bound in a clever way, does not have this limitation.

Corollary 3.6: PAC-Bayes-λ bound (Thiemann et al., 2017)

With probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, for all $Q \in \mathcal{M}(\mathcal{H})$ and all $\lambda \in (0,2)$ *simultaneously*,

$$R_{\mathcal{D}}(G_Q) \leq \frac{\widehat{R}_S(G_Q)}{1-\frac{\lambda}{2}} + \frac{\mathrm{KL}(Q\|P) + \ln\frac{2\sqrt{m}}{\delta}}{\lambda(1-\frac{\lambda}{2})m}. \tag{3.6}$$

Proof. Let $\hat{q} = \widehat{R}_S(G_Q)$, $p = R_{\mathcal{D}}(G_Q)$ and $\varepsilon = \frac{1}{m}[\mathrm{KL}(Q\|P) + \ln\frac{2\sqrt{m}}{\delta}]$. On the event of Corollary 3.2, if $p > \hat{q}$, the refined Pinsker inequality gives $(p-\hat{q})^2 \leq 2p\,\varepsilon$, that is $p - \hat{q} \leq \sqrt{2p\varepsilon}$. For any $\lambda > 0$, $\sqrt{2p\varepsilon} = \sqrt{\lambda p \cdot \frac{2\varepsilon}{\lambda}} \leq \frac{\lambda p}{2} + \frac{\varepsilon}{\lambda}$ by the inequality between arithmetic and geometric means. Hence $p(1-\frac{\lambda}{2}) \leq \hat{q} + \frac{\varepsilon}{\lambda}$. Since this is a deterministic consequence of (3.2), it holds for all λ at once. □

The PAC-Bayes-λ bound keeps the linear structure of the previous one, but it is a deterministic consequence of Seeger's bound. Because it holds for all λ simultaneously, λ can be optimized freely, at no cost. For a fixed Q, the optimal value is

$$\lambda^\star = \frac{2}{\sqrt{\frac{2m\widehat{R}_S(G_Q)}{\mathrm{KL}(Q\|P)+\ln\frac{2\sqrt{m}}{\delta}} + 1} + 1}, \tag{3.7}$$

and for this value the PAC-Bayes-λ bound is smaller than the relaxation $\hat{q} + \sqrt{2\hat{q}\varepsilon} + 2\varepsilon$ of Seeger's bound given by Lemma 2.5, but larger than Seeger's bound itself, from which it is derived. Thiemann et al. (2017) show moreover that (3.6) is jointly quasiconvex in (Q,λ) under mild conditions, which is the basis of the alternating minimization algorithm of Section 7.2.2.

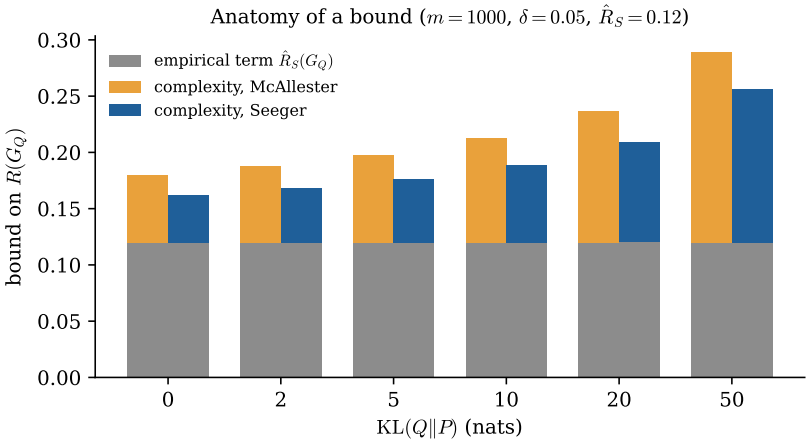


Figure 21: Decomposition of McAllester's and Seeger's bounds into the empirical term (grey) and the complexity term, for $m = 1000$, $\delta = 0.05$ and $\widehat{R}_S(G_Q) = 0.12$, as the KL divergence grows. [code: `make_figures.py: fig_bound_anatomy`]

3.4 Reading and comparing the bounds

We now have five bounds; to choose among them, we must learn to read them. All these bounds share the same anatomy. The unknown true risk is bounded by a function of two quantities: the empirical risk of the posterior, which measures how well it fits the data, and the complexity per example $\frac{1}{m}[\mathrm{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}]$, which measures how far the posterior has moved away from the prior. The first term decreases when Q concentrates on good voters, the second increases at the same time, and the bounds only differ in the way these two ingredients are combined. The confidence term is almost negligible: dividing δ by ten only adds $\ln 10 \approx 2.3$ nats to the numerator.

Figure 21 and Table 2 make this concrete. We fix $m = 1000$, $\delta = 0.05$ and an empirical Gibbs risk of 0.12, and we let the KL divergence grow from 0 to 50 nats. With $\mathrm{KL} = 0$, the complexity per example is only 0.007, and all bounds lie between 0.16 and 0.18. The complexity term of McAllester's bound is already 0.06, half the empirical risk, while Seeger's bound adds only 0.04: the square root of Pinsker's inequality is expensive when the risk is far from $1/2$.

As the KL grows, all bounds deteriorate, but slowly: 50 nats, which is the cost of a posterior putting all its mass on one voter among $e^{50} \approx 5 \times 10^{21}$, still leaves a non-trivial certificate of 0.26. Catoni's bound optimized on a grid and Seeger's bound are almost indistinguishable, the former being slightly smaller for small KL. The PAC-Bayes-λ bound, finally, is tighter than McAllester's bound when the KL is small but not when it is large.

Figure 22 takes the point of view of the user who wants a certificate of a given quality. For an empirical Gibbs risk of 0.1, a certificate below 0.15 requires about 640 examples when the posterior is equal to the prior, 1100 when $\mathrm{KL}(Q\|P) = 5$ and more than 5000 when $\mathrm{KL}(Q\|P) = 50$: the KL divergence is literally paid in examples, and the right panel shows that, beyond a few nats, the price is roughly linear, each nat costing about a hundred examples.

Figure 23 shows the same comparison from two other angles. As a function of the sample size, the difference between the kl-based bounds and McAllester's bound is

Table 2: Values of the bounds for $\widehat{R}_S(G_Q) = 0.12$, $m = 1000$, $\delta = 0.05$. Catoni's bound is optimized on a grid of 30 values of C with a union bound.

$\mathrm{KL}(Q\|P)$	$\frac{1}{m}[\mathrm{KL} + \ln \frac{2\sqrt{m}}{\delta}]$	McAllester	PAC-Bayes-λ	Catoni (grid)	Seeger
0	0.0071	0.180	0.169	0.160	0.162
2	0.0091	0.188	0.177	0.166	0.168
5	0.0121	0.198	0.187	0.175	0.177
10	0.0171	0.213	0.204	0.187	0.189
20	0.0271	0.236	0.232	0.207	0.209
50	0.0571	0.289	0.307	0.256	0.256

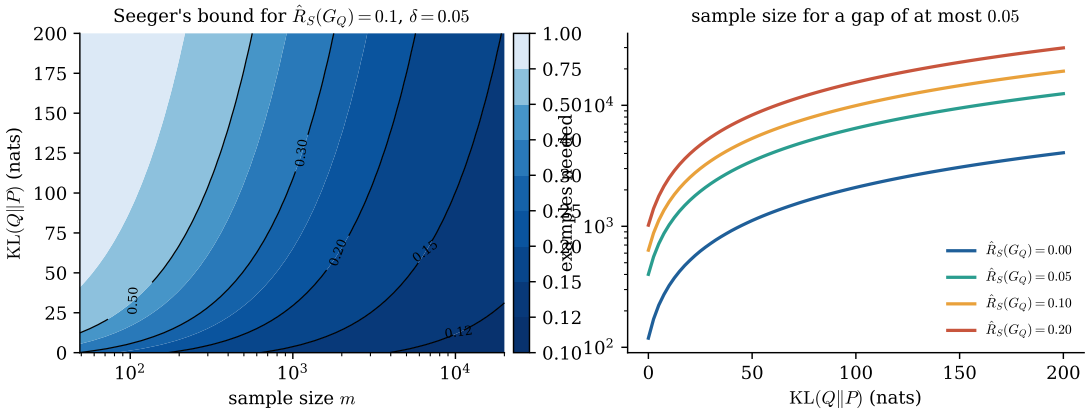


Figure 22: Left: Seeger's bound for $\widehat{R}_S(G_Q) = 0.1$ and $\delta = 0.05$, as a function of the sample size and of the KL divergence (level lines at 0.12, 0.15, 0.2, 0.3 and 0.5). Right: number of examples needed for the certificate to exceed the empirical Gibbs risk by at most 0.05. [code: `make_figures.py: fig_seeger_landscape`]

most visible when the empirical risk is zero: the former decrease like $1/m$, the latter like $1/\sqrt{m}$. As a function of the empirical risk, the bounds coincide near $1/2$ and separate when the risk is small, which is precisely the regime of good classifiers.

The following table summarizes the section. The first column gives the comparison function used in Theorem 3.1, and the second the bound on its exponential moment that makes the theorem explicit.

Bound	$\Delta(q,p)$	$\ln \mathcal{I}_\Delta(m) \leq$	Main features
Seeger (3.2)	$\mathrm{kl}(q\|p)$	$\ln 2\sqrt{m}$	among the tightest; implicit
McAllester (3.3)	$2(q-p)^2$	$\ln 2\sqrt{m}$	readable; slow rate $1/\sqrt{m}$
Catoni (3.4)	$\mathcal{F}_C(p) - Cq$	0	tight on a grid of C
Hoeffding (3.5)	$\lambda(p-q)$	$\lambda^2 m/8$	linear; λ fixed in advance
PAC-Bayes-λ (3.6)	(from Seeger)	$\ln 2\sqrt{m}$	linear, all λ at once

Recent work: which comparator is the best? Theorem 3.1 leaves the choice of the comparison function Δ open. Hellström and Guedj (2024) answer the question: if the loss is dominated (in the sense of its moment generating function) by a family of distributions of mean p, the best convex comparator, up to the

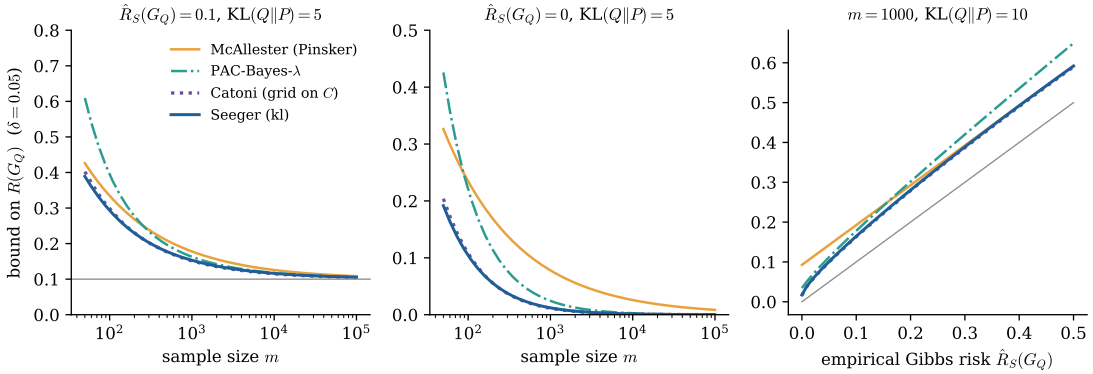


Figure 23: Comparison of the bounds for $\delta = 0.05$. Left: as a function of m, for $\widehat{R}_S(G_Q) = 0.1$ and $\mathrm{KL}(Q\|P) = 5$. Middle: same with $\widehat{R}_S(G_Q) = 0$: the kl-based bounds (Seeger, and Catoni and PAC-Bayes-λ which approximate it) decrease in $1/m$ whereas McAllester's bound decreases in $1/\sqrt{m}$. Right: as a function of the empirical risk for $m = 1000$ and $\mathrm{KL} = 10$. [code: `make_figures.py: fig_bounds_comparison`]

logarithmic term $\ln \mathcal{I}_\Delta(m)$, is the *Cramér function* of this family, $\Delta^\star(q,p) = \sup_t\{tq - \ln \mathbb{E}e^{tZ_p}\}$, the convex conjugate of the cumulant generating function of the variable Z_p of mean p. For losses in $[0,1]$, dominated by Bernoulli variables, $\Delta^\star = \mathrm{kl}$: Seeger's bound is optimal among all the bounds of this table. For Gaussian losses of variance σ^2 the optimal comparator is $(q-p)^2/(2\sigma^2)$, and for Poisson losses $p - q + q\ln\frac{q}{p}$. Lugosi and Neu (2023) propose another unifying view, in which a generalization bound is the *regret* of an online learner playing against the sample; the exponential weights algorithm then gives back the linear bound (3.5).

Example 3.1 (Three posteriors on the same ensemble). Consider an ensemble of $n = 100$ voters with the uniform prior P, $m = 1000$ training examples and $\delta = 0.05$, so that $\ln\frac{2\sqrt{m}}{\delta} = \ln(1264.9) \approx 7.14$. Suppose first that Q is uniform over the 10 voters with the smallest empirical risk, and that its empirical Gibbs risk is 0.12. The divergence is $\mathrm{KL}(Q\|P) = \ln\frac{100}{10} \approx 2.30$, so that $\varepsilon = \frac{2.30+7.14}{1000} \approx 0.0094$. McAllester's bound gives $0.12 + \sqrt{0.0094/2} \approx 0.189$, the PAC-Bayes-$\lambda$ bound (with $\lambda^\star \approx 0.33$) gives 0.178, and Seeger's bound gives $\overline{\mathrm{kl}}^{-1}(0.12, 0.0094) \approx 0.169$. If instead we keep $Q = P$, the KL vanishes but the Gibbs risk is larger, say 0.20, and Seeger's bound gives 0.251. At the other extreme, the Dirac on the best voter, with empirical risk 0.10, costs $\mathrm{KL} = \ln 100 \approx 4.61$ and Seeger's bound gives 0.152: this is the Occam bound.

In this example, the best *Gibbs* certificate is thus obtained by the Dirac, because $\ln 100$ nats is a small price for $m = 1000$ examples. But the Gibbs risk is not the end of the story: the *majority vote* of the 10 best voters can be much more accurate than any of them, and certifying it requires spread-out posteriors, as we will see in Section 5. In general, the best trade-off between empirical risk and KL is given by the Gibbs posterior of Section 4.1.

Running example Let us certify the Gibbs classifier of our forest, that is a tree drawn at random, with the uniform prior and the uniform posterior, so that

$\mathrm{KL}(Q\|P) = 0$. We need examples that the trees have not seen; for now, let us admit that each tree may be evaluated on the examples left out of its bootstrap sample (its *out-of-bag* examples), a construction justified in Section 7.2. The average out-of-bag error of the trees is 0.174, computed with at least 747 examples per tree. With $\delta = 0.05$, the complexity per example is $\ln(2\sqrt{747}/0.05)/747 = 0.0094$, McAllester's bound gives 0.243 and Seeger's bound 0.230. The test Gibbs risk, which the bound did not see, is 0.190: the certificate is valid and only four points above. The difficulty is elsewhere: the vote has a test risk of 0.044, and nothing in this section says anything about it. This is the object of Section 5.

Remark (Bernstein-type bounds) When the variance of the loss is small, the kl bound can be further improved by PAC-Bayes-Bernstein inequalities, which involve an empirical variance term (Tolstikhin and Seldin, 2013; Mhammedi et al., 2019). The same idea is used for majority votes in Section 5.8 (Wu et al., 2021).

To remember from Section 3

A generalization bound for a data-dependent predictor must be uniform, otherwise the selection bias makes it meaningless. All the classical PAC-Bayesian bounds follow from Theorem 3.1, whose only probabilistic step uses the independence between the prior and the sample; the posterior can therefore depend on the data arbitrarily. The complexity term is $\mathrm{KL}(Q\|P)/m$: a posterior close to the prior is cheap. Seeger's kl bound and Catoni's bound optimized on a grid are the tightest, McAllester's bound is the easiest to read, and linear bounds such as PAC-Bayes-λ are the easiest to minimize.

Check your understanding

Short questions to answer without computation; answers in Appendix A.5.

Q3.1. Which step of the proof of Theorem 3.1 requires the prior to be independent of the sample?

Q3.2. Why may the posterior depend on the data without invalidating the bound?

Q3.3. Which bound has a rate in $1/m$ when the empirical risk is zero, McAllester's or Seeger's?

Q3.4. Why can λ be optimized freely in the PAC-Bayes-λ bound, but not in the Hoeffding-based bound?

Exercises

Exercise 3.1. Starting from Theorem 3.1 with $\Delta(q,p) = 2(q-p)^2$, prove McAllester's bound directly, without going through the kl.

Exercise 3.2. For $m = 500$, $\delta = 0.05$, $\widehat{R}_S(G_Q) = 0.05$ and $\mathrm{KL}(Q\|P) \in \{0, 5, 20\}$, compute McAllester's and Seeger's bounds, and compare with Table 2.

Exercise 3.3. Using the linear form of Catoni's bound given after Corollary 3.4, find the value of C that minimizes it when $\widehat{R}_S(G_Q) = 0.12$, $\mathrm{KL}(Q\|P) = 5$, $m = 1000$ and $\delta = 0.05$, and compare the result with Table 2.

Exercise 3.4. Reproduce the experiment of Fig. 18 for $m \in \{100, 1000, 10000\}$. Show that the failure frequency of the naive bound does not go to zero when m grows, and explain why by comparing $\sqrt{\ln(1/\delta)/(2m)}$ with the typical deviation of the minimum of 1000 empirical risks.

Exercise 3.5. **(Comprehension)** In the proof of Theorem 3.1, which step uses the joint convexity of Δ, which step uses the independence of the prior from the sample, and which step is the only probabilistic one? Why does the bound hold for all posteriors although Markov's inequality is applied only once?

Exercise 3.6. **(Comprehension)** True or false? Justify briefly. (a) For the same $\widehat{R}_S(G_Q)$, $\mathrm{KL}(Q\|P)$, m and δ, Seeger's bound is never larger than McAllester's bound. (b) Doubling the sample size m halves the complexity term of McAllester's bound. (c) In Catoni's bound, the parameter C may be chosen after seeing the data, since the bound holds for all posteriors. (d) Dividing δ by 10 adds about 2.3 nats to the numerator of the complexity term.

Exercise 3.7. **(Application)** An ensemble of $n = 50$ voters has a uniform prior. With $m = 500$ examples and $\delta = 0.05$, the posterior Q uniform over the 5 best voters has an empirical Gibbs risk of 0.08. Compute $\mathrm{KL}(Q\|P)$ and the complexity per example $\varepsilon = \frac{1}{m}[\mathrm{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}]$, then McAllester's bound, the PAC-Bayes-λ bound at $\lambda^\star$, Seeger's bound (with `pbtools`), and Catoni's bound for $C \in \{0.5, 1, 2\}$. Which value of C is legitimate to report?

Exercise 3.8. **(Application)** A posterior has zero empirical Gibbs risk on $m = 200$ examples, with $\mathrm{KL}(Q\|P) = 3$ and $\delta = 0.05$. Compute Seeger's bound (using its closed form $1 - e^{-\varepsilon}$ when $\widehat{R}_S(G_Q) = 0$) and McAllester's bound. How many examples does each of them need to certify a Gibbs risk below 0.05 (with the same KL)?

Exercise 3.9. **(Proof)** Show that the binary kl is *jointly* convex in $(q, p) \in [0,1] \times (0,1)$, as required by Theorem 3.1. *Hint:* for $a > 0$, $b > 0$, write $a \ln \frac{a}{b} = a\, f\big(\frac{b}{a}\big)$ with the convex function $f(u) = -\ln u$, and show that the perspective $(a, b) \mapsto a\, f(b/a)$ of a convex function is jointly convex.

Exercise 3.10. **(Proof)** Show that, for fixed $\hat{q} > 0$ and $\varepsilon > 0$, the function $\lambda \mapsto \frac{\hat{q}}{1-\lambda/2} + \frac{\varepsilon}{\lambda(1-\lambda/2)}$ is minimized on $(0, 2)$ at $\lambda^\star = 2/\big(\sqrt{1 + 2\hat{q}/\varepsilon} + 1\big)$, which is (3.7), and that its minimal value is $\big(\sqrt{\hat{q} + \varepsilon/2} + \sqrt{\varepsilon/2}\big)^2$.

4 Posteriors, priors and the computation of the bounds

About this section

Guiding question. *Since the bounds hold for every prior and every posterior, which ones should we choose, and how do we evaluate the bound in practice?*

The bounds of Section 3 hold for any prior and any posterior. This section explains how to choose them. We show that the posterior minimizing a linear bound is the Gibbs posterior, a soft selection of the good voters that also contains Bayesian inference as a special case, and we run it on a real dataset. We then discuss the choice of the prior, which is the main lever to obtain tight certificates, and we measure on an experiment how to split the data between the prior and the bound. We finally explain how to compute a bound when the Gibbs risk has no closed form, and how PAC-Bayes can say something about deterministic predictors.

4.1 The optimal posterior: Gibbs posterior and Bayesian inference

A bound that holds for every posterior invites an obvious question: which posterior makes it smallest? For linear bounds the answer is explicit. The bounds (3.5) and (3.6) depend on Q only through the objective

$$\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)}{\lambda m} = \mathop{\mathbb{E}}_{h\sim Q}\Big[\widehat{R}_S(h)\Big] + \frac{1}{\lambda m}\,\mathrm{KL}(Q\|P), \tag{4.1}$$

which balances the fit of the posterior to the data against its distance to the prior; the parameter λ plays the role of an inverse temperature. We already know how to minimize such an objective: this is exactly the variational formula (2.3) of Section 2.

Proposition 4.1: Gibbs posterior (Catoni, 2007; Zhang, 2006)

The minimizer over $\mathcal{M}(\mathcal{H})$ of (4.1) is the **Gibbs posterior**

$$Q_\lambda(h) = \frac{P(h)\,e^{-\lambda m \widehat{R}_S(h)}}{\mathbb{E}_{h'\sim P}\Big[e^{-\lambda m \widehat{R}_S(h')}\Big]}, \tag{4.2}$$

and the minimal value is $-\frac{1}{\lambda m}\ln \mathbb{E}_{h\sim P}\, e^{-\lambda m \widehat{R}_S(h)}$.

Proof. Multiplying the objective by $-\lambda m < 0$ turns the minimization into a maximization, to which the variational formula applies:

$$\begin{aligned}\min_Q \Big\{\mathbb{E}_Q \widehat{R}_S + \frac{\mathrm{KL}(Q\|P)}{\lambda m}\Big\} &= -\frac{1}{\lambda m}\max_Q \Big\{\mathbb{E}_Q\big[-\lambda m \widehat{R}_S\big] - \mathrm{KL}(Q\|P)\Big\} \\ &\qquad \downarrow \text{ variational formula (2.3) with } \phi = -\lambda m \widehat{R}_S \\ &= -\frac{1}{\lambda m}\ln \mathop{\mathbb{E}}_{h\sim P}\Big[e^{-\lambda m \widehat{R}_S(h)}\Big],\end{aligned}$$

and the maximum is attained at the Gibbs distribution $Q_\phi \propto P\,e^{\phi} = P\,e^{-\lambda m \widehat{R}_S}$, which is Q_λ. □

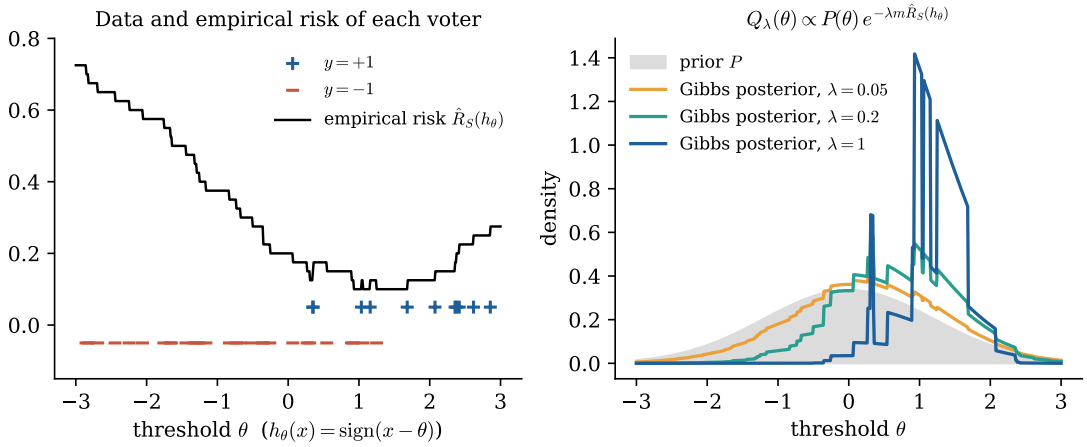


Figure 24: Gibbs posteriors on a family of threshold classifiers $h_\theta(x) = \mathrm{sign}(x - \theta)$ with a Gaussian prior on θ. As λ increases, the posterior concentrates on the empirical risk minimizers; for small λ it stays close to the prior. [code: `make_figures.py: fig_prior_posterior`]

In words, the Gibbs posterior starts from the prior and reweights it exponentially according to the empirical risk: a voter that makes k more training errors than another sees its relative weight divided by $e^{\lambda k}$. The temperature controls how sharp this selection is. When $\lambda \to 0$, the posterior stays equal to the prior: there is no complexity to pay, but nothing is learned. When $\lambda \to \infty$, the posterior concentrates on the empirical risk minimizers; for a finite class with the uniform prior, its KL tends to $\ln|\mathcal{H}|$ minus the logarithm of the number of minimizers, and we recover the Occam bound. Figure 24 shows this transition on a family of threshold classifiers $h_\theta(x) = \mathrm{sign}(x - \theta)$ with a Gaussian prior on θ: as λ grows, the posterior moves from the prior towards the interval of thresholds that minimize the empirical risk.

Example 4.1 (The Gibbs posterior on decision stumps). Let us see the temperature at work on the `breast cancer` dataset. We build 300 decision stumps (one feature, one threshold, one orientation). Their thresholds are the quantiles of the features computed on 100 examples that are then set aside, so that the set of voters, and hence the uniform prior, is independent of the $m = 120$ examples on which the bound is computed. We then compute Q_λ for 80 values of λ between 10^{-3} and 10^2, and Seeger's bound for each of them.

The right panel of Fig. 25 shows the trade-off. For small λ, the posterior is uniform, its empirical Gibbs risk is 0.5 (each stump comes with its opposite), and the bound is vacuous. As λ grows, the empirical risk drops quickly while the KL increases slowly. The bound reaches its minimum 0.256 for $\lambda \approx 1.3$, where the empirical Gibbs risk is 0.084 and $\mathrm{KL}(Q_\lambda \| P) = 5.5$ nats; the Gibbs risk measured on the 349 test examples is 0.114, well below the certificate. Beyond this point, the KL keeps increasing towards $\ln 300 \approx 5.7$ while the empirical risk no longer decreases.

With only $m = 120$ examples, a certificate of 0.256 may seem loose. It is however a *guarantee*, valid with probability 0.95 and obtained without any test set. The example also shows that the Gibbs posterior is essentially as good as the best stump in terms of certificate, while being more stable. What it certifies, though, is a stump drawn at random; we will see in Section 5 how to certify the *vote* of these stumps instead.

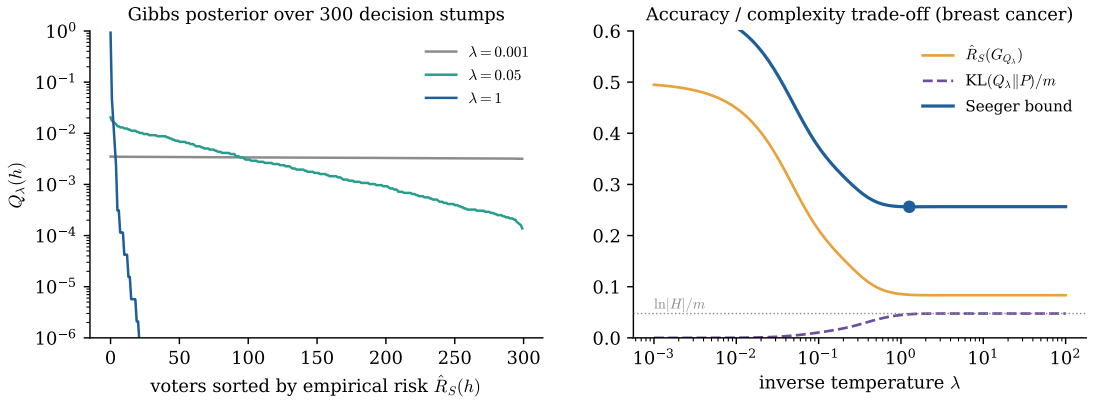


Figure 25: Gibbs posterior over 300 decision stumps on the `breast cancer` dataset (thresholds at quantiles computed on 100 separate examples, uniform prior, bound computed on $m = 120$ other examples). Left: posterior weights for three temperatures. Right: as λ grows, the empirical Gibbs risk decreases while $\mathrm{KL}(Q_\lambda \| P)$ increases, up to $\ln|\mathcal{H}|$ (Occam); Seeger's bound is minimized for an intermediate temperature. [code: `make_figures.py: fig_temperature`]

> **Running example** The same recipe applies to the 100 trees of our forest, with their out-of-bag errors in place of the empirical risks. Choosing the temperature by the procedure of Section 7.2.2, the Gibbs posterior lowers Seeger's certificate on the Gibbs risk from 0.230 (uniform weights) to 0.193. This looks like progress, but there is a catch, which we will only understand in Section 5: the Gibbs posterior puts almost all its weight on the two best trees, and the test risk of the *vote* jumps from 0.044 to 0.150. A better Gibbs classifier does not always make a better vote.

Link with Bayesian inference. The Gibbs posterior is not only an algorithmic device: it contains Bayesian inference as a special case. Take a probabilistic model $p(y \,|\, x, \theta)$ indexed by $\theta \in \mathcal{H}$, and the (unbounded) negative log-likelihood loss $\ell(\theta, (x, y)) = -\ln p(y \,|\, x, \theta)$. For $\lambda = 1$, the Gibbs posterior is

$$Q_1(\theta) \propto P(\theta)\, e^{-\sum_{i=1}^m \ell(\theta,(x_i,y_i))} = P(\theta) \prod_{i=1}^m p(y_i \,|\, x_i, \theta),$$

which is exactly the Bayesian posterior. Germain et al. (2016) exploited this correspondence to obtain PAC-Bayesian guarantees for Bayesian inference, and to show that minimizing the bound amounts to maximizing the *marginal likelihood* $\mathbb{E}_{\theta\sim P} \prod_i p(y_i|x_i,\theta)$, which justifies model selection by the evidence. Since the log-loss is unbounded, the bound requires additional assumptions on the tails of the loss (sub-Gaussian or sub-gamma). Other values of λ give *tempered*, or "generalized", posteriors, and Alquier et al. (2016) study variational approximations of Gibbs posteriors when they cannot be computed exactly.

The formulas coincide, but the two frameworks read them differently. In the Bayesian view the prior encodes beliefs and the posterior is the result of conditioning, whereas in PAC-Bayes the prior is a reference measure, the posterior is any distribution we like, and the guarantee does not require the model to be correct.

In a nutshell PAC-Bayes explains *why* averaging with exponential weights is a good idea: it is the optimal way to trade empirical risk against a KL penalty. In a finite ensemble (Section 7), the Gibbs posterior simply assigns to each voter a weight proportional to $P(h)\, e^{-\lambda m \widehat{R}_S(h)}$, a soft version of "keep the best voters".

4.2 Choosing the prior

The Gibbs posterior tells us how to choose Q once P is given. But the prior matters just as much, since the whole complexity term is measured relative to it. The only rule is that the prior must not depend on the sample used to evaluate the bound; otherwise it can be anything. Since the bound pays $\mathrm{KL}(Q\|P)$, it is tight only if the prior already puts mass where the posterior will be. Much of the art of PAC-Bayes thus consists in finding priors that are independent of the sample and yet well informed. We review the main strategies, from the most naive to the most data-efficient.

The simplest priors are *data-independent*: the uniform distribution on a finite set of voters, a Gaussian $\mathcal{N}(0, \sigma^2 I)$ on the weights of linear classifiers or neural networks, or the Fourier transform of a kernel for random features (Letarte et al., 2019b). Their parameters, such as the scale σ, can be selected on a grid at the price of a union bound. They are often too vague, however: nothing tells them in which region of $\mathcal{H}$ the good hypotheses lie.

To make the prior better informed without breaking the rules, a more effective strategy is to learn the prior on a part of the data that is not used in the bound (Ambroladze et al., 2006; Parrado-Hernández et al., 2012). The sample is split into $S = S_{\text{prior}} \cup S_{\text{bound}}$; the prior is learned on S_{prior}, for instance a Gaussian centred at an SVM trained on it, and the bound is computed on S_{bound} only, with $m = |S_{\text{bound}}|$. This is the approach behind the tightest certificates known for neural networks (Dziugaite and Roy, 2017; Pérez-Ortiz et al., 2021). The split involves a trade-off: a larger S_{prior} gives a better prior, hence a smaller KL, but it leaves fewer examples to compute the bound. Figure 26 shows the first effect on Gaussian posteriors over linear classifiers: the posterior is the same in both panels, but measured from a prior centred at the origin it costs 4.66 nats, and measured from a prior centred at a classifier learned on S_{prior} it costs only 0.21 nat. How should we balance the two effects?

Example 4.2 (How much data for the prior?). We measure this trade-off on `breast cancer` (Fig. 28). A fraction of the data is used to build a set of 200 shallow decision trees on bootstrap samples. The uniform distribution over these trees plays the role of the prior, which is legitimate since it only depends on S_{prior}. The remaining examples are split into a sample S_{bound} (70%), on which we minimize Seeger's bound over Gibbs posteriors, and a test sample (30%).

When only 5% of the data are used for the prior, the trees are weak, but the bound is computed on 378 examples and the certificate is 0.142. Increasing the fraction to 10–30% improves the trees (the test Gibbs risk decreases from 0.080 to 0.067) and the certificate stays around 0.135. Beyond 50%, the trees keep improving but the bound is computed on too few examples: with 85% of the data for the prior, only 60 examples remain, and the certificate degrades to 0.225 although the voters are the best ones. On this dataset, a fifth to a third of the data for the prior is a good compromise. The best

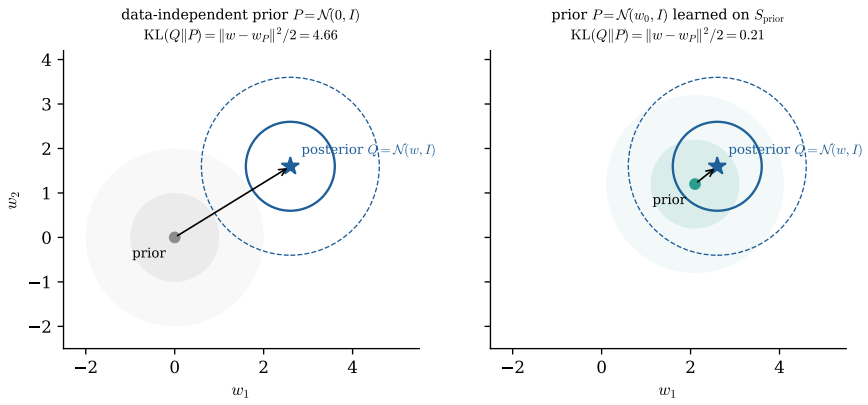


Figure 26: The same posterior $Q=\mathcal{N}(w,I)$ on linear classifiers (blue, circles at one and two standard deviations) measured against two priors. The KL divergence only depends on the distance between the centres, so a prior centred at a classifier w_0 learned on separate data makes it twenty times smaller. [code: `make_figures.py: fig_prior_choice`]

proportion depends on the problem; like any hyperparameter of the certificate, it must be fixed in advance or selected on a grid with a union bound.

Recent work: using the data twice, recursively Dziugaite et al. (2021) studied this trade-off in general: the best prior that does not depend on the data (the "oracle" prior $\mathbb{E}_S[Q_S]$) can leave the bound vacuous, whereas spending a fraction α of the sample on the prior and computing the kl bound on the rest, $\mathrm{kl}\left(\widehat{R}_{S\setminus S_\alpha}(G_Q)\,\|\,R_{\mathcal{D}}(G_Q)\right) \leq \frac{\mathrm{KL}(Q\|P(S_\alpha))+\ln(2\sqrt{(1-\alpha)m}/\delta)}{(1-\alpha)m}$, can make it non-vacuous. The drawback is that the examples of S_α no longer contribute to the confidence. Wu et al. (2024) remove it with *Recursive PAC-Bayes*. The sample is split into chunks $S_1,\dots,S_T$, and the posterior π_t learned on $S_1\cup\cdots\cup S_t$ becomes the prior of the next step. The key is the decomposition, for $\gamma_t\in(0,1)$,

$$\mathbb{E}_{\pi_t}[R_{\mathcal{D}}(h)] = \underbrace{\mathbb{E}_{\pi_t}\left[R_{\mathcal{D}}(h)-\gamma_t\,\mathbb{E}_{\pi_{t-1}}R_{\mathcal{D}}(h')\right]}_{\text{excess loss, certified on } S_t\cup\cdots\cup S_T} + \gamma_t \underbrace{\mathbb{E}_{\pi_{t-1}}[R_{\mathcal{D}}(h')]}_{\text{bounded at step } t-1},$$

which gives a recursive certificate $B_t(\pi_t)=\mathcal{E}_t(\pi_t,\gamma_t)+\gamma_t B_{t-1}(\pi_{t-1})$ (Fig. 27). Each chunk is used to learn a prior *and* to certify, and the excess loss, which takes four values, is bounded with the split-kl inequality of Section 5.8. On MNIST, the certificate on the Gibbs risk of a stochastic network goes from 0.457 with a data-independent prior and 0.408 with a data-dependent prior to 0.158 with $T=8$ steps.

Any split, however well chosen, sacrifices part of the sample. Two further ideas avoid this waste altogether. In bagging, each voter is trained on a subsample and can be evaluated on its out-of-bag examples; this *split construction* lets every example serve both for building some voters and for evaluating the others (Thiemann et al., 2017; Lorenzen et al., 2019), and it is the backbone of the self-bounding algorithms of Section 7.2. Finally, the prior may depend on the distribution $\mathcal{D}$ without depending on the sample, for instance $P\propto e^{-\beta R_{\mathcal{D}}(h)}$; the bound then involves unknown quantities which can be controlled (Catoni, 2007). Dziugaite and Roy (2018) go one step further

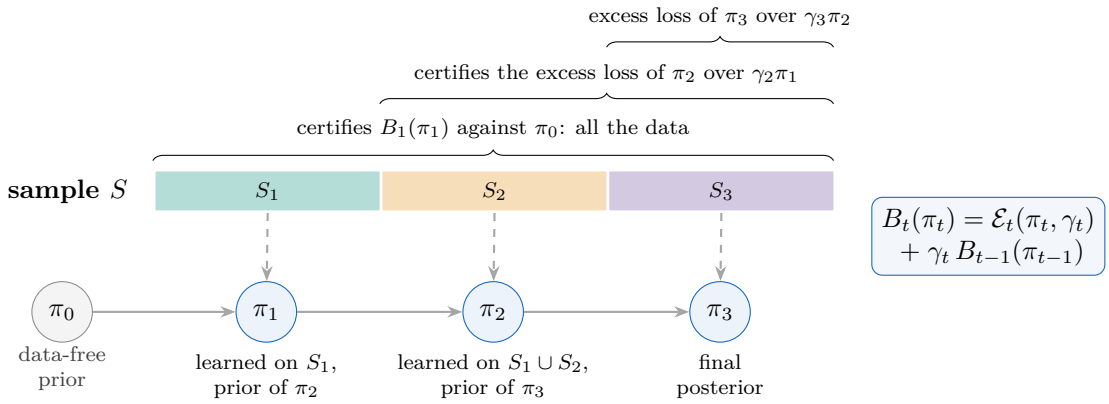


Figure 27: Recursive PAC-Bayes (Wu et al., 2024) with $T = 3$ chunks. Each posterior is the prior of the next one; the first term is certified with all the data against a data-free prior, and each subsequent excess loss with the chunks that were not used to learn the previous prior. [code: `tikz/recursive_pb.tex`]

and learn the prior from S itself, using differential privacy to limit how much it depends on each example.

Once the prior and the posterior are chosen, one practical step remains: evaluating the bound numerically.

4.3 Computing the bound in practice

A certificate is only useful if we can compute it. When $\mathcal{H}$ is a finite ensemble of n voters, $\widehat{R}_S(G_Q) = \sum_h Q(h)\widehat{R}_S(h)$ and $\mathrm{KL}(Q\|P)$ are finite sums, and computing a bound only requires one kl inversion. When $\mathcal{H}$ is infinite, for instance with a Gaussian posterior on the weights of a neural network, the empirical Gibbs risk is an integral that may not have a closed form. Langford and Caruana (2001) and Dziugaite and Roy (2017) estimate it by Monte Carlo: they draw $h_1, \dots, h_N \sim Q$ independently and apply a second kl inversion. This is legitimate because, conditionally on S, the $\widehat{R}_S(h_j)$ are i.i.d. variables in $[0,1]$ with mean $\widehat{R}_S(G_Q)$. With probability at least $1-\delta'$ over the draws,

$$\widehat{R}_S(G_Q) \le \overline{\mathrm{kl}}^{-1}\Big(\frac{1}{N}\sum_{j=1}^{N}\widehat{R}_S(h_j),\ \frac{\ln(2/\delta')}{N}\Big),$$

and plugging this upper bound into Seeger's bound gives a certificate valid with probability at least $1-\delta-\delta'$. In other words, sampling costs us a little confidence and a slightly larger bound, but never the validity of the certificate.

Example 4.3 (Certifying a Gaussian posterior by sampling). How large is this extra cost? We take a linear classifier $\mathbf{w}$ learned by a linear SVM on `breast cancer` ($m = 398$ training examples, standardized features), rescaled to $\|\mathbf{w}\| = 8$, and the posterior $Q = \mathcal{N}(\mathbf{w}, I)$ with prior $P = \mathcal{N}(\mathbf{0}, I)$, so that $\mathrm{KL}(Q\|P) = \|\mathbf{w}\|^2/2 = 32$. For this family, the empirical Gibbs risk has a closed form (Section 7.4), equal to 0.076, and Seeger's bound gives 0.245. Figure 29 shows the certificate obtained when the Gibbs risk is instead estimated from N sampled classifiers, with $\delta = \delta' = 0.025$. With $N = 100$ samples, the certificate is 0.375, and it reaches 0.26 for $N = 10^4$: the price of the Monte

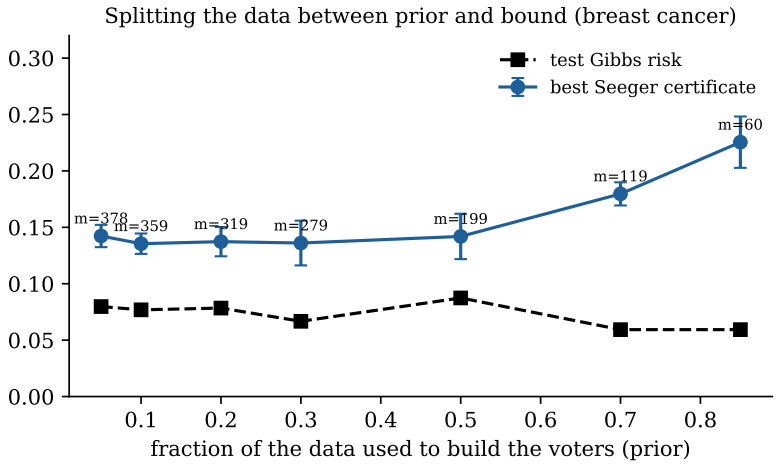


Figure 28: Splitting the data between the construction of the voters (prior) and the computation of the bound, on `breast cancer` (mean and standard deviation over 5 splits; m is the size of the sample used by the bound). [code: `make_figures.py: fig_ddprior`]

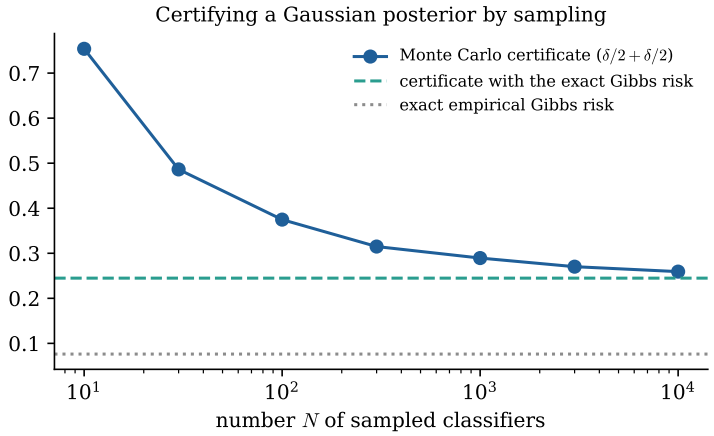


Figure 29: Certificate of a Gaussian posterior on linear classifiers (`breast cancer`), when the empirical Gibbs risk is computed exactly (dashed) or estimated with N sampled classifiers and a second kl inversion. [code: `make_figures.py: fig_montecarlo`]

Carlo estimation decreases quickly with N, and a few thousand samples are enough in practice.

4.4 From stochastic to deterministic predictors

Everything so far certifies a *stochastic* predictor. A PAC-Bayesian bound controls the *average* risk of a randomly drawn voter, whereas in practice we usually deploy a single deterministic predictor. There are three ways to bridge this gap. The first one, which is the object of Section 5, is to bound the risk of the majority vote B_Q using the Gibbs risk and second-order quantities describing the correlations between voters. The second one exploits a margin: if Q is concentrated around a hypothesis h with a large margin, then h and the Gibbs classifier make similar predictions, and the bound on the latter transfers to the former up to a margin term (Langford and Shawe-Taylor, 2002;

McAllester, 2003b; Biggs et al., 2022). The third one consists in *disintegrated* bounds, which hold with high probability for a single hypothesis $h \sim Q$ drawn once, rather than on average over Q (Blanchard and Fleuret, 2007; Rivasplata et al., 2020; Viallard et al., 2024b); we come back to them in Section 8. For ensemble methods the first route is the natural one, and it is the subject of the next section.

Recent work: Gibbs distributions for any complexity measure Viallard et al. (2024a) combine the last route with the Gibbs posterior. For any complexity measure $\mu(h, S)$ chosen by the user (a norm of the weights, a sharpness measure…), they draw a *single* hypothesis from $\rho_S(h) \propto e^{-\mu(h,S)}$ and prove a disintegrated bound: with the uniform prior on a bounded $\mathcal{H}$ and a loss in $[0,1]$, with probability at least $1-\delta$ over S, $h' \sim P$ and $h \sim \rho_S$, $\mathrm{kl}\left(\widehat{R}_S(h) \,\|\, R_{\mathcal{D}}(h)\right) \leq \frac{1}{m}\left[\mu(h', S) - \mu(h, S) + \ln \frac{8\sqrt{m}}{\delta^2}\right]$. The complexity measure thus enters the bound directly, in place of a KL divergence.

To remember from Section 4

Linear bounds are minimized by the Gibbs posterior $Q_\lambda \propto P\, e^{-\lambda m \widehat{R}_S}$, a soft selection of the good voters whose temperature trades fit against complexity, and which contains the Bayesian posterior as a special case. The prior is the main lever for tight bounds: it can be learned from data, provided those data are not used by the bound, and on our example a fifth to a third of the sample was a good investment. When the Gibbs risk is not computable exactly, a Monte Carlo estimate and a second kl inversion give a valid certificate at a small extra cost.

Check your understanding

Short questions to answer without computation; answers in Appendix A.5.

Q4.1. What happens to the Gibbs posterior when $\lambda \to 0$, and when $\lambda \to \infty$?

Q4.2. Why can a prior learned on part of the data be legitimate, and on which part must the bound then be computed?

Q4.3. Why does a Monte Carlo estimate of the Gibbs risk require a second kl inversion?

Q4.4. Name three ways to obtain a guarantee for a deterministic predictor from a PAC-Bayesian bound.

Exercises

Exercise 4.1. For a finite class and a uniform prior, show that $\mathrm{KL}(Q_\lambda \| P) \to \ln \frac{|\mathcal{H}|}{k}$ when $\lambda \to \infty$, where k is the number of empirical risk minimizers.

Exercise 4.2. Explain why using the same sample to learn the centre of a Gaussian prior and to compute the empirical term of the bound is not allowed. What happens if we do it anyway?

Exercise 4.3. In the Monte Carlo certificate, how should δ and δ' be split to minimize the final bound when $N = 100$? when $N = 10^4$?

Exercise 4.4. **(Comprehension)** (a) What is the Gibbs posterior Q_λ when all the voters have the same empirical risk? (b) With the uniform prior, voter h makes 3 more training errors

than voter h'; what is the ratio $Q_\lambda(h)/Q_\lambda(h')$? (c) True or false: a prior $P \propto e^{-\beta R_{\mathcal{D}}(h)}$, which depends on $\mathcal{D}$, is forbidden.

Exercise 4.5. **(Application)** Four voters have empirical risks 0.10, 0.12, 0.15 and 0.30 on $m = 100$ examples, and the prior is uniform. For $\lambda \in \{0.1, 0.5, 2\}$, compute the Gibbs posterior Q_λ, its empirical Gibbs risk, $\mathrm{KL}(Q_\lambda \| P)$ and Seeger's bound with $\delta = 0.05$. For $\lambda = 0.5$, check numerically that the objective (4.1) at Q_λ equals $-\frac{1}{\lambda m} \ln \mathbb{E}_{h\sim P} e^{-\lambda m \widehat{R}_S(h)}$. Compare with the bounds of the uniform posterior and of the Dirac on the best voter.

Exercise 4.6. **(Application)** A practitioner has 500 examples and 100 voters. Option A uses the 500 examples for the bound with a data-independent uniform prior; the best posterior found has an empirical Gibbs risk of 0.15 and $\mathrm{KL}(Q\|P) = 2$. Option B learns a prior on 150 examples and computes the bound on the remaining 350; the best posterior has an empirical Gibbs risk of 0.10 and $\mathrm{KL}(Q\|P) = 1$. Compare Seeger's bounds ($\delta = 0.05$). Can the practitioner compute both and report the smaller one?

Exercise 4.7. **(Application)** In the setting of the Gaussian posterior example ($m = 398$, $\mathrm{KL}(Q\|P) = 32$), suppose that $N = 1000$ classifiers sampled from Q have an average empirical risk of 0.08. With $\delta = \delta' = 0.025$, compute the Monte Carlo upper bound on $\widehat{R}_S(G_Q)$ and the final certificate. Compare with the certificate that would be obtained if $\widehat{R}_S(G_Q) = 0.08$ were known exactly.

Exercise 4.8. **(Proof)** Prove the formula $\mathrm{KL}\left(\mathcal{N}(\mu_1, \sigma^2 I_d) \,\|\, \mathcal{N}(\mu_0, \sigma^2 I_d)\right) = \frac{\|\mu_1 - \mu_0\|^2}{2\sigma^2}$, and deduce $\mathrm{KL}(Q_{\mathbf{w}}\|P) = \frac{1}{2}\|\mathbf{w}\|^2$ for $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$ and $P = \mathcal{N}(\mathbf{0}, I_d)$.

Exercise 4.9. **(Proof)** Let P be a prior on a finite set of parameters θ and $\ell(\theta, (x, y)) = -\ln p(y \,|\, x, \theta)$. Using Proposition 4.1 with $\lambda = 1$, show that the Bayesian posterior minimizes $\mathbb{E}_{\theta\sim Q} \sum_i \ell(\theta, (x_i, y_i)) + \mathrm{KL}(Q\|P)$ over Q, and that the minimal value is minus the log marginal likelihood $-\ln \mathbb{E}_{\theta\sim P} \prod_i p(y_i \,|\, x_i, \theta)$.

5 From the Gibbs classifier to the majority vote

About this section

Guiding question. *Why is the vote better than its voters, and how can we certify the vote itself rather than a voter drawn at random?*

The bounds of Section 3 concern the Gibbs classifier, whose risk is the risk of an average voter. Ensemble methods, however, predict with the majority vote, whose whole point is to be better than the average voter. This section relates the two. We show that the risk of the vote is a tail probability of the proportion of erring voters, and we control this tail with moments, which leads to the first-order bound, the tandem bound and the C-bound. We compare them with Condorcet's jury theorem and on random forests, prove which one is the tightest in which regime, and show how PAC-Bayes estimates each of them from data.

Throughout this section, the voters take values in $\{-1, +1\}$ and we use the 0-1 loss.

5.1 Two random variables that describe the vote

To compare the vote with its voters, we need a common language. A single random variable will do: on each example, the proportion of voters that err. For an example (x, y), let

$$W_Q(x,y) = \mathop{\mathbb{E}}_{h\sim Q}[\mathbb{1}[h(x) \neq y]] \in [0,1],$$
$$M_Q(x,y) = y \mathop{\mathbb{E}}_{h\sim Q}[h(x)] = 1 - 2W_Q(x,y) \in [-1,1].$$

The variable W_Q is the Q-weight of the voters that make a mistake on (x, y), and M_Q is the *margin* of the vote, already met in Section 1 for AdaBoost. The vote errs exactly when at least half of the weight is on the wrong side, so that, with our convention for ties,

$$R_{\mathcal{D}}(B_Q) = \mathbb{P}_{(x,y)\sim\mathcal{D}}(M_Q(x,y) \leq 0) = \mathbb{P}_{(x,y)\sim\mathcal{D}}\Big(W_Q(x,y) \geq \tfrac{1}{2}\Big), \tag{5.1}$$

whereas the Gibbs risk is its expectation, $R_{\mathcal{D}}(G_Q) = \mathbb{E}_{\mathcal{D}} W_Q$ (Fig. 30). In other words, the risk of the majority vote is a *tail probability* of the random variable W_Q, and the Gibbs risk is its *first moment*. All the bounds of this section control this tail with moments of W_Q, exactly as Markov's, Chebyshev's and Cantelli's inequalities control the tail of a random variable with its first two moments.

Figure 31 shows the distribution of W_Q for a random forest of 200 trees on the `digits` dataset (digits 0–4 against 5–9). This forest is larger than the one of our running example and is evaluated on a different split, hence the slightly different values. Most test points are classified correctly by a large majority of trees, so that the histogram is concentrated near zero; its mean, the Gibbs risk, is 0.164. Only the points in the red area, where at least half of the trees err, are misclassified by the vote, and they represent 3.5% of the test sample. The vote is thus almost five times more accurate than an average tree, and a good bound on the vote must be able to see this. Let us check how far each moment inequality can go, starting with the crudest one.

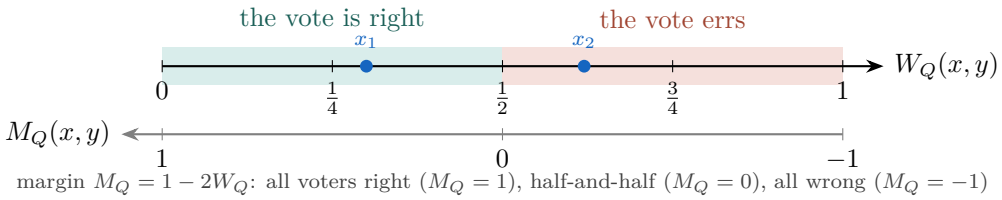


Figure 30: On an example (x, y), the vote errs if and only if the weight $W_Q(x, y)$ of the erring voters is at least $\frac{1}{2}$, that is if and only if the margin $M_Q(x, y) = 1 - 2W_Q(x, y)$ is non-positive. The example x_1 is correctly classified by the vote although 30% of the voters err on it; x_2 is misclassified. [code: `tikz/wq_line.tex`]

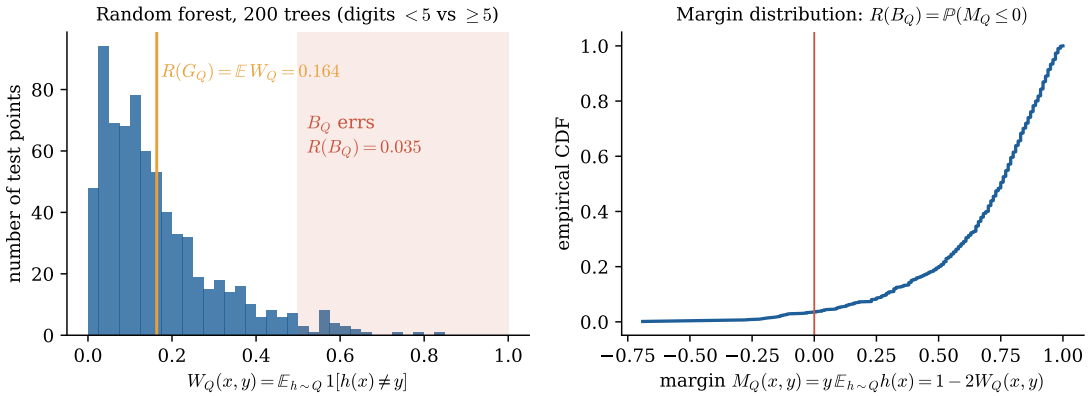


Figure 31: Random forest of 200 trees on `digits` (classes 0–4 vs 5–9). Left: histogram of W_Q on the test set; the Gibbs risk is the mean of this distribution (0.164), while the majority vote only errs when $W_Q \geq 1/2$ (red area, 3.5% of the points). Right: distribution of the margins $M_Q = 1 - 2W_Q$. [code: `make_figures.py: fig_wq_distribution`]

5.2 The first-order bound

Since we already know how to bound the Gibbs risk, the first moment of W_Q, the simplest idea is to turn it into a bound on the tail with Markov's inequality.

Proposition 5.1: First-order bound (folklore, e.g. Langford and Shawe-Taylor, 2002)

For any $Q \in \mathcal{M}(\mathcal{H})$, $R_{\mathcal{D}}(B_Q) \leq 2\, R_{\mathcal{D}}(G_Q)$.

Proof. The variable W_Q is non-negative, so

$$R_{\mathcal{D}}(B_Q) = \mathbb{P}\left(W_Q \geq \tfrac{1}{2}\right)$$

$$\downarrow \text{ Markov's inequality (Lemma 2.2) with } a = \frac{1}{2}$$

$$\leq \frac{\mathbb{E} W_Q}{1/2} = 2 R_{\mathcal{D}}(G_Q). \qquad \square$$

In words, the vote is never worse than twice an average voter: whatever the ensemble, voting cannot go badly wrong. Combined with Seeger's bound, the first-order

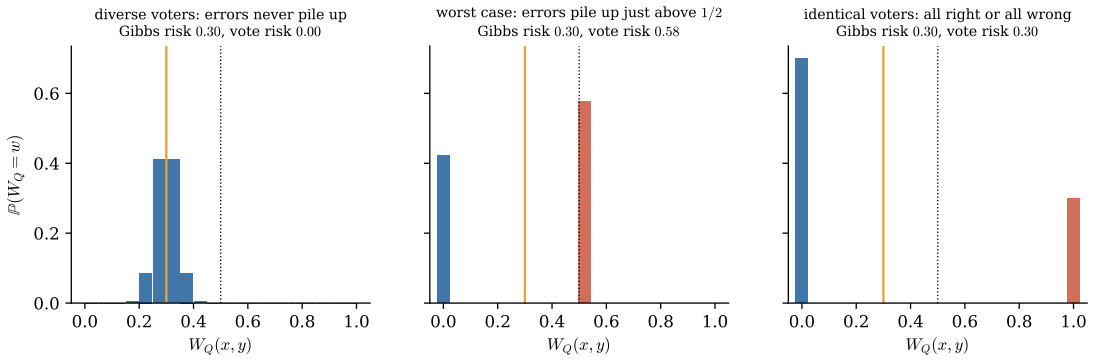


Figure 32: Three distributions of W_Q with the same mean, i.e. the same Gibbs risk 0.3 (orange line). Left: the voters are diverse and their errors never pile up, so the vote is always right. Middle: the errors pile up just above $\frac{1}{2}$ on 58% of the examples, so the vote errs there; this is the worst case of Markov's inequality. Right: identical voters, the vote is as good as one voter. [code: `make_figures.py: fig_wq_scenarios`]

bound gives a first PAC-Bayesian certificate for the majority vote: with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, for all Q,

$$R_{\mathcal{D}}(B_Q) \leq 2\,\overline{\mathrm{kl}}^{-1}\Big(\widehat{R}_S(G_Q), \frac{\mathrm{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}}{m}\Big). \tag{5.2}$$

The factor 2 cannot be improved in general: if W_Q only takes the values 0 and $\frac{1}{2}$, the vote errs on every example where $W_Q = \frac{1}{2}$ and the inequality is an equality. But this worst case is precisely the one where voting is useless, and the first-order bound completely ignores the benefit of the vote. Figure 32 makes the point with three ensembles that have the *same* Gibbs risk 0.3: depending on how the errors of the voters are distributed over the examples, the risk of the vote can be 0, 0.3, or almost 0.6. The Gibbs risk alone cannot distinguish these situations, so any bound that only uses it must cover the worst one.

On the forest of Fig. 31, it gives 0.327, twice the risk of an average tree, for a vote whose risk is ten times smaller (0.035). Worse, as we will see in Section 7.2.2, *minimizing* this bound pushes Q towards the few best voters and destroys the diversity of the ensemble. To do better, we must understand where the benefit of voting comes from.

5.3 Condorcet's jury theorem

To understand what a good bound on the vote should capture, consider the most favourable situation, known since Condorcet (1785): the voters err *independently*.

Example 5.1 (Condorcet's jury theorem). Let Q be uniform over $n = 2k+1$ voters that err independently, each with probability $p < 1/2$. Then $nW_Q \sim \mathrm{Bin}(n, p)$, the Gibbs risk is p and $R_{\mathcal{D}}(B_Q) = \mathbb{P}(\mathrm{Bin}(n,p) \geq k+1)$, which tends to 0 exponentially fast when $n \to \infty$. With $n = 3$ and $p = 0.3$, $R_{\mathcal{D}}(B_Q) = 3p^2(1-p) + p^3 = 0.216$; with $n = 25$, it is already 0.017, and with $n = 101$ it is about 10^{-5}.

The left panel of Fig. 33 plots this risk together with the bounds of this section. The first-order bound stays at $2p = 0.6$ whatever the number of voters. The tandem

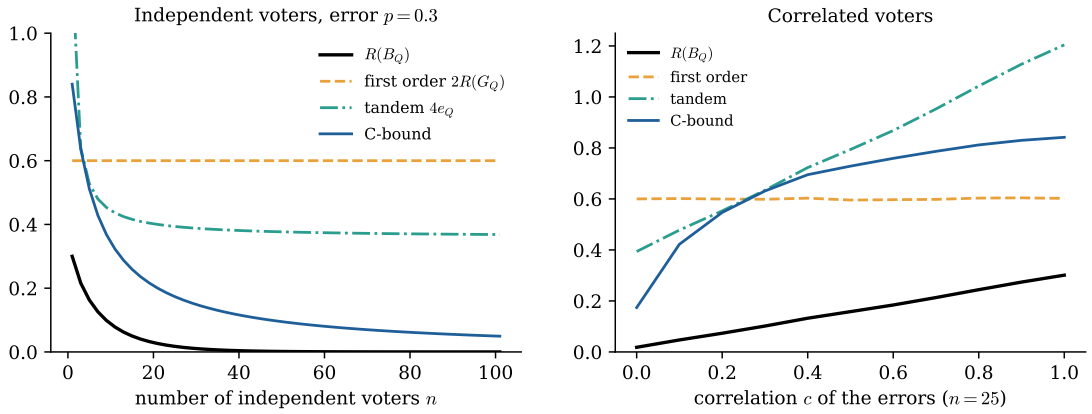


Figure 33: Risk of the majority vote and oracle bounds for voters of individual error $p = 0.3$. Left: independent errors, as a function of the number of voters. Right: 25 voters whose errors are correlated (each voter copies the errors of a common leader with probability c); 20 000 simulated examples. [code: `make_figures.py: fig_condorcet`]

bound, introduced below, improves on it but converges to $4p^2 = 0.36$. Only the C-bound follows the risk of the vote down to zero, at the rate $1/n$.

Independence is of course unrealistic. In the right panel, the 25 voters copy the errors of a common "leader" with probability c and err independently otherwise, so that c measures the correlation of their errors. When c increases, the vote loses its advantage, its risk goes from 0.018 to 0.30, and all the bounds deteriorate; for strongly correlated voters, the first-order bound becomes the best of the three. The rest of this section explains these observations.

5.4 Second-order quantities: joint error and disagreement

Condorcet's example contains the key idea: what makes a vote good is not only the quality of each voter, but the fact that the voters do not all make their mistakes on the same examples. A single voter cannot tell us this; we need to look at *pairs* of voters $(h, h') \sim Q^2 = Q \otimes Q$ drawn independently from the posterior.

Definition 5.2: Joint error and disagreement

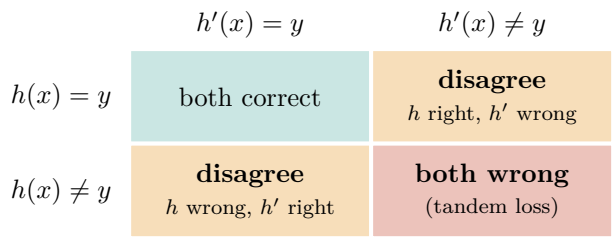
The **joint error** (or *tandem loss*) and the **disagreement** of Q are

$$e_{\mathcal{D}}(Q) = \mathop{\mathbb{E}}_{(h,h')\sim Q^2}\Big[\mathbb{P}_{(x,y)\sim\mathcal{D}}\big(h(x) \neq y \ \wedge \ h'(x) \neq y\big)\Big] = \mathbb{E}_{\mathcal{D}}[W_Q^2],$$

$$d_{\mathcal{D}}(Q) = \mathop{\mathbb{E}}_{(h,h')\sim Q^2}\big[\mathbb{P}_{x\sim\mathcal{D}_{\mathcal{X}}}\big(h(x) \neq h'(x)\big)\big] = \mathbb{E}_{\mathcal{D}}[2W_Q(1 - W_Q)],$$

and their empirical counterparts $\widehat{e}_S(Q)$, $\widehat{d}_S(Q)$ are defined on S.

Figure 34 shows the four outcomes of a pair of voters. The expressions in terms of W_Q follow from Fubini's theorem: for a fixed example (x, y), two independent draws $h, h' \sim Q$ both err with probability W_Q^2, and exactly one of them errs with probability $2W_Q(1 - W_Q)$, which, for binary voters, is exactly the probability that they disagree.



Averaging over $h, h' \sim Q$ independently and over $(x, y) \sim \mathcal{D}$:

- $e_{\mathcal{D}}(Q) = \mathbb{E}\,\mathbb{1}[h(x) \neq y]\,\mathbb{1}[h'(x) \neq y]$
- $d_{\mathcal{D}}(Q) = \mathbb{E}\,\mathbb{1}[h(x) \neq h'(x)]$

and the Gibbs risk counts one "wrong" voter:
$R_{\mathcal{D}}(G_Q) = e_{\mathcal{D}}(Q) + \frac{1}{2} d_{\mathcal{D}}(Q)$.

Figure 34: The four possible outcomes of two voters h, h' on an example. The joint error $e_{\mathcal{D}}(Q)$ is the probability of the red cell and the disagreement $d_{\mathcal{D}}(Q)$ the probability of the two orange cells, when h, h' are drawn independently from Q and (x, y) from $\mathcal{D}$. [code: `tikz/tandem_table.tex`]

The joint error is therefore the second moment of W_Q, and the Gibbs risk, the joint error and the disagreement are linked by a simple identity.

Proposition 5.3: Decomposition of the Gibbs risk

For any Q,

$$R_{\mathcal{D}}(G_Q) = e_{\mathcal{D}}(Q) + \tfrac{1}{2} d_{\mathcal{D}}(Q), \qquad \text{and} \qquad 0 \leq d_{\mathcal{D}}(Q) \leq 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q)).$$

Proof. Pointwise, $W_Q = W_Q^2 + \frac{1}{2} \cdot 2W_Q(1 - W_Q)$; taking expectations gives the identity. For the inequality,

$$\begin{aligned} d_{\mathcal{D}}(Q) &= 2(R_{\mathcal{D}}(G_Q) - e_{\mathcal{D}}(Q)) \\ &\quad \downarrow \text{ Jensen: } e_{\mathcal{D}}(Q) = \mathbb{E} W_Q^2 \geq (\mathbb{E} W_Q)^2 = R_{\mathcal{D}}(G_Q)^2 \\ &\leq 2(R_{\mathcal{D}}(G_Q) - R_{\mathcal{D}}(G_Q)^2) = 2R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q)). \end{aligned}$$

□

This decomposition says that, for a fixed Gibbs risk, a larger disagreement means a smaller joint error: the voters make their mistakes on *different* examples, which is exactly what makes a vote effective. On the forest of Fig. 31, the disagreement is 0.236 and the joint error is only 0.0457. Note also that the disagreement does not depend on the labels: it can be estimated on unlabeled data, which are often more plentiful, an observation exploited in semi-supervised learning and domain adaptation (Germain et al., 2020). With these second-order quantities in hand, we can revisit Markov's inequality.

5.5 The tandem bound

The first-order bound was loose because it only used the mean of W_Q. Applying Markov's inequality to W_Q^2 instead of W_Q gives a bound based on the second moment, which sees the joint errors.

Theorem 5.4: Tandem (second-order) bound (Masegosa et al., 2020)

For any $Q \in \mathcal{M}(\mathcal{H})$, $R_{\mathcal{D}}(B_Q) \leq 4\, e_{\mathcal{D}}(Q)$.

Proof.

$$R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \geq \tfrac{1}{2}) = \mathbb{P}(W_Q^2 \geq \tfrac{1}{4})$$

$$\downarrow \text{ Markov's inequality applied to the non-negative variable } W_Q^2$$

$$\leq \frac{\mathbb{E}W_Q^2}{1/4} = 4e_{\mathcal{D}}(Q). \qquad \square$$

In words, wherever the vote errs, at least half of the weight is on the wrong side, so two voters drawn independently from Q both err there with probability at least $\frac{1}{4}$; the frequency of such joint errors therefore controls the risk of the vote. On the forest of Fig. 31, the tandem bound is $4 \times 0.0457 = 0.183$, almost twice smaller than the first-order bound. By Proposition 5.3, $4e_{\mathcal{D}}(Q) = 2R_{\mathcal{D}}(G_Q) - 2(d_{\mathcal{D}}(Q) - R_{\mathcal{D}}(G_Q))$, so the tandem bound improves on the first-order bound exactly when $d_{\mathcal{D}}(Q) \geq R_{\mathcal{D}}(G_Q)$, that is when the ensemble is diverse enough. In the worst case of identical voters, $d_{\mathcal{D}}(Q) = 0$ and the tandem bound is twice the first-order bound.

To turn the tandem bound into a certificate, note that the joint error is the Gibbs risk of the *pair voter* $(h, h') \mapsto$ "both err" under the posterior Q^2 on $\mathcal{H}^2$. Applying Seeger's bound on $\mathcal{H}^2$ with prior P^2, and using $\mathrm{KL}(Q^2\|P^2) = 2\,\mathrm{KL}(Q\|P)$, we obtain the following corollary.

Corollary 5.5: PAC-Bayesian tandem bound (Masegosa et al., 2020)

With probability at least $1-\delta$ over $S \sim \mathcal{D}^m$, for all $Q \in \mathcal{M}(\mathcal{H})$,

$$R_{\mathcal{D}}(B_Q) \leq 4\,\overline{\mathrm{kl}}^{-1}\Big(\widehat{e}_S(Q),\ \frac{2\,\mathrm{KL}(Q\|P) + \ln\frac{2\sqrt{m}}{\delta}}{m}\Big). \tag{5.3}$$

The price of the second-order information is a doubled KL term, and a factor 4 instead of 2 in front of the kl inversion. The tandem certificate therefore pays off when the ensemble is diverse and the sample is large enough for the complexity term to be small.

5.6 The C-bound

The first-order bound uses the first moment of W_Q, the tandem bound its second moment. Using both at once should do better, and this is what the C-bound achieves through the Chebyshev–Cantelli inequality (Lemma 2.6).

Theorem 5.6: C-bound (Lacasse et al., 2006; Germain et al., 2015)

For any Q such that $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$ (equivalently $\mathbb{E}_{\mathcal{D}} M_Q > 0$),

$$R_{\mathcal{D}}(B_Q) \leq \mathcal{C}_Q := 1 - \frac{\left(\mathbb{E}_{\mathcal{D}} M_Q\right)^2}{\mathbb{E}_{\mathcal{D}} M_Q^2} = 1 - \frac{\left(1 - 2R_{\mathcal{D}}(G_Q)\right)^2}{1 - 2d_{\mathcal{D}}(Q)}. \tag{5.4}$$

Proof. Let $\mu = \mathbb{E}M_Q > 0$. Then

$$R_{\mathcal{D}}(B_Q) = \mathbb{P}(M_Q \leq 0) = \mathbb{P}(M_Q - \mu \leq -\mu)$$

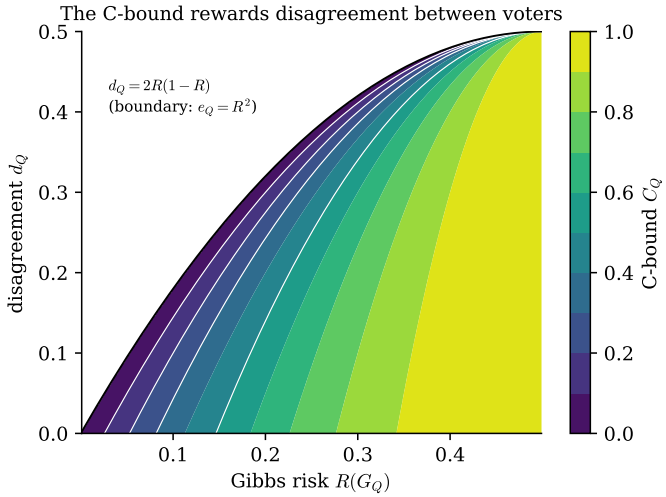


Figure 35: Value of the C-bound as a function of the Gibbs risk and the disagreement, on the feasible domain $d_{\mathcal{D}} \leq 2R(1-R)$. For a fixed Gibbs risk, increasing the disagreement decreases the bound: the C-bound formalizes the *strength–diversity* trade-off of ensemble methods (Breiman, 2001). [code: `make_figures.py: fig_cbound_landscape`]

$$
\begin{aligned}
&\downarrow \text{ Cantelli's inequality (Lemma 2.6) with } a = \mu \\
&\leq \frac{\operatorname{Var} M_Q}{\operatorname{Var} M_Q + \mu^2} \\
&\downarrow \text{ König–Huygens: } \operatorname{Var} M_Q = \mathbb{E}M_Q^2 - \mu^2 \\
&= \frac{\mathbb{E}M_Q^2 - \mu^2}{\mathbb{E}M_Q^2} = 1 - \frac{(\mathbb{E}M_Q)^2}{\mathbb{E}M_Q^2}.
\end{aligned}
$$

Finally, $\mathbb{E}M_Q = 1 - 2\mathbb{E}W_Q = 1 - 2R_{\mathcal{D}}(G_Q)$ and $\mathbb{E}M_Q^2 = \mathbb{E}(1-2W_Q)^2 = 1 - 4\mathbb{E}W_Q + 4\mathbb{E}W_Q^2 = 1 - 2d_{\mathcal{D}}(Q)$ by Proposition 5.3. □

The first expression of the C-bound has a nice interpretation: it is small when the margin is large on average and has a small variance, i.e. when the vote is correct with a comfortable and stable majority. The second expression, in terms of the Gibbs risk and the disagreement, makes the classical intuition on ensembles precise; Fig. 35 shows its landscape. For a fixed Gibbs risk, the C-bound decreases when the disagreement increases: *a good vote needs voters that are individually better than random and that disagree*. This is exactly what bagging and random forests try to achieve by randomizing the training of the trees, and what Breiman (2001) described with the *strength* and the *correlation* of the trees; in regression, the same idea is the *ambiguity decomposition* of Krogh and Vedelsby (1995), and Kuncheva and Whitaker (2003) and Brown et al. (2005) review the many measures of diversity proposed for classifier ensembles. On the forest of Fig. 31, the C-bound is 0.143 (with the unrounded values of the Gibbs risk and of the disagreement), the smallest of the three bounds. Is this always the case? The next subsection answers precisely.

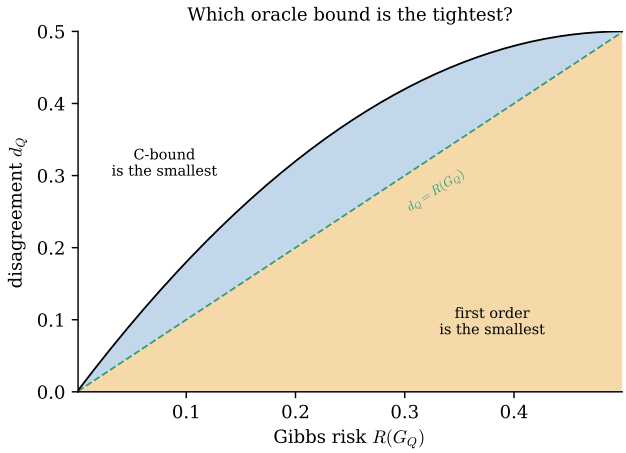


Figure 36: The smallest of the three oracle bounds in the plane (Gibbs risk, disagreement). Above the line $d_Q = R(G_Q)$, the C-bound is the smallest; below, the first-order bound is. The tandem bound is never strictly the smallest. [code: `make_figures.py: fig_bound_regions`]

5.7 Which bound is the tightest?

The three oracle bounds only depend on the pair $(R_{\mathcal{D}}(G_Q), d_{\mathcal{D}}(Q))$, since $4e_{\mathcal{D}}(Q) = 4R_{\mathcal{D}}(G_Q) - 2d_{\mathcal{D}}(Q)$. They can therefore be compared once and for all, and the comparison turns out to be remarkably simple.

Proposition 5.7: Comparison of the oracle bounds

Let $R = R_{\mathcal{D}}(G_Q) < \frac{1}{2}$ and $d = d_{\mathcal{D}}(Q)$. Then the C-bound is never larger than the tandem bound, $\mathcal{C}_Q \le 4e_{\mathcal{D}}(Q)$, and

$$d \ge R \implies \mathcal{C}_Q \le 4e_{\mathcal{D}}(Q) \le 2R, \qquad d \le R \implies 2R \le \mathcal{C}_Q \le 4e_{\mathcal{D}}(Q).$$

The three bounds are equal on the line $d = R$.

Proof. The inequality $\mathcal{C}_Q \le 4e_{\mathcal{D}}(Q)$ follows from Theorem 5.8 below: the tandem bound is the member $\mu = 0$ of a family whose minimum is the C-bound. We have seen that $4e_{\mathcal{D}}(Q) \le 2R$ if and only if $d \ge R$. For the C-bound, since $1 - 2R > 0$ and $1 - 2d > 0$,

$$\mathcal{C}_Q \le 2R \iff \frac{(1-2R)^2}{1-2d} \ge 1 - 2R \iff 1 - 2R \ge 1 - 2d \iff d \ge R.$$

On the line $d = R$, all three expressions equal $2R$. □

In plain words: when two random voters disagree more often than an average voter errs, the C-bound is the best of the three; otherwise, the simple first-order bound wins. Figure 36 draws the two regions.

The proposition explains the observations made on Condorcet's example: independent voters have a large disagreement, $d = 2p(1-p)(1-1/n)$, which exceeds p as soon as $n \ge 5$ when $p = 0.3$, so that the C-bound is the best; strongly correlated voters rarely disagree, and the first-order bound becomes the best. The proposition also shows that, as *oracle* quantities, the C-bound dominates the tandem bound. The situation

changes once the bounds have to be *estimated* from data: the tandem bound needs a single kl inversion on one quantity, while the C-bound needs two, and the ratio in its definition amplifies the estimation errors. In practice the two certificates are often close, as we will see in Section 7.

PAC-Bayesian C-bound. Let us see how the C-bound is estimated. Since $\mathcal{C}_Q$ is increasing in $R_{\mathcal{D}}(G_Q)$ on $[0, \frac{1}{2})$ and decreasing in $d_{\mathcal{D}}(Q)$, an upper bound on the first and a lower bound on the second give an upper bound on $\mathcal{C}_Q$. With a union bound (confidence $\delta/2$ for each), with probability at least $1-\delta$, for all Q,

$$R_{\mathcal{D}}(B_Q) \le 1 - \frac{\big(1-2\,\overline{r}\big)^2}{1-2\,\underline{d}}, \qquad \text{with} \tag{5.5}$$

$$\overline{r} = \overline{\mathrm{kl}}^{-1}\Big(\widehat{R}_S(G_Q), \tfrac{\mathrm{KL}(Q\|P)+\ln\frac{4\sqrt{m}}{\delta}}{m}\Big), \qquad \underline{d} = \underline{\mathrm{kl}}^{-1}\Big(\widehat{d}_S(Q), \tfrac{2\,\mathrm{KL}(Q\|P)+\ln\frac{4\sqrt{m'}}{\delta}}{m'}\Big),$$

provided $\overline{r} < \frac{1}{2}$, where m' is the number of (possibly unlabeled) examples used to estimate the disagreement. Lacasse et al. (2006) and Germain et al. (2015) give tighter versions that bound the pair $(R_{\mathcal{D}}(G_Q), d_{\mathcal{D}}(Q))$ jointly, and Viallard et al. (2021) derive C-bounds that can be minimized directly by gradient descent. Note that Vignault et al. (2023) corrected one of the results of Germain et al. (2015) on the variance of the margin (their Proposition 15), and showed that when the labels are noisy the C-bound cannot be made arbitrarily small, even for a very diverse ensemble.

5.8 A unifying view: the Chebyshev–Cantelli family

We have met three bounds, each obtained from $\mathbb{P}(W_Q \ge \frac{1}{2})$ with a moment inequality. Are they really different objects? Wu et al. (2021) noticed that the last two belong to a single parametric family.

Theorem 5.8: Parametric second-order bound (Wu et al., 2021)

For any Q and any $\mu < \frac{1}{2}$,

$$R_{\mathcal{D}}(B_Q) \le \frac{\mathbb{E}_{\mathcal{D}}\big[(W_Q-\mu)^2\big]}{\big(\frac{1}{2}-\mu\big)^2} = \frac{e_{\mathcal{D}}(Q) - 2\mu R_{\mathcal{D}}(G_Q) + \mu^2}{\big(\frac{1}{2}-\mu\big)^2}. \tag{5.6}$$

For $\mu = 0$ it is the tandem bound, and, when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$, its minimum over $\mu < \frac{1}{2}$ is reached at $\mu^\star = \frac{R_{\mathcal{D}}(G_Q)/2 - e_{\mathcal{D}}(Q)}{1/2 - R_{\mathcal{D}}(G_Q)}$ and equals the C-bound $\mathcal{C}_Q$.

Proof. For $\mu < \frac{1}{2}$, $\{W_Q \ge \frac{1}{2}\} \subseteq \{(W_Q-\mu)^2 \ge (\frac{1}{2}-\mu)^2\}$; apply Markov's inequality. The minimization over μ is left as an exercise (TD, Exercise 8); it reproduces the classical proof of Cantelli's inequality. □

In words, instead of measuring how far W_Q is from 0, as the tandem bound does, we measure how far it is from a well-chosen centre μ; the best centre gives back the C-bound. Figure 37 draws this family on a random forest of 100 trees trained on a simulated problem, for which the Gibbs risk is 0.247, the disagreement 0.314 and the

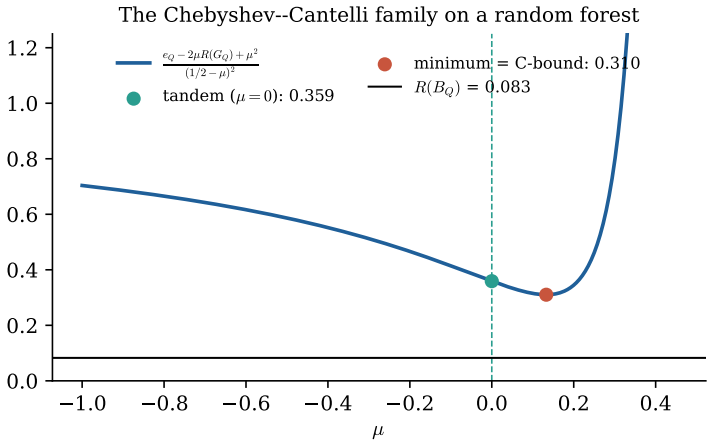


Figure 37: The Chebyshev–Cantelli family (5.6) as a function of μ, for a random forest of 100 trees on a simulated problem (oracle values computed on the test sample). [code: `make_figures.py: fig_cctnd`]

joint error 0.090. The curve starts from the tandem bound 0.359 at $\mu = 0$, decreases to the C-bound 0.310 at $\mu^\star = 0.13$, and blows up when μ approaches $\frac{1}{2}$. The vote itself has a risk of 0.083: even the best member of the family is loose, which is the price of using only two moments of W_Q.

If the minimum is the C-bound, why keep the whole family? The interest of the parametric form is statistical. The quantity $\mathbb{E}(W_Q - \mu)^2$ has a small variance when μ is well chosen, and it can be estimated with a PAC-Bayes-Bennett inequality that exploits this small variance. The resulting *CCPBB* bound was the tightest of the bounds of this section in the experiments of Wu et al. (2021) when enough data are available. Related refinements exploit the fact that the excess losses involved take only three values (split-kl inequality, Wu and Seldin, 2022).

Recent work: the split-kl inequality The quantities met in this section, such as $(W_Q - \mu)^2$ estimated through pairs of voters, or the excess losses of Section 4.2, are not Bernoulli variables but take a few values, and their variance is often small. The kl inequality, optimal for Bernoulli variables, cannot see this. Wu and Seldin (2022) split a variable $Z \in [a, b]$ around a fixed μ into its positive and negative parts, $Z = \mu + (Z - \mu)_+ - (Z - \mu)_-$, and apply a kl inequality to each part: with probability at least $1 - \delta$,

$$\mathbb{E}Z \le \mu + (b - \mu)\,\overline{\mathrm{kl}}^{-1}\Big(\frac{\hat{p}^+}{b - \mu}, \frac{\ln \frac{2}{\delta}}{n}\Big) - (\mu - a)\,\underline{\mathrm{kl}}^{-1}\Big(\frac{\hat{p}^-}{\mu - a}, \frac{\ln \frac{2}{\delta}}{n}\Big),$$

where $\hat{p}^\pm$ are the empirical means of the two parts; the PAC-Bayesian version replaces $\ln \frac{2}{\delta}$ by $\mathrm{KL}(Q\|P) + \ln \frac{4\sqrt{n}}{\delta}$. For a ternary variable $Z \in \{-1, 0, 1\}$ which is zero most of the time, the gain over the kl inequality applied to $(Z + 1)/2$ is a factor of about two (Fig. 38). This inequality is the basis of the tightest second-order certificates for majority votes (Wu and Seldin, 2022) and of the Recursive PAC-Bayes of Section 4.2. Biggs and Guedj (2023) follow a related idea: they subtract from the loss of each example a reference loss computed online on the previous examples, which gives low-variance, signed losses, and a

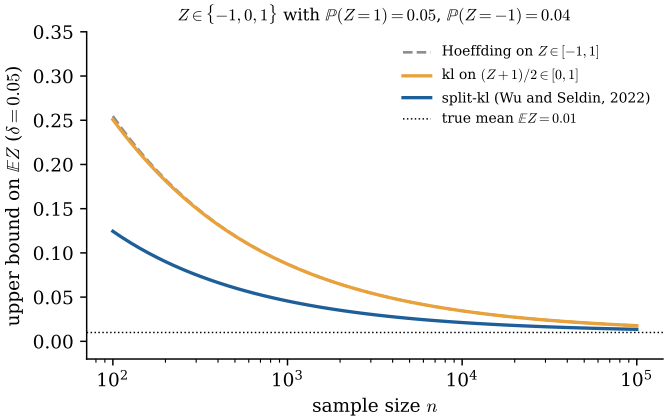


Figure 38: Upper bounds on the mean of a ternary variable $Z \in \{-1, 0, 1\}$ with mean 0.01 and small variance, as a function of the sample size (empirical frequencies equal to the true ones). Hoeffding's and the kl inequality applied to $(Z+1)/2$ almost coincide; the split-kl inequality is about twice tighter. [code: `make_figures.py: fig_split_kl`]

kl-type inequality adapted to them.

5.9 Margins and boosting

All the bounds so far summarize the margin distribution by one or two moments. The *margin theory* of boosting (Schapire et al., 1998), which we met in Fig. 6, uses the whole distribution instead, and Equation (5.1) is the bridge between the two viewpoints. For any $\theta > 0$, $R_{\mathcal{D}}(B_Q) = \mathbb{P}(M_Q \leq 0) \leq \mathbb{P}(M_Q \leq \theta)$, and the classical margin bound for a finite class of voters states that, with probability at least $1 - \delta$,

$$\mathbb{P}_{\mathcal{D}}(M_Q \leq 0) \leq \mathbb{P}_S(M_Q \leq \theta) + O\left(\sqrt{\frac{\ln|\mathcal{H}| \, \ln m}{m\,\theta^2} + \frac{\ln(1/\delta)}{m}}\right).$$

This explains why AdaBoost may keep improving after its training error has reached zero: it keeps increasing the margins, and the empirical term $\mathbb{P}_S(M_Q \leq \theta)$ decreases for every θ.

A PAC-Bayesian proof of such bounds is obtained by approximating B_Q by a vote of N voters drawn from Q (Langford and Shawe-Taylor, 2002; McAllester, 2003b), and Biggs et al. (2022) sharpened margin bounds for voting classifiers using Dirichlet posteriors. The C-bound can be read as a margin bound that uses the first two moments of the margin distribution instead of its whole empirical distribution function. Outside PAC-Bayes, Høgsgaard Møller and Larsen (2025) recently removed a superfluous $\ln m$ factor from the classical margin bound, obtaining an almost optimal margin bound for voting classifiers.

Recent work: bounding the vote itself, exactly All the bounds of this section go from the Gibbs risk to the risk of the vote through an inequality. Leblanc and Germain (2025) propose an *identity* instead. For a fixed deterministic classifier h, for instance $h = B_Q$, let $b^Q(h)$ be the Gibbs risk restricted to the

examples on which h is right, and $c^Q(h)$ the Gibbs risk restricted to the examples on which h errs. Since the Gibbs risk is the average of these two conditional risks, $R_{\mathcal{D}}(G_Q) = (1 - R_{\mathcal{D}}(h))\, b^Q(h) + R_{\mathcal{D}}(h)\, c^Q(h)$, hence

$$R_{\mathcal{D}}(h) = \frac{R_{\mathcal{D}}(G_Q) - b^Q(h)}{c^Q(h) - b^Q(h)} \qquad \text{when } c^Q(h) \neq b^Q(h).$$

For the majority vote, $c^Q(B_Q) \geq \frac{1}{2}$ since the vote errs only where half of the voters err, and $b^Q(B_Q) \leq \frac{1}{2}$. Bounding each of the three quantities with a PAC-Bayes-kl bound gives a certificate for the vote which, according to the authors, can be up to twice tighter than the usual alternatives.

5.10 Back to the running example

We now have all the tools to return to the two questions raised by our forest in Section 1.

Running example We can now answer the first question of Section 1. On the test sample, the trees of our forest have a Gibbs risk of 0.190 and a disagreement of 0.266: two trees drawn at random disagree on more than a quarter of the digits, and they both err on only 5.7% of them. Since the disagreement exceeds the Gibbs risk, Proposition 5.7 tells us that the C-bound is the smallest oracle bound, and indeed the first-order bound is 0.381, the tandem bound 0.229 and the C-bound 0.179, for a vote whose risk is 0.044. The vote is good *because the trees are diverse*: they are individually mediocre, but their errors are spread over different digits.
The second question, certifying the vote without a test set, receives a more modest answer. Estimating the three quantities on the out-of-bag examples (as in the running example of Section 3), with $\delta = 0.05$, gives the certificates 0.460 (first order), 0.387 (tandem) and 0.559 (C-bound). They are valid, but far from the test risk: the factor 4 of the tandem bound and the doubled KL term are expensive, and the C-bound suffers from its two kl inversions. Improving these certificates by *learning* the weights of the trees is the program of Part III.

5.11 Summary and illustration on random forests

To close the section, Fig. 39 puts all the pieces together on random forests of 100 trees, where we vary the number of features tried at each split (`max_features`), the main diversity parameter of the algorithm. With a single feature per split, the trees are weak (Gibbs risk 0.30) but very diverse (disagreement 0.38); with all 20 features, they are stronger (Gibbs risk 0.17) but much more similar (disagreement 0.19). The risk of the vote is smallest in between, around 4 to 8 features, where it reaches 0.077.

The first-order bound only sees the strength of the trees and decreases monotonically, from 0.60 to 0.34, missing the optimum. The tandem bound and the C-bound see both strength and diversity: they are minimal at 8 features, close to the best vote, with values 0.29 and 0.28. Since the disagreement always exceeds the Gibbs risk in this experiment, Proposition 5.7 predicts the order C-bound $\leq$ tandem $\leq$ first order, which is what we observe.

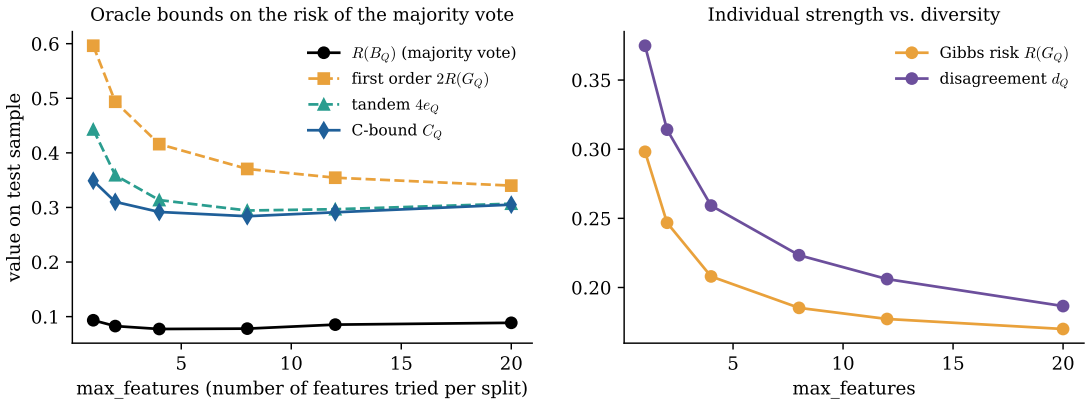


Figure 39: Random forests of 100 trees on a simulated problem, for an increasing number of features tried at each split (`max_features`). Left: oracle values of the bounds computed on a large test sample. Right: stronger trees (small Gibbs risk) are also less diverse (small disagreement). The C-bound and the tandem bound track the risk of the vote much better than the first-order bound. [code: `make_figures.py: fig_mv_bounds_diversity`]

The following table summarizes the bounds of the section: which moment of W_Q they use, and how PAC-Bayes estimates them.

Bound	Oracle form	Moment used	PAC-Bayesian estimation
First order	$2R_{\mathcal{D}}(G_Q)$	$\mathbb{E}W_Q$	Seeger on $\widehat{R}_S(G_Q)$, KL
Tandem	$4e_{\mathcal{D}}(Q)$	$\mathbb{E}W_Q^2$	Seeger on $\widehat{e}_S(Q)$, 2 KL
C-bound	$1-\frac{(1-2R_{\mathcal{D}}(G_Q))^2}{1-2d_{\mathcal{D}}(Q)}$	$\mathbb{E}W_Q$, $\mathbb{E}W_Q^2$	two kl inversions
Chebyshev–Cantelli	$\frac{e_{\mathcal{D}}-2\mu R_{\mathcal{D}}(G_Q)+\mu^2}{(1/2-\mu)^2}$	$\mathbb{E}(W_Q-\mu)^2$	PAC-Bayes-Bennett

To remember from Section 5

The risk of a vote is the tail probability $\mathbb{P}(W_Q \geq \frac{1}{2})$ of the proportion of erring voters, whereas the Gibbs risk is its mean. The average quality of the voters is not enough to control this tail: one also needs to know how their errors are correlated, which is measured by the disagreement, through the identity $R_{\mathcal{D}}(G_Q) = e_{\mathcal{D}}(Q) + \frac{1}{2}d_{\mathcal{D}}(Q)$. The first-order bound $2R_{\mathcal{D}}(G_Q)$ ignores the correlations, the tandem bound $4e_{\mathcal{D}}(Q)$ and the C-bound take them into account, and the C-bound, which is the minimum of the Chebyshev–Cantelli family, is the smallest as soon as $d_{\mathcal{D}}(Q) \geq R_{\mathcal{D}}(G_Q)$. PAC-Bayes estimates the second-order quantities on pairs of voters, with a doubled KL term.

Check your understanding

Short questions to answer without computation; answers in Appendix A.5.

Q5.1. Express the risk of the vote and the Gibbs risk in terms of the random variable W_Q.

Q5.2. Why does the first-order bound ignore the benefit of voting?

Q5.3. Which quantity of the ensemble can be estimated without labels?

Q5.4. In which regime is the first-order bound smaller than the C-bound?

Exercises

Exercise 5.1. Give an example of three voters and a distribution on three points for which $R_{\mathcal{D}}(B_Q) = 0$ while $R_{\mathcal{D}}(G_Q) = \frac{1}{3}$. Compute the three oracle bounds.

Exercise 5.2. Show that the C-bound is always smaller than 1 when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$, and that it equals $4R_{\mathcal{D}}(G_Q)(1 - R_{\mathcal{D}}(G_Q))$ when $d_{\mathcal{D}}(Q) = 0$. Compare with the first-order bound.

Exercise 5.3. ($\star$) Prove that the minimum over μ of the Chebyshev–Cantelli family equals the C-bound (TD, Exercise 8).

Exercise 5.4. In Condorcet's model with n independent voters of error p, compute $e_{\mathcal{D}}(Q)$ and $d_{\mathcal{D}}(Q)$, and show that the C-bound decreases like $1/n$ while the tandem bound tends to $4p^2$.

Exercise 5.5. **(Comprehension)** True or false? Justify briefly. (a) If $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$, then $R_{\mathcal{D}}(B_Q) < \frac{1}{2}$. (b) If $d_{\mathcal{D}}(Q) = 0$, all the voters in the support of Q make the same predictions on almost every x. (c) Estimating the joint error $e_{\mathcal{D}}(Q)$ requires labels, but estimating the disagreement $d_{\mathcal{D}}(Q)$ does not. (d) For a fixed Gibbs risk, the tandem bound decreases when the disagreement increases.

Exercise 5.6. **(Application)** Three voters are evaluated on five equally likely examples $z_1, \dots, z_5$. Voter h_1 errs only on z_1, h_2 errs on z_2 and z_4, and h_3 errs on z_2 and z_3; the posterior is $Q = (0.4, 0.35, 0.25)$. Compute $W_Q(z_j)$ and $M_Q(z_j)$ for each example, the risk of the vote, the Gibbs risk, $e_{\mathcal{D}}(Q)$ and $d_{\mathcal{D}}(Q)$ (check the identity of Proposition 5.3), and the three oracle bounds. Which one is the smallest, and is this consistent with Proposition 5.7?

Exercise 5.7. **(Application)** In Condorcet's model with $n = 5$ independent voters of error $p = 0.3$, compute the exact risk of the vote, $e_{\mathcal{D}}(Q)$, $d_{\mathcal{D}}(Q)$ and the three oracle bounds. Check the regime predicted by Proposition 5.7.

Exercise 5.8. **(Application)** A posterior with $\mathrm{KL}(Q\|P) = 1$ has an empirical Gibbs risk 0.15 on $m = 1000$ labelled examples and an empirical disagreement 0.25 on $m' = 1000$ examples. With $\delta = 0.05$, compute $\overline{r}$, $\underline{d}$ and the PAC-Bayesian C-bound (5.5), the first-order certificate (5.2) and the tandem certificate (5.3) (the empirical joint error follows from Proposition 5.3, assuming the disagreement was measured on the same 1000 labelled examples). Compare with the value of the C-bound computed at the empirical values. What changes if $m' = 10\,000$ unlabeled examples are available?

Exercise 5.9. **(Proof)** For voters with values in $\{-1, +1\}$ and a fixed example (x, y), show that $\mathbb{P}_{(h,h')\sim Q^2}\big(h(x) \neq h'(x)\big) = 2W_Q(x,y)\big(1 - W_Q(x,y)\big)$. Deduce that $\mathbb{E}_{\mathcal{D}} M_Q^2 = 1 - 2d_{\mathcal{D}}(Q)$ and $R_{\mathcal{D}}(G_Q) = e_{\mathcal{D}}(Q) + \frac{1}{2}d_{\mathcal{D}}(Q)$.

Exercise 5.10. **(Proof)** In Condorcet's model with $n = 2k + 1$ independent voters of error $p < \frac{1}{2}$, use Hoeffding's inequality (Lemma 2.3) to show that $R_{\mathcal{D}}(B_Q) \leq e^{-2n(1/2-p)^2}$. Evaluate this bound for $p = 0.3$ and $n \in \{5, 25, 101\}$, and compare it with the exact risks and with the C-bound.

Exercise 5.11. **(Application)** Consider three ensembles with the same Gibbs risk 0.3, described by the distribution of W_Q over the examples (compare with Fig. 32): (i) $W_Q = 0.3$ on every example; (ii) $W_Q = 0.52$ with probability $\frac{0.3}{0.52}$ and $W_Q = 0$ otherwise; (iii) $W_Q = 1$ with probability 0.3 and $W_Q = 0$ otherwise. For each of them, compute the risk of the vote, the joint error, the disagreement, and the three oracle bounds $2R_{\mathcal{D}}(G_Q)$, $4e_{\mathcal{D}}(Q)$ and $\mathcal{C}_Q$. Which bound is the best in each case, and is this consistent with Proposition 5.7?

Part III

Self-bounding learning

Because PAC-Bayesian bounds hold for all posteriors simultaneously, they can be used as learning objectives: instead of certifying a vote after learning it, we learn the vote that has the best certificate. This part develops this *self-bounding* point of view: its principles and its rules (what may depend on the data), its comparison with the hold-out approach, the optimization tools, and the main self-bounding algorithms for ensembles, linear classifiers, kernels and neural networks. Along the way, we will see that the choice of the certificate matters: a bound that ignores the diversity of the voters leads to a bad vote.

6 Self-bounding learning

About this section

Guiding question. *If a bound holds for every posterior, why not choose the posterior that makes it smallest, and what does this change to the certificate?*

Up to now, PAC-Bayesian bounds were used after learning, to certify a posterior produced by some algorithm. This section reverses the roles: since the bounds hold for all posteriors, we can minimize *them, and obtain a predictor together with its own risk certificate. We explain why this is legitimate, what may and may not depend on the data, how this approach compares with the classical hold-out certificate, and which optimization tools are used by all self-bounding algorithms. Each point is illustrated by running the algorithms on random forests.*

The bounds of Sections 3 to 5 have a remarkable property: they hold *simultaneously for all posteriors*. So far we have used them to evaluate a posterior produced by another algorithm, such as the uniform weights of a random forest; in our running example, the uniform forest received a tandem certificate of 0.387. But nothing prevents us from choosing the posterior *because* its bound is small. Instead of minimizing an empirical risk and then trying to certify the result, we can directly minimize the bound: the algorithm then outputs a predictor *and* a guarantee on its risk. This idea goes back to Freund (1998) and Langford and Blum (2003), who coined the term *self-bounding learning algorithm*. PAC-Bayes turned it into a general and practical methodology (Germain et al., 2009, 2015; Thiemann et al., 2017; Masegosa et al., 2020; Viallard et al., 2021; Pérez-Ortiz et al., 2021).

6.1 From empirical risk minimization to bound minimization

To see what self-bounding learning changes, let us first recall how we usually learn and evaluate a model. In the first part of the course, learning followed a three-step pipeline. A model is learned by minimizing a regularized empirical risk on a training sample, its hyperparameters are tuned by cross-validation, and it is finally evaluated on a test sample that was never used before. The test error is an *estimate* of the true risk, which a concentration inequality turns into a *guarantee* (the test-set bound of Section 6.4). This last step has a cost: the examples of the test sample do not contribute to learning.

A self-bounding algorithm merges the three steps into one. Suppose that we have a PAC-Bayesian bound of the form

$$\mathbb{P}_{S\sim\mathcal{D}^m}\Big(\forall Q\in\mathcal{M}(\mathcal{H}),\quad R_{\mathcal{D}}(Q)\le\mathcal{B}(Q,S,\delta)\Big)\ge 1-\delta,$$

where $R_{\mathcal{D}}(Q)$ denotes the risk we want to control (the Gibbs risk, or the risk of the vote) and $\mathcal{B}(Q,S,\delta)$ is a quantity that can be computed from the sample. The algorithm learns

$$Q_S\in\operatorname*{arg\,min}_{Q\in\mathcal{Q}}\ \mathcal{B}(Q,S,\delta),\qquad\text{and returns the certificate}\qquad\mathcal{B}(Q_S,S,\delta),\tag{6.1}$$

where $\mathcal{Q}\subseteq\mathcal{M}(\mathcal{H})$ is the family of posteriors in which we search, for instance all distributions on a finite set of voters, or Gaussian distributions on the weights of a model.

Definition 6.1: Self-bounding learning algorithm

A *self-bounding learning algorithm* is a procedure which, given a sample $S \sim \mathcal{D}^m$ and a confidence level $\delta \in (0, 1]$, returns a predictor f_S and a number $\mathcal{B}_S \in [0, 1]$, both computed from S only, such that

$$\mathbb{P}_{S\sim\mathcal{D}^m}\Big(R_{\mathcal{D}}(f_S) \le \mathcal{B}_S\Big) \ge 1 - \delta.$$

The number $\mathcal{B}_S$ is called the *risk certificate* of f_S.

In plain words, a self-bounding algorithm hands us two things computed from the same data: a predictor, and a number that we can trust, with high probability, as an upper bound on its true risk. Many algorithms of this course fit this definition: the Gibbs posterior Q_λ of Section 4.1 with its Seeger certificate, the forest weighted by Algorithm 3 with its first-order certificate, or PBGD with Catoni's bound (Section 7.4). A random forest with uniform weights, returned together with its out-of-bag tandem certificate, is also self-bounding in the sense of the definition; the certificate is simply not used to learn. One question must be settled before we go further: if the posterior is chosen by looking at its own bound, is that bound still trustworthy?

6.2 Why the certificate remains valid

The following result is almost trivial, but it is the reason why the whole approach works, so it deserves to be stated carefully.

Theorem 6.2: Validity of self-bounding certificates

Let $\mathcal{B}$ be such that $\mathbb{P}_S(\forall Q \in \mathcal{Q},\ R_{\mathcal{D}}(Q) \le \mathcal{B}(Q, S, \delta)) \ge 1 - \delta$. Let $S \mapsto Q_S \in \mathcal{Q}$ be *any* learning rule (deterministic, or using a source of randomness independent of the data). Then

$$\mathbb{P}_S\Big(R_{\mathcal{D}}(Q_S) \le \mathcal{B}(Q_S, S, \delta)\Big) \ge 1 - \delta.$$

In particular this holds for Q_S minimizing $\mathcal{B}(\cdot, S, \delta)$ or any other objective, and for any stopping time of an iterative optimizer.

Proof. Let $E = \{S : \forall Q \in \mathcal{Q},\ R_{\mathcal{D}}(Q) \le \mathcal{B}(Q, S, \delta)\}$. For every $S \in E$, the inequality holds in particular for the posterior Q_S. Hence $\{S : R_{\mathcal{D}}(Q_S) \le \mathcal{B}(Q_S, S, \delta)\} \supseteq E$ and its probability is at least $\mathbb{P}(E) \ge 1 - \delta$. If Q_S also depends on an independent random seed ξ, apply the argument conditionally on ξ and integrate. □

In words: since the bad event does not depend on which posterior we pick, no way of picking one, however clever or data-driven, can make us fall into it. The key is the quantifier "for all Q" inside the probability. A bound which only holds for a *fixed* predictor, such as Hoeffding's inequality for a fixed h or a test-set bound, does not remain valid if the predictor is chosen by looking at the data used in the bound.

We saw in Fig. 18 how badly such a bound fails: after selecting the best of 1000 random classifiers, the naive Hoeffding bound was wrong in 99.8% of the runs. The

Occam bound remained valid because it pays $\ln \frac{1}{P(h)} = \ln 1000$, which is exactly the price of having *selected* h with the data. PAC-Bayes extends this accounting to any soft selection: the price of choosing a posterior Q with the data is $\mathrm{KL}(Q\|P)$. Because the price is already included in the bound, we are allowed to choose Q in any way we like, and in particular by minimizing the bound. This freedom concerns the posterior only, and the next subsection makes precise where it stops.

6.3 What may depend on the data?

Theorem 6.2 allows the posterior to depend arbitrarily on S. Everything else in the bound — the prior, the confidence level δ, the form of the bound and its internal parameters — must be fixed before seeing the sample used in the bound. When we nevertheless want to choose one of these ingredients with the data, we pay for it with a union bound, as the following proposition shows.

Proposition 6.3: Combining several bounds

Let $\mathcal{B}_1, \dots, \mathcal{B}_K$ be K bounds (they may use different priors, different parameters, different forms, or even different sets of voters), each valid in the sense that for every δ', $\mathbb{P}_S(\forall Q,\ R_{\mathcal{D}}(Q) \le \mathcal{B}_k(Q, S, \delta')) \ge 1 - \delta'$. Let $\pi_1, \dots, \pi_K > 0$ with $\sum_k \pi_k = 1$. Then

$$\mathbb{P}_S\Big(\forall k,\ \forall Q,\quad R_{\mathcal{D}}(Q) \le \mathcal{B}_k(Q, S, \pi_k \delta)\Big) \ge 1 - \delta.$$

Consequently, $\min_k \mathcal{B}_k(Q_S, S, \pi_k\delta)$ is a valid certificate for any learning rule Q_S, and the index k may be chosen with the data.

Proof. The probability that at least one of the K events fails is at most $\sum_k \pi_k \delta = \delta$ (union bound). □

In other words, we may try several bounds and keep the best one, provided we share the confidence budget δ among them in advance. With $\pi_k = \frac{1}{K}$, choosing among K candidates amounts to replacing δ by δ/K, that is to adding $\ln K$ to the numerator of the complexity term. This is a very small price for a moderate K: choosing among ten candidates costs 2.3 nats, the same as dividing δ by ten. With a countable family, one can take $\pi_k = \frac{6}{\pi^2 k^2}$, which costs $2\ln k + \ln \frac{\pi^2}{6}$ for the k-th candidate.

The line between what is free and what must be paid for is subtle, so the table below gathers the most common cases. The underlying rule is simple to state. A parameter is free when it only indexes a family of *posteriors*, all compared with the same prior in the same bound: the parameter C of PBGD, for instance, selects a weight vector $\mathbf{w}$, hence a posterior $Q_{\mathbf{w}}$, and Seeger's bound holds for all of them at once. A parameter is not free when it changes the *probabilistic event* on which the bound holds: the parameter C of Catoni's bound changes the function Δ of Theorem 3.1, and each value gives a different event.

Free (no correction)	**Union bound** ($\delta \to \delta/K$)	**Forbidden**
the posterior Q and the algorithm producing it; the temperature of a Gibbs posterior; λ in the PAC-Bayes-λ bound; C in PBGD; the learning rate, initialization and number of iterations (early stopping on the bound)	C in Catoni's bound; λ in the Hoeffding-type bound; the variance of a Gaussian prior chosen on a grid; the choice among first-order / tandem / C-bound; the choice among K ensembles built with different hyperparameters	a prior learned on the examples used in the bound; choosing δ after seeing the bound; repeating the experiment with many seeds and reporting the best certificate (unless all of them are accounted for)

Self-bounding model selection. Because a choice among K candidates costs only $\ln K$, Proposition 6.3 offers a principled alternative to cross-validation. We build K candidate models, for instance forests with different tree depths, learn a posterior for each of them by minimizing its bound computed with δ/K, and keep the candidate with the smallest certificate. The selected model comes with a valid certificate, and no example was set aside for validation.

Example 6.1 (Choosing the depth of the trees with the certificate). On `digits`, we build six forests of 100 trees, with maximal depths 1, 2, 3, 5, 8 and unlimited, each tree being trained on a bootstrap sample of half the training set. For each forest, the posterior minimizes the out-of-bag tandem bound of Section 7.2, computed with $\delta/6$. Figure 40 reports the certificates and the test risks over five random splits. The certificate decreases steadily with the depth, from 0.96 for stumps to 0.375 for unpruned trees, and so does the test risk of the vote, from 0.19 to 0.031. The certificate thus selects the unpruned trees, which is also the best choice for the test risk, and the price of the selection ($\ln 6 \approx 1.8$ nats) is invisible at this scale. (These values are averaged over five stratified splits, which explains the small differences with the single split used in Section 7.) In general, the certificate remains a pessimistic proxy for the test risk: when two candidates are close, it may select a slightly sub-optimal one.

6.4 Self-bounding versus hold-out

A self-bounding certificate is only interesting if it compares well with the simplest alternative. This natural competitor is the *test-set bound* (Langford, 2005), which certifies a predictor with an independent sample.

Proposition 6.4: Test-set bound

Let f be a predictor learned on S_1 and $T \sim \mathcal{D}^n$ a sample independent of S_1. Then, with probability at least $1-\delta$ over T, $R_{\mathcal{D}}(f) \leq \overline{\mathrm{kl}}^{-1}\left(\widehat{R}_T(f), \frac{\ln(1/\delta)}{n}\right)$.

Proof. Conditionally on S_1, f is fixed and $n\widehat{R}_T(f) \sim \mathrm{Bin}(n, R_{\mathcal{D}}(f))$; apply Chernoff's bound (Lemma 2.4). Equivalently, it is the Occam bound with a single hypothesis ($P = \delta_f$, $\mathrm{KL} = 0$). □

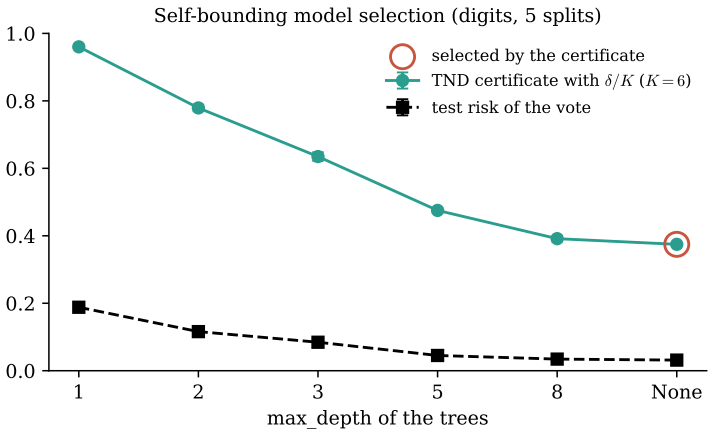


Figure 40: Self-bounding model selection. Six forests (100 trees, bootstrap samples of size $m/2$) with increasing tree depth are learned on `digits`; for each of them the posterior minimizes the tandem bound and the certificate is computed with $\delta/6$ (Proposition 6.3). The certificate decreases with the depth like the test risk; here it selects the model with the best test risk (means and standard deviations over 5 splits). [code: `make_figures.py: fig_selfbounding_selection`]

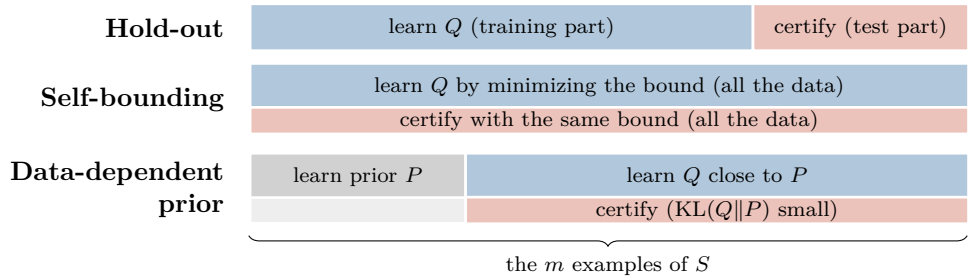


Figure 41: Three ways of using the sample. The hold-out approach learns on one part and certifies on the other; the self-bounding approach uses every example twice, to learn and to certify, which is legitimate because the bound holds for all posteriors; a data-dependent prior sits in between, the examples used for the prior being excluded from the bound. [code: `tikz/data_usage.tex`]

The test-set bound certifies *any* predictor, in particular the majority vote itself, with no factor 2 or 4 and no KL term, so it is very tight. Its drawback is that the n test examples do not contribute to learning. Self-bounding learning makes the opposite trade-off: all the data serve both to learn and to certify, but the certificate relies on a PAC-Bayesian bound, which is looser. Figure 41 compares how the two approaches, and the intermediate one of data-dependent priors (Section 4.2), use the m examples. Which trade-off wins is an empirical question, which the next example examines.

Example 6.2 (Which approach gives the better vote, and the better certificate?). We compare the two approaches on a simulated problem, for sample sizes m from 200 to 4000 (Fig. 42). The self-bounding approach builds a forest of 100 trees on the m examples and minimizes its out-of-bag tandem bound; the hold-out approach builds the same forest on 70% of the examples and computes the test-set bound of its vote on the remaining 30%. As expected, using all the data gives a better vote for every m: with $m = 1000$, its test risk is 0.099 against 0.110. But the hold-out certificate is much tighter: 0.17 against 0.63 for $m = 1000$, and 0.10 against 0.39 for $m = 4000$. For

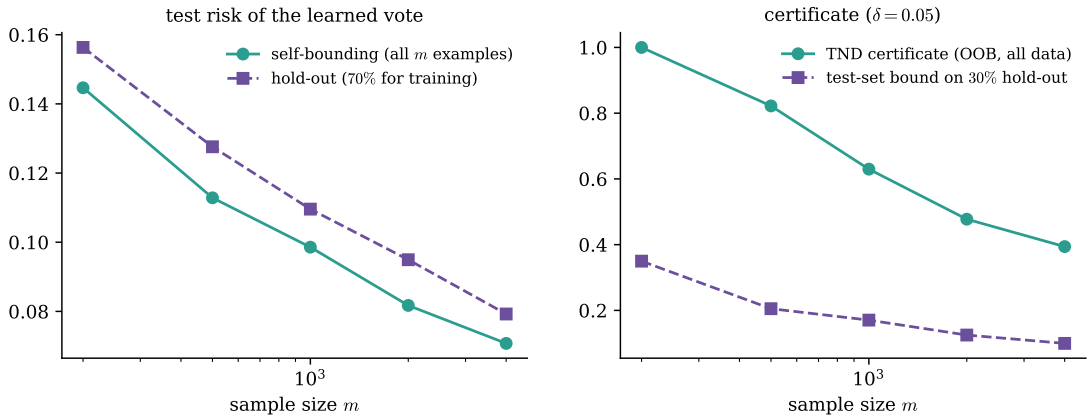


Figure 42: Self-bounding (forest learned on the m examples, posterior minimizing the OOB tandem bound, TND certificate) versus hold-out (forest learned on 70% of the examples, test-set bound on the remaining 30%), on a simulated problem, as a function of m (average over 3 repetitions). Using all the data gives a better vote (left), but for majority votes the hold-out certificate remains much tighter (right). [code: `make_figures.py: fig_selfbounding_vs_holdout`]

majority votes, the factor 4 of the tandem bound and the second-order KL term cost more than the examples withheld by the hold-out.

This experiment should be read honestly. If the only goal is to report a tight guarantee on a fixed vote, the hold-out is hard to beat. The value of self-bounding learning lies elsewhere: it uses all the data, it provides a learning objective with a guarantee, and it makes model selection almost free.

The gap in tightness is also specific to majority votes, where the passage from the Gibbs risk to the vote costs a constant factor; for stochastic neural networks, self-bounding certificates with data-dependent priors are tight enough to be informative, for instance on MNIST (Pérez-Ortiz et al., 2021). The following table summarizes the comparison.

	Hold-out (test-set bound)	Self-bounding (PAC-Bayes)
Data used to learn	part of the sample	the whole sample
Predictor certified	any (vote, deep net…)	a posterior (Gibbs, vote via Section 5)
Tightness	excellent: a single kl inversion, no KL term	depends on the KL and on the Gibbs-to-vote relation
Guides learning?	no	yes: the certificate is the objective
Model selection	needs another split	free or $\ln K$ (Proposition 6.3)

Best of both worlds: data-dependent priors. We do not have to choose between the two approaches: they can be combined. We split $S = S_1 \cup S_2$, use S_1 to learn a *prior* P_{S_1} (a Gaussian centred on a network trained on S_1, or voters built on S_1), and compute the PAC-Bayesian bound on S_2.

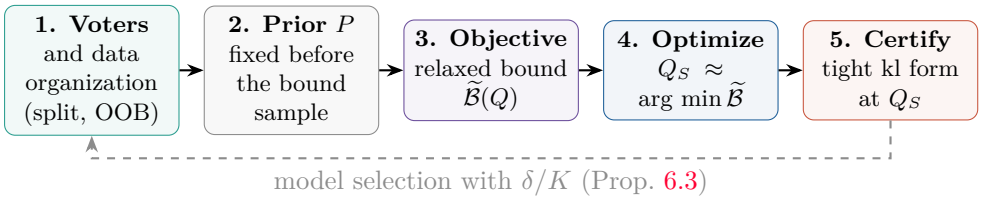


Figure 43: The five ingredients of a self-bounding algorithm. [code: `tikz/sb_pipeline.tex`]

Proposition 6.5: Self-bounding with a data-dependent prior

Let P_{S_1} be a prior learned on S_1 and $\mathcal{B}$ a PAC-Bayesian bound computed on $S_2 \sim \mathcal{D}^{m_2}$, independent of S_1. Then, with probability at least $1-\delta$ over (S_1, S_2), for all posteriors Q — including posteriors learned on the whole sample $S_1 \cup S_2$ — we have $R_{\mathcal{D}}(Q) \leq \mathcal{B}(Q, S_2, P_{S_1}, \delta)$.

Proof. Conditionally on S_1, the prior is fixed and S_2 is an i.i.d. sample, so the bound holds with probability at least $1-\delta$ over S_2, for all Q simultaneously. Integrate over S_1. □

Put simply, the prior may look at S_1 as long as the bound only looks at S_2. Only the *empirical term* of the bound must be computed on S_2; the posterior itself may use all the data. This is the strategy behind the tightest certificates for neural networks (Dziugaite and Roy, 2017; Pérez-Ortiz et al., 2021) and SVMs (Ambroladze et al., 2006; Parrado-Hernández et al., 2012), and we measured its trade-off in Fig. 28.

The two approaches of this subsection appear as the two ends of this spectrum. The Recursive PAC-Bayes of Wu et al. (2024), presented in Section 4.2, narrows the gap further by letting every example serve both to build a prior and to certify. The test-set bound is the extreme case where the posterior is a Dirac on the prior ($\mathrm{KL} = 0$), and a pure self-bounding algorithm is the other extreme ($S_1 = \emptyset$). The out-of-bag construction of Section 7.2 can be seen as a clever way to obtain a data-dependent prior without splitting the sample once and for all.

6.5 Anatomy of a self-bounding algorithm

We now know what is allowed; it remains to turn these rules into algorithms. Fortunately, all the self-bounding algorithms of this course follow the same template, summarized in Fig. 43 and Algorithm 2. The first decision is how to organize the data: which examples build the voters or the prior, and which ones are used in the bound. The prior is then fixed, independently of the examples of the bound. The third step chooses an objective, which may be the bound itself or a relaxation that is easier to optimize, and the fourth step minimizes it with one of the tools of Section 6.6. Finally, the tightest valid bound is computed for the posterior that was found.

Two features of this template are worth stressing. First, the objective need not be the certificate. By Theorem 6.2, the certificate is valid whatever the way Q_S was obtained, so we may optimize a convenient relaxation (McAllester's bound, the PAC-Bayes-λ bound, a surrogate loss) and report the tightest valid bound at the end; if we report the minimum of several bounds, Proposition 6.3 applies. Second, the optimizer

Algorithm 2: Generic self-bounding learning

Input: sample S, confidence δ, family of posteriors $\{Q_\theta\}$

1 Organize the data: which examples build the voters / the prior, which ones are used in the bound;

2 Fix the prior P (independent of the examples used in the bound);

3 Choose a *relaxed* objective $\widetilde{\mathcal{B}}(\theta) \geq$ (or $\approx$) the bound, easy to optimize;

4 $\theta_S \leftarrow$ approximately minimize $\widetilde{\mathcal{B}}(\theta)$ (closed form, alternating or gradient-based);

5 Compute the *tight* certificate $\mathcal{B}(Q_{\theta_S}, S, \delta)$ with the kl inversion (Monte Carlo if needed);

Output: Q_{θ_S} and its certificate

need not find the global minimum: every posterior is certified, and a poor optimizer only yields a looser certificate, never a wrong one. Step 4 is therefore a matter of efficiency rather than validity, and the next subsection reviews the tools available for it.

6.6 Optimization toolbox

The right optimizer depends on the shape of the objective. We go from the easiest case, where the minimizer is explicit, to the most general one, where we differentiate through the kl inversion itself.

6.6.1 Closed form: linear objectives and the Gibbs posterior

When the objective is linear in the empirical term and in the KL, $\widehat{R}_S(G_Q) + \frac{\mathrm{KL}(Q\|P)}{\lambda m}$, as for Hoeffding-type or PAC-Bayes-λ bounds, Proposition 4.1 gives the minimizer in closed form, $Q_\lambda(h) \propto P(h)e^{-\lambda m \widehat{R}_S(h)}$. The only remaining parameter is the temperature, which is free since it indexes posteriors: we can scan a grid of values of λ and keep the best certificate, as in Fig. 25.

6.6.2 Alternating minimization

When the temperature is not scanned but optimized, a closed form is available in each variable separately. The PAC-Bayes-λ bound (3.6) is minimized in Q by the Gibbs posterior when λ is fixed, and in λ by the closed form (3.7) when Q is fixed. Alternating the two steps (Algorithm 3) converges to the global minimum under the conditions of Thiemann et al. (2017); we will see in Table 3 that a few iterations suffice in practice.

6.6.3 Gradient-based minimization

For richer objectives, such as second-order bounds, C-bounds or bounds on neural networks, we use gradient descent on a parametrization of the posterior. For a finite set of voters, the simplest parametrization is the softmax $\rho = \mathrm{softmax}(\theta)$, $\rho_t = e^{\theta_t} / \sum_s e^{\theta_s}$, which removes the simplex constraint. If $g = \nabla_\rho \mathcal{B}$ is the gradient with respect to the weights, the chain rule gives

$$\nabla_\theta \mathcal{B} = \rho \odot (g - (\rho^\top g)\mathbf{1}), \qquad \text{with, for the KL term,} \qquad \nabla_\rho \mathrm{KL}(\rho\|\pi) = \ln \frac{\rho}{\pi} + \mathbf{1}. \quad (6.2)$$

For Gaussian posteriors $Q = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$ on the parameters $\mathbf{w}$ of a model, the KL to a Gaussian prior is explicit, and the expected loss is differentiated with the *reparametrization trick*: writing $\mathbf{w} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, I)$, we have $\nabla_{\boldsymbol{\mu},\boldsymbol{\sigma}} \mathbb{E}_{\mathbf{w}\sim Q} \ell(\mathbf{w}) = \mathbb{E}_{\boldsymbol{\varepsilon}} \nabla_{\boldsymbol{\mu},\boldsymbol{\sigma}} \ell(\boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon})$, which is estimated with a few samples of $\boldsymbol{\varepsilon}$ (Dziugaite and Roy, 2017; Pérez-Ortiz et al., 2021).

In both cases a difficulty remains: the 0-1 loss is not differentiable. Either the Gibbs risk has a smooth closed form, as the probit loss of linear classifiers (Eq. (7.5)), or the 0-1 loss is replaced during learning by a bounded surrogate such as a clipped cross-entropy. The final certificate is always computed with the 0-1 loss.

6.6.4 Differentiating through the kl inversion

The tightest bounds are expressed with $\overline{\mathrm{kl}}^{-1}$, which has no closed form. It can nevertheless be differentiated, which allows us to minimize Seeger-type certificates directly (Reeb et al., 2018; Viallard et al., 2021).

Proposition 6.6: Derivatives of the kl inversion

Let $q \in (0,1)$, $\varepsilon > 0$ and $p = \overline{\mathrm{kl}}^{-1}(q, \varepsilon)$ (or $p = \underline{\mathrm{kl}}^{-1}(q, \varepsilon)$), so that $\mathrm{kl}(q\|p) = \varepsilon$ and $p \neq q$. Then

$$\frac{\partial p}{\partial q} = \frac{p(1-p)}{p-q} \ln \frac{p(1-q)}{q(1-p)}, \qquad \frac{\partial p}{\partial \varepsilon} = \frac{p(1-p)}{p-q}.$$

Proof. Let $F(q,p) = \mathrm{kl}(q\|p) - \varepsilon$. We have $F_q = \partial_q \mathrm{kl}(q\|p) = \ln \frac{q(1-p)}{p(1-q)}$ and $F_p = \partial_p \mathrm{kl}(q\|p) = \frac{p-q}{p(1-p)} \neq 0$. By the implicit function theorem, p is a differentiable function of (q,ε) with $\frac{\partial p}{\partial q} = -\frac{F_q}{F_p}$ and $\frac{\partial p}{\partial \varepsilon} = \frac{1}{F_p}$. □

So, although $\overline{\mathrm{kl}}^{-1}$ has no closed form, its gradient only requires the value p that we compute anyway. As expected, $\frac{\partial p}{\partial \varepsilon}$ is positive for the upper inversion and negative for the lower one. These derivatives also tell us how sensitive a certificate is to its ingredients. At $\hat{q} = 0.1$ and $\varepsilon = 0.05$, where $p \approx 0.220$, we find $\frac{\partial p}{\partial \varepsilon} \approx 1.43$: with $m = 1000$ examples, one more nat of KL increases ε by 0.001 and the certificate by about 0.0014.

For the tandem certificate of a finite ensemble, $\mathcal{B}_{\mathrm{TND}}(\rho) = 4\,\overline{\mathrm{kl}}^{-1}(\rho^\top \widehat{\mathbf{L}} \rho, \varepsilon(\rho))$ with $\varepsilon(\rho) = \frac{2\,\mathrm{KL}(\rho\|\pi) + \ln(2\sqrt{n_0}/\delta)}{n_0}$, where n_0 is the number of examples on which the tandem losses are computed, the chain rule gives

$$\nabla_\rho \mathcal{B}_{\mathrm{TND}} = 4\Big[\frac{\partial p}{\partial q}\, 2\widehat{\mathbf{L}}\rho + \frac{\partial p}{\partial \varepsilon}\, \frac{2}{n}\Big(\ln \frac{\rho}{\pi} + \mathbf{1}\Big)\Big], \tag{6.3}$$

to be combined with (6.2). The function `certificate_and_grad` of `pbtools.py` implements this formula.

Example 6.3 (Watching a self-bounding algorithm learn). Figure 44 runs this gradient descent (Adam, 400 iterations, softmax parametrization, uniform initialization) on the out-of-bag first-order and tandem certificates of a forest of 100 trees on `digits`, and records the certificate, the test risk of the vote and the effective number of trees $1/\sum_t \rho_t^2$ along the iterations. The first-order certificate decreases from 0.46 to 0.39, but

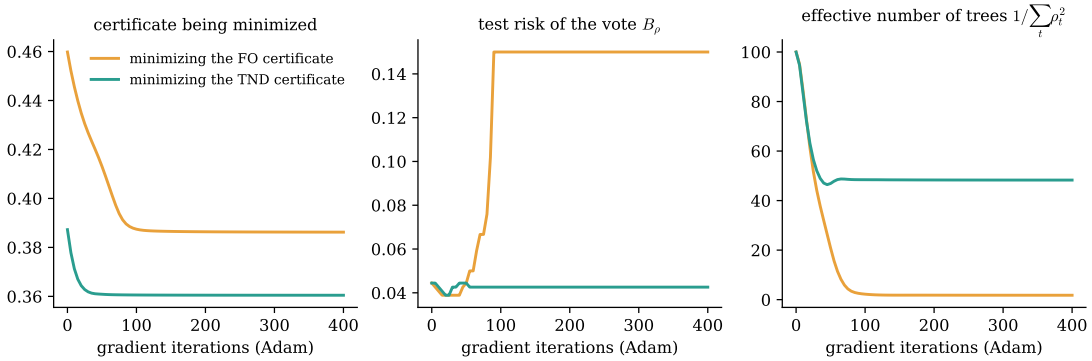


Figure 44: Direct minimization (Adam, softmax parametrization, gradient of Proposition 6.6) of the OOB first-order and tandem certificates of a forest of 100 trees on `digits`. Left: value of the certificate; middle: test risk of the vote; right: effective number of trees. [code: `make_figures.py: fig_selfbounding_dynamics`]

it achieves this by concentrating the posterior on fewer and fewer trees: after 200 iterations, fewer than two trees carry the whole weight, and the test risk of the vote jumps from 0.044 to 0.15. The tandem certificate decreases from 0.39 to 0.36 while keeping an effective number of about 48 trees, and the vote remains as accurate as the uniform forest. This is the phenomenon announced in Section 5 and analysed in Section 7.2.2: an objective that ignores the correlations between voters destroys the diversity that makes the vote work.

6.6.5 Relaxations of Seeger's bound used as objectives

Differentiating through $\overline{\mathrm{kl}}^{-1}$ is not always convenient. When a differentiable closed form is preferred, one of the following upper bounds on $\overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon)$ can be minimized instead (Pérez-Ortiz et al., 2021). All of them follow from the Pinsker-type inequalities of Lemma 2.5. The PAC-Bayes-λ bound, optimized in λ, coincides with the "quadratic" bound, but its linear form is more convenient for alternating minimization.

Name	Objective	Comment
McAllester ("classic")	$\hat{q} + \sqrt{\varepsilon/2}$	simplest, loose when $\hat{q}$ is small
PAC-Bayes-λ	$\frac{\hat{q}}{1-\lambda/2} + \frac{\varepsilon}{\lambda(1-\lambda/2)}$	linear; quadratic at $\lambda^\star$
"Quadratic"	$\left(\sqrt{\hat{q} + \varepsilon/2} + \sqrt{\varepsilon/2}\right)^2$	from $(p-\hat{q})^2 \le 2p\varepsilon$
Seeger (kl)	$\overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon)$	tightest; gradient by Prop. 6.6

Recent work: new ways to minimize a certificate Three recent directions address the optimization of PAC-Bayesian objectives. Clerico et al. (2022) avoid the surrogate loss for stochastic neural networks: with a Gaussian distribution on the weights of the last layer, the output of the network is Gaussian conditionally on the hidden layer, so that the expected 0-1 loss has a closed,

differentiable form (a probit, as in PBGD), and Seeger's bound itself can be minimized. Picard-Weibel et al. (2024) address models whose empirical risk is expensive to evaluate: they replace $\widehat{R}_S$ by its projection on a low-dimensional family of functions under the current posterior, for which the Gibbs posterior of Section 4.1 has a closed form, and iterate; the minimizer of the Catoni-type objective is a fixed point of the resulting algorithm. Viallard et al. (2023) replace the KL divergence by the Wasserstein distance between the posterior and priors learned on parts of the data; for Lipschitz losses the resulting bound, of the form $\mathbb{E}_Q[R_{\mathcal{D}} - \widehat{R}_S] \lesssim \sum_i \frac{2|S_i|L}{m} W(Q, P_i) + \sqrt{\ln(K/\delta)/m}$, remains finite for deterministic (Dirac) posteriors and can be minimized by gradient descent.

6.7 Pitfalls and good practice

Self-bounding learning is simple, but its guarantees are only as good as the care with which the rules of Section 6.3 are applied. Three mistakes are particularly common.

Pitfall: a prior that has seen the data Building the voters on the whole sample and then using a uniform prior over them, with empirical risks computed on the same sample, is *not* valid: the voters, hence the prior, depend on the examples of the bound. Use a split, or the out-of-bag construction of Section 7.2.

Pitfall: hidden selections Trying several bounds, several priors or several random seeds and reporting the best certificate is a selection with the data, which must be accounted for with Proposition 6.3.

Pitfall: certifying the wrong predictor A bound on the Gibbs risk is not a bound on the risk of the vote (use Section 5), nor on the risk of a single network drawn from the posterior (use disintegrated bounds). State precisely what is certified.

Turning these warnings around, we obtain a short checklist. In practice, a sound self-bounding experiment starts by deciding what is certified: a Gibbs classifier, a majority vote or a deterministic model. The prior is then fixed before looking at the examples used in the bound, and every choice made with the data is accounted for, at the price of $\ln K$ for K candidates. A convenient relaxation can be optimized, but the reported certificate should be the tight kl form. For majority votes of diverse voters, second-order bounds should be preferred. Finally, a certificate should always be reported with its ingredients (empirical term, KL, m, δ) and next to the test error, since a bound is a guarantee and not an estimate.

To remember from Section 6

A PAC-Bayesian bound holds for all posteriors: minimizing it is legitimate and yields a predictor together with its risk certificate (Theorem 6.2). The posterior, and anything that only indexes posteriors, may depend freely on the data; the prior, δ and the parameters of the bound may not, and a choice among K candidates costs $\ln K$. Hold-out certificates are tighter for majority votes, but

self-bounding learning uses all the data, makes the certificate a learning objective and makes model selection almost free; data-dependent priors interpolate between the two. The generic recipe is always the same: organize the data, fix the prior, optimize a relaxed bound (closed form, alternating minimization, or gradient through the kl inversion), and certify with the kl form.

Check your understanding

Short questions to answer without computation; answers in Appendix A.5.

Q6.1. Why does minimizing a PAC-Bayesian bound not invalidate the certificate of the minimizer?

Q6.2. Is the temperature of a Gibbs posterior a free parameter? And the parameter C of Catoni's bound?

Q6.3. What is the price of choosing the best of K candidate models with their certificates?

Q6.4. Give one advantage of the hold-out certificate and one advantage of the self-bounding certificate.

Exercises

Exercise 6.1. Reproduce the experiment of Fig. 18 (draw 1000 random classifiers and $m = 100$ labels). How often is the "fixed-h" Hoeffding bound violated? And the Occam bound? What happens to the Occam bound with 10^6 classifiers?

Exercise 6.2. Show that the value of λ minimizing the Hoeffding-type bound (3.5) depends on $\mathrm{KL}(Q\|P)$, and explain why using it directly is forbidden, whereas using $\lambda^\star$ in the PAC-Bayes-λ bound is allowed.

Exercise 6.3. Using Proposition 6.6, compute $\partial p/\partial q$ at $\hat{q} = 0.1$, $\varepsilon = 0.05$ (with $p \approx 0.220$, see Fig. 12). Which ingredient of the certificate is it more profitable to decrease: the empirical risk by 0.01, or the KL by 5 nats with $m = 1000$?

Exercise 6.4. ($\star$) Derive the "quadratic" relaxation of the table above from the refined Pinsker inequality. Writing $a = \sqrt{\varepsilon/2}$, show that it is smaller than McAllester's bound iff $2a+2\sqrt{\hat{q}+a^2} \leq 1$; in particular, for small ε it is the case as soon as $\hat{q} < \frac{1}{4}$.

Exercise 6.5. **(Comprehension)** For each of the following practices, say whether it is free, requires a union bound, or is forbidden, with a one-sentence justification. (a) Stopping the gradient descent on the certificate when the certificate stops decreasing. (b) Choosing the scale σ_0 of a Gaussian prior among 5 values after computing the five certificates. (c) Computing the certificate with $\delta = 0.01$, reporting it with a 99% claim if it is below a target value, and otherwise reporting the certificate computed with $\delta = 0.05$ with a 95% claim. (d) Choosing the temperature of a Gibbs posterior by cross-validation on the sample used in the bound.

Exercise 6.6. **(Comprehension)** True or false? Justify briefly. (a) A self-bounding certificate is an unbiased estimate of the true risk of the learned predictor. (b) If the optimizer of a self-bounding algorithm is stuck in a poor local minimum, the reported certificate may be wrong. (c) A test-set bound remains valid if the test set is also used to choose among several predictors.

Exercise 6.7. **(Application)** In the model-selection example, the tandem certificates are computed with $\delta/6$. For a forest with an OOB empirical joint error $\widehat{e}_S(Q) = 0.05$, $\mathrm{KL}(Q\|P) = 2$ and $n_0 = 423$, compute the tandem certificate (5.3) with $\delta = 0.05$, with $\delta/6$, and with $\delta/100$.

With a countable family and $\pi_k = \frac{6}{\pi^2 k^2}$, what is the extra cost, in nats, for the candidates $k = 1$ and $k = 10$?

Exercise 6.8. **(Application)** With $m = 1000$ examples, the hold-out approach trains a forest on 700 examples and its vote errs on 30 of the 300 remaining ones. The self-bounding approach trains a forest on the 1000 examples and obtains an OOB empirical joint error of 0.06 with $\mathrm{KL}(Q\|P) = 1.5$ and $n_0 = 370$. With $\delta = 0.05$, compute the test-set bound (Proposition 6.4) and the tandem certificate. Below which value of the empirical joint error would the tandem certificate beat the hold-out one? What is the tandem certificate when $\widehat{e}_S(Q) = 0$?

Exercise 6.9. **(Proof)** Prove Proposition 6.3 in detail: write the failure event of each bound, apply the union bound, and explain why the certificate $\min_k \mathcal{B}_k(Q_S, S, \pi_k\delta)$ is valid for any learning rule and any data-dependent choice of k. Check that $\pi_k = \frac{6}{\pi^2 k^2}$, $k \geq 1$, is a valid choice of weights.

Exercise 6.10. **(Proof)** Let $\rho = \mathrm{softmax}(\theta)$ with $\theta \in \mathbb{R}^n$. Show that $\frac{\partial \rho_s}{\partial \theta_t} = \rho_s(\mathbb{1}[s = t] - \rho_t)$, deduce the formula (6.2) for $\nabla_\theta \mathcal{B}$, and show that the components of $\nabla_\theta \mathcal{B}$ sum to zero. Interpret this last fact.

7 Self-bounding algorithms for ensembles and beyond

About this section

Guiding question. *How do we learn the weights of a forest, of a vote of boosted voters, of a linear classifier or of a neural network by minimizing a certificate?*

This section applies the recipe of Section 6 to the ensemble methods of the course, and then to linear classifiers, kernels and neural networks. For each algorithm we specify the five ingredients of Algorithm 2: the voters and the organization of the data, the prior, the objective, the optimizer and the certificate. The central tool for bagging is the out-of-bag construction, which lets every example serve both to build some voters and to certify the others. We run the algorithms on real data and look at what they learn, including when they fail.

7.1 Warm-up: a finite ensemble with an independent sample

Before tackling bagging, where voters and bound share the same data, we start with the simplest situation, the one of the decision stumps of Fig. 25. A set of n voters is built on a sample S_1, for instance trees learned on bootstrap samples of S_1, or stumps whose thresholds are chosen on S_1. A second sample S_2 of size m is used to compute the empirical risks $\widehat{R}_{S_2}(h_t)$, and the prior is uniform over the n voters, which is legitimate since it only depends on S_1.

The remaining ingredients follow the toolbox of Section 6.6. The objective can be the PAC-Bayes-λ bound, minimized by the Gibbs posterior $\rho_t \propto e^{-\lambda m \widehat{R}_{S_2}(h_t)}$; the temperature λ can be scanned on a grid, since it only indexes posteriors, or optimized by alternating with the closed-form $\lambda^\star$ (Algorithm 3 below). The certificate is Seeger's bound on S_2 for the Gibbs classifier, or one of the bounds of Section 5 for the vote, the tandem losses being computed on S_2.

On `breast cancer`, with $n = 300$ stumps and $m = 120$ examples in S_2, we obtained in Section 4 a Seeger certificate of 0.256 for a test Gibbs risk of 0.114. The certificate would be exactly the same if λ had been chosen by cross-validation, or by any other method: every choice of λ is certified. The drawback of this construction is that the examples of S_1 only serve to build the voters, and those of S_2 only to weight and certify them. Can every example play both roles? The bagging construction below shows that it can.

7.2 Bagging and random forests: out-of-bag PAC-Bayes

7.2.1 The split construction

The prior must not depend on the data used to evaluate the bound. With bagging this looks problematic, since every tree is learned from S. Yet each tree h_t is learned on a bootstrap sample S_t only, and the *out-of-bag* (OOB) examples $T_t = S \setminus S_t$ were not used to build it: from the point of view of h_t, T_t is a fresh test sample (Fig. 45). The idea of Thiemann et al. (2017) is to use the *indices* of the trees as hypotheses, with a prior on the indices fixed in advance, and to evaluate each tree on its own OOB sample.

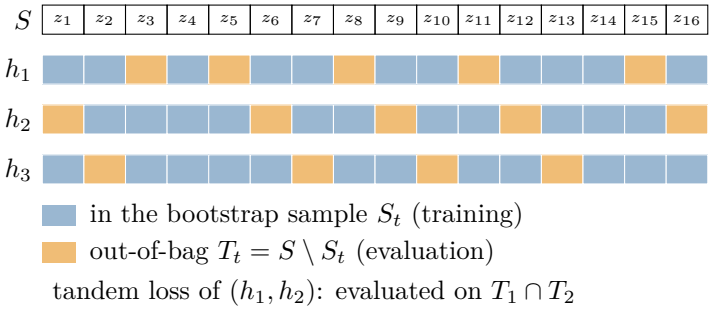


Figure 45: Out-of-bag construction. Each tree is trained on its bootstrap sample (blue) and its loss is estimated on its OOB sample (orange), which is independent of the tree. The joint error of two trees is estimated on the examples that are out-of-bag for both. [code: `tikz/oob_sets.tex`]

Theorem 7.1: PAC-Bayes with a split construction (Thiemann et al., 2017; Lorenzen et al., 2019)

Let $h_1, \dots, h_n$ be voters, h_t being learned on $S_t \subset S$ by any procedure, and let $T_t = S \setminus S_t$ with $|T_t| \geq n_0$ for all t. Let $\widehat{L}_t = \frac{1}{|T_t|} \sum_{(x,y)\in T_t} \mathbb{1}[h_t(x) \neq y]$ be the OOB loss of h_t and π a distribution on $\{1, \dots, n\}$ fixed in advance (e.g. uniform). Then, with probability at least $1-\delta$ over $S \sim \mathcal{D}^m$ (and the randomness of the bootstrap), for all $\rho \in \mathcal{M}(\{1, \dots, n\})$,

$$\operatorname{kl}\Big(\sum_t \rho_t \widehat{L}_t \,\Big\|\, \sum_t \rho_t R_{\mathcal{D}}(h_t)\Big) \leq \frac{\operatorname{KL}(\rho\|\pi) + \ln \frac{2\sqrt{n_0}}{\delta}}{n_0}. \tag{7.1}$$

The same holds for the tandem losses $\widehat{L}_{tt'}$ computed on $T_t \cap T_{t'}$, with $\operatorname{KL}(\rho^2\|\pi^2) = 2\operatorname{KL}(\rho\|\pi)$ and $n_0 \leq \min_{t,t'} |T_t \cap T_{t'}|$.

Proof sketch. We follow the proof of Theorem 3.1, the "hypotheses" being the indices t, on which π is a legitimate data-independent prior. Only the third step changes: for each t, conditionally on S_t the voter h_t is fixed and T_t is an i.i.d. sample independent of it, so Maurer's lemma gives $\mathbb{E}\, e^{|T_t|\operatorname{kl}(\widehat{L}_t\|R_{\mathcal{D}}(h_t))} \leq 2\sqrt{|T_t|}$. Since $|T_t| \geq n_0$, Jensen's inequality yields $\mathbb{E}\, e^{n_0 \operatorname{kl}(\widehat{L}_t\|R_{\mathcal{D}}(h_t))} \leq (2\sqrt{|T_t|})^{n_0/|T_t|} \leq 2\sqrt{n_0}$. □

In plain words, each tree is graded only on examples it has never seen, and the bound behaves as if we had an independent sample of size n_0; the posterior over trees is then free, exactly as in Section 6. The price is that the certificate is computed with the size n_0 of the *smallest* OOB set, so it is important to make these sets large. With the standard bootstrap, each OOB set contains about $(1-1/m)^m \approx e^{-1} \approx 37\%$ of the sample, and the intersection of two OOB sets about $e^{-2} \approx 14\%$. Drawing bootstrap samples of size $m/2$ instead of m raises these proportions to about $e^{-1/2} \approx 61\%$ and $e^{-1} \approx 37\%$, at the price of slightly weaker trees; this is the choice made in the figures and notebooks of this course.

On the `digits` experiments below, the sizes are $m = 1257$, $n_1 = 747$, $n_2 = 423$: the first-order bound is computed with n_1 examples and the tandem bound with n_2 examples, instead of the m training examples.

7.2.2 Minimizing the first-order bound

With the OOB losses in hand, the most natural first attempt is to minimize the first-order bound. With the PAC-Bayes-λ form (3.6) of the bound, the optimal posterior for a fixed λ is the Gibbs posterior on the OOB losses (Proposition 4.1), and the optimal λ for a fixed posterior is given by (3.7). This suggests an alternating minimization, in which each step solves one half of the problem exactly.

Algorithm 3: PAC-Bayes-λ minimization for a forest (Thiemann et al., 2017)

Input: OOB losses $\widehat{L}_1, \dots, \widehat{L}_n$, prior π, n_0, δ

$\lambda \leftarrow 1$;

repeat

$\quad \rho_t \leftarrow \dfrac{\pi_t \, e^{-\lambda n_0 \widehat{L}_t}}{\sum_s \pi_s \, e^{-\lambda n_0 \widehat{L}_s}}$ for $t = 1, \dots, n$;

$\quad \lambda \leftarrow \dfrac{2}{\sqrt{2n_0 \, \rho^\top \widehat{L} / \big(\mathrm{KL}(\rho\|\pi) + \ln \frac{2\sqrt{n_0}}{\delta} \big) + 1} + 1}$;

until *convergence of* λ;

Output: ρ and the certificate $2\,\overline{\mathrm{kl}}^{-1}\big(\rho^\top \widehat{L}, \frac{\mathrm{KL}(\rho\|\pi) + \ln(2\sqrt{n_0}/\delta)}{n_0}\big)$ on $R_{\mathcal{D}}(B_\rho)$

Thiemann et al. (2017) proved that, under mild conditions, the PAC-Bayes-λ bound is strongly quasiconvex and that this procedure converges to its global minimum. Table 3 follows it on a forest of 100 trees on `digits`. Starting from $\lambda = 1$, the first Gibbs posterior already has an empirical Gibbs risk of 0.128 and a KL of 4.2 nats; λ then drops to 0.37 and is stable after three iterations, and the bound on the Gibbs risk goes from 0.207 to 0.206. The algorithm is fast and does exactly what it promises: it produces an excellent bound on the *Gibbs* risk.

The last column of the table tells a different story: the test risk of the vote is 0.150, whereas the uniform forest had a test risk of 0.044. As observed by Lorenzen et al. (2019) and Masegosa et al. (2020), the vote learned by minimizing a first-order bound is often much *worse* than the uniform vote. The Gibbs posterior concentrates on the few trees with the smallest OOB loss; the left panel of Fig. 46 shows that it gives essentially all its weight to two trees. These trees are good individually, but a vote of two trees is not a forest, and the benefit of voting disappears. This is a direct consequence of the first-order nature of the bound, which only sees the average quality of the voters and ignores the correlations of their errors. To protect the vote, the objective must see these correlations.

7.2.3 Minimizing the tandem bound

The tandem bound sees exactly what the first-order bound misses, since its empirical term counts the errors made jointly by pairs of voters. Combined with the PAC-Bayes-λ relaxation, the tandem bound (5.3) gives the objective

$$\mathcal{B}_{\mathrm{TND}}(\rho, \lambda) = 4 \left[\frac{\rho^\top \widehat{\mathbf{L}} \rho}{1 - \frac{\lambda}{2}} + \frac{2\,\mathrm{KL}(\rho\|\pi) + \ln \frac{2\sqrt{n_0}}{\delta}}{\lambda(1 - \frac{\lambda}{2}) n_0} \right], \qquad \widehat{\mathbf{L}} = (\widehat{L}_{tt'})_{t,t'}, \tag{7.2}$$

Table 3: Iterations of Algorithm 3 on a forest of 100 trees on `digits` ($\delta = 0.05$). On row k, ρ is the Gibbs posterior for the value of λ shown, and the PB-λ bound is evaluated at the updated value of λ (the one of row $k+1$). The last column is the test risk of the majority vote B_ρ; the uniform forest has a test risk of 0.044. Later iterations are identical.

iter.	λ	$\rho^\top \widehat{L}$	$\mathrm{KL}(\rho\|\pi)$	PB-λ bound on $R(G_\rho)$	FO cert.	test $R(B_\rho)$
1	1.0000	0.1283	4.18	0.2070	0.3866	0.150
2	0.3802	0.1293	3.76	0.2064	0.3863	0.150
3	0.3735	0.1294	3.74	0.2064	0.3863	0.150
4	0.3732	0.1294	3.74	0.2064	0.3863	0.150
5	0.3732	0.1294	3.74	0.2064	0.3863	0.150

which is quadratic in ρ, plus the convex KL term. It has no closed-form minimizer, but it is easily minimized by gradient descent on the softmax parametrization $\rho = \mathrm{softmax}(\theta)$, alternated with the closed-form update of λ (Masegosa et al., 2020). The gradient with respect to ρ is

$$\nabla_\rho \mathcal{B}_{\mathrm{TND}} = 4\left[\frac{2\widehat{\mathbf{L}}\rho}{1-\frac{\lambda}{2}} + \frac{2\big(\ln\frac{\rho}{\pi}+\mathbf{1}\big)}{\lambda(1-\frac{\lambda}{2})n_0}\right], \tag{7.3}$$

to be combined with (6.2).

Algorithm 4: Tandem bound minimization for a forest (Masegosa et al., 2020)

Input: OOB tandem losses $\widehat{\mathbf{L}}$ (on OOB intersections), prior π, $n_0 = \min_{t,t'} |T_t \cap T_{t'}|$, δ

$\theta \leftarrow \ln \pi$, $\lambda \leftarrow 1$;

repeat

  $\theta \leftarrow$ minimize $\mathcal{B}_{\mathrm{TND}}(\mathrm{softmax}(\theta), \lambda)$ with L-BFGS, using (7.3) and (6.2);

  $\rho \leftarrow \mathrm{softmax}(\theta)$;

  $\lambda \leftarrow \dfrac{2}{\sqrt{2n_0\,\rho^\top\widehat{\mathbf{L}}\rho\,/\,(2\,\mathrm{KL}(\rho\|\pi) + \ln\frac{2\sqrt{n_0}}{\delta}) + 1} + 1}$;

until *convergence of* λ *(or a maximal number of rounds)*;

Output: ρ and the certificate $4\,\overline{\mathrm{kl}}^{-1}\big(\rho^\top\widehat{\mathbf{L}}\rho, \frac{2\,\mathrm{KL}(\rho\|\pi)+\ln(2\sqrt{n_0}/\delta)}{n_0}\big)$ on $R_{\mathcal{D}}(B_\rho)$

In words, the quadratic term $\rho^\top\widehat{\mathbf{L}}\rho$ penalizes pairs of voters that err *together*, so the optimal posterior has no reason to concentrate on a few similar trees: diversity is rewarded instead of ignored. Alternatively, the kl-form certificate itself can be minimized with the gradient (6.3) of Section 6.6; the dynamics of this approach were shown in Fig. 44.

7.2.4 Comparing the learned posteriors

We can now put the two algorithms side by side, together with the uniform weights and a C-bound minimizer. Table 4 and Fig. 46 compare, on the same forest, four posteriors: the uniform one, the minimizers of the first-order and of the tandem bounds, and the minimizer of the PAC-Bayesian C-bound (5.5), obtained by a direct numerical minimization. The first-order posterior has the smallest Gibbs risk and the smallest first-order certificate, but it keeps two effective trees and its vote is three times worse than the others.

Table 4: Four posteriors on the same forest of 100 trees on `digits`: test risks of the vote and of the Gibbs classifier, the three OOB certificates on the risk of the vote ($\delta = 0.05$), and the effective number of trees $1/\sum_t \rho_t^2$.

posterior	$R(B_\rho)$	$R(G_\rho)$	FO cert.	TND cert.	C-bound cert.	eff. trees
uniform	0.044	0.190	0.460	0.387	0.559	100.0
FO-optimized	0.150	0.158	0.386	0.666	0.596	2.0
TND-optimized	0.043	0.183	0.436	0.361	0.538	58.1
C-bound-optimized	0.048	0.176	0.419	0.369	0.531	30.8

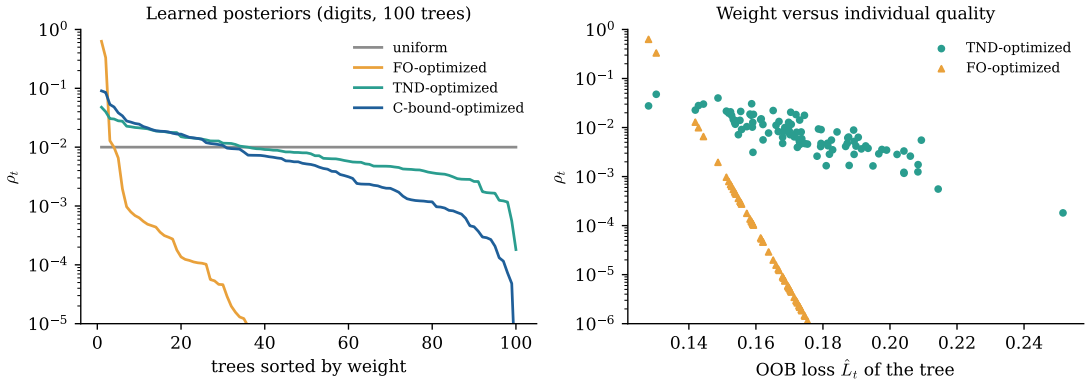


Figure 46: Weights learned on a forest of 100 trees on `digits`. Left: sorted weights of the four posteriors (log scale). Right: weight of each tree as a function of its OOB loss for the first-order and tandem posteriors: the first-order posterior is a hard selection of the two best trees, the tandem posterior a soft one. [code: `make_figures.py: fig_fo_iterations_and_weights`]

The tandem posterior keeps an effective number of 58 trees, spread over the good and the average trees (right panel of Fig. 46): its vote is as good as the uniform one (0.043), and it has the smallest certificate of all, 0.361. The C-bound posterior is in between, with 31 effective trees and a vote of comparable quality, but its certificate (0.531) is looser than the tandem certificate, because it needs two kl inversions and a ratio. This illustrates the remark following Proposition 5.7: the C-bound is the best *oracle* bound, but not necessarily the best *certificate*.

Is this behaviour specific to `digits`? Figure 47 repeats the comparison between the uniform, first-order and tandem posteriors on four datasets. The picture is consistent: minimizing the first-order bound always improves the first-order certificate, but it degrades the vote on all four datasets, sometimes severely (from 0.044 to 0.150 on `digits`, from 0.074 to 0.184 on the simulated data). Minimizing the tandem bound never degrades the vote and always yields the best tandem certificate. The certificates remain far from the test risks — this is the price of certifying a vote with a second-order Markov inequality and a doubled KL — but they are valid and obtained without any test data.

7.3 Boosting-like algorithms: learning the weights of a vote

Bagging builds its voters independently and only asks us to weight them. Boosting builds them sequentially and learns their weights at the same time; we now see how the PAC-Bayesian viewpoint, and in particular the C-bound, shapes algorithms of this

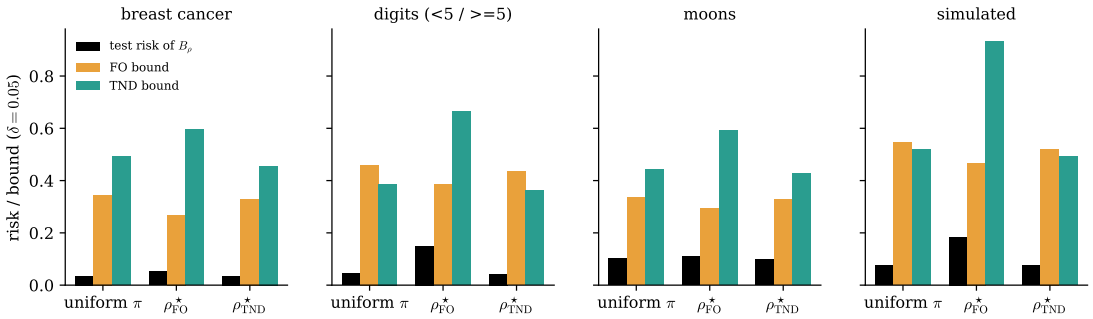


Figure 47: Forests of 100 trees (bootstrap samples of size $m/2$), on four datasets. For the uniform posterior, the posterior minimizing the first-order (FO) bound, and the posterior minimizing the tandem (TND) bound, we report the test risk of the majority vote and the two OOB certificates ($\delta = 0.05$). Generated with `pbtools.py` (notebook 4). [code: `make_figures.py: fig_rf_optimisation`]

kind.

7.3.1 AdaBoost from the PAC-Bayesian viewpoint

AdaBoost greedily minimizes the exponential loss $\frac{1}{m}\sum_i e^{-y_i f(x_i)}$ of $f = \sum_t \alpha_t h_t$, an upper bound on the training error of the vote (Freund and Schapire, 1997). In PAC-Bayesian terms, it learns a posterior $Q \propto (\alpha_t)_t$ whose margins M_Q on S are large, the normalized margins being $y_i f(x_i)/\|\alpha\|_1$; this is the explanation of Schapire et al. (1998), illustrated in Fig. 6. But nothing in AdaBoost controls the divergence $\mathrm{KL}(Q\|P)$ or the disagreement of the voters. The algorithms of this subsection instead learn the weights of a set of voters, possibly produced by boosting, by minimizing a C-bound.

7.3.2 MinCq: minimizing the C-bound

The C-bound rewards votes whose margin is large on average and varies little. Recall that $\mathcal{C}_Q = 1 - \frac{(\mathbb{E}M_Q)^2}{\mathbb{E}M_Q^2}$. For a fixed value μ of the first moment of the margin, minimizing the C-bound amounts to minimizing its second moment. Roy et al. (2011) and Germain et al. (2015) proposed the algorithm

$$\textbf{MinCq:} \qquad \min_Q \ \widehat{\mathbb{E}}_S[M_Q^2] \quad \text{s.t.} \quad \widehat{\mathbb{E}}_S[M_Q] = \mu, \quad Q \text{ quasi-uniform}, \tag{7.4}$$

where $\mathcal{H}$ is *self-complemented* ($h \in \mathcal{H} \Rightarrow -h \in \mathcal{H}$) and a *quasi-uniform* posterior satisfies $Q(h) + Q(-h) = \frac{1}{|\mathcal{H}|/2}$. Since M_Q is linear in Q, $\widehat{\mathbb{E}}_S[M_Q^2]$ is a quadratic form in the weights and (7.4) is a quadratic program.

The restriction to quasi-uniform posteriors is the key trick. On the one hand, it does not reduce the set of votes that can be expressed, since any vote can be represented by a quasi-uniform posterior with the same predictions. On the other hand, PAC-Bayesian bounds hold for quasi-uniform posteriors *without the KL term* (Germain et al., 2015), so that the empirical C-bound itself is controlled.

The hyperparameter μ, the targeted first moment of the margin, plays the role of a regularization parameter and is selected by cross-validation. MinCq has been extended

to incorporate prior knowledge on the weights (P-MinCq, Bellet et al., 2014) and to a column-generation algorithm that adds voters one at a time (CqBoost, Roy et al., 2016).

7.3.3 CB-Boost: greedy C-bound minimization

MinCq weights a fixed set of voters. To grow the vote one voter at a time, as AdaBoost does, Bauvin et al. (2020) proposed CB-Boost, a boosting-like algorithm that builds the vote greedily by minimizing the empirical C-bound. At each round, it adds a voter and chooses its weight so as to decrease the empirical C-bound; since this quantity is a ratio of two quadratic functions of the weights, the optimal weight of the new voter has a closed form, which makes each round as cheap as a round of AdaBoost. The empirical C-bound decreases at each iteration, and the final vote can be certified with a PAC-Bayesian C-bound.

7.3.4 Direct minimization of PAC-Bayesian C-bounds

MinCq and CB-Boost minimize the *empirical* C-bound and certify afterwards. A fully self-bounding alternative is to minimize the certificate itself: Viallard et al. (2021) proposed to minimize by (stochastic) gradient descent the *PAC-Bayesian* C-bounds themselves, that is the right-hand side of (5.5) and tighter joint versions, over the posterior $\rho = \mathrm{softmax}(\theta)$. The C-bound (5.5) is a composition of $\overline{\mathrm{kl}}^{-1}$ (for the Gibbs risk), $\underline{\mathrm{kl}}^{-1}$ (for the disagreement) and a smooth function, so its gradient follows from Proposition 6.6 and the chain rule: with $\bar{r} = \overline{\mathrm{kl}}^{-1}(\rho^\top \widehat{L}, \varepsilon_1)$, $\underline{d} = \underline{\mathrm{kl}}^{-1}(\rho^\top \widehat{D}\rho, \varepsilon_2)$ and $\mathcal{C} = 1 - \frac{(1-2\bar{r})^2}{1-2\underline{d}}$,

$$\frac{\partial \mathcal{C}}{\partial \bar{r}} = \frac{4(1-2\bar{r})}{1-2\underline{d}}, \qquad \frac{\partial \mathcal{C}}{\partial \underline{d}} = -\frac{2(1-2\bar{r})^2}{(1-2\underline{d})^2}.$$

The learned vote comes with its certificate by construction.

Zantedeschi et al. (2021) go one step further and consider *stochastic majority votes*, whose weights are themselves drawn from a Dirichlet posterior. For such votes the expected risk can be computed, or tightly approximated, and bounded directly, which avoids the loose Gibbs-to-vote relations of Section 5.

7.4 Linear classifiers and kernel methods

The recipe is not limited to finite ensembles. Linear classifiers are the natural next step: they are the one case where every term of the bound can be written explicitly.

7.4.1 Gaussian posteriors on linear classifiers

Consider linear voters $h_{\mathbf{v}}(x) = \mathrm{sign}(\langle \mathbf{v}, x \rangle)$, $\mathbf{v} \in \mathbb{R}^d$, a prior $P = \mathcal{N}(\mathbf{0}, I_d)$ and posteriors $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$. By symmetry of the Gaussian around $\mathbf{w}$, the majority vote is

$$B_{Q_{\mathbf{w}}}(x) = \mathrm{sign}\left(\mathbb{E}_{\mathbf{v}} \,\mathrm{sign}\langle \mathbf{v}, x \rangle\right) = \mathrm{sign}\langle \mathbf{w}, x \rangle :$$

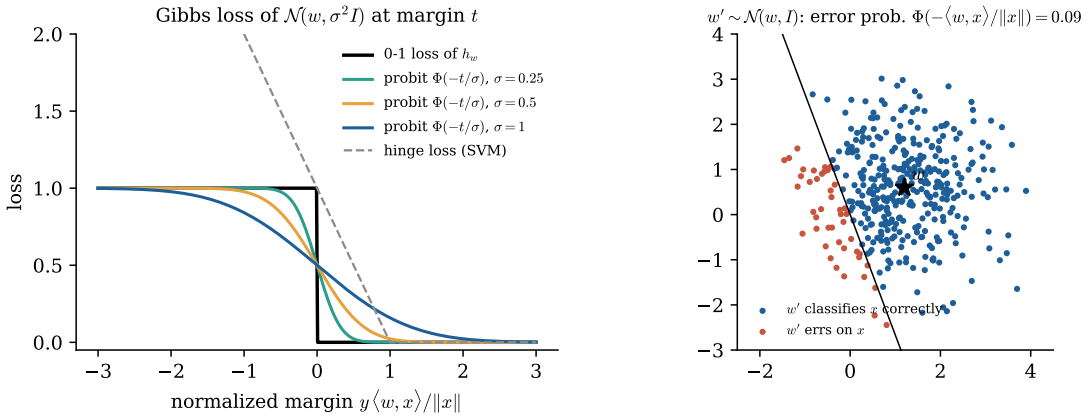


Figure 48: The probit loss. Left: loss of the Gibbs classifier of $\mathcal{N}(\mathbf{w}, \sigma^2 I)$ as a function of the normalized margin $t = y\langle \mathbf{w}, x\rangle / \|x\|$, for three values of σ (the notes use $\sigma = 1$), compared with the 0-1 and hinge losses. Right: 400 classifiers $\mathbf{v} \sim \mathcal{N}(\mathbf{w}, I)$ in the plane; those on the wrong side of the line $\langle \mathbf{v}, x\rangle = 0$ (red) err on x. [code: `make_figures.py: fig_probit_loss`]

the vote of an infinite number of linear classifiers is the linear classifier $\mathbf{w}$ itself. For $\mathbf{v} \sim Q_{\mathbf{w}}$, the random variable $y\langle \mathbf{v}, x\rangle$ is Gaussian with mean $y\langle \mathbf{w}, x\rangle$ and variance $\|x\|^2$, so the empirical Gibbs risk is

$$\widehat{R}_S(G_{Q_{\mathbf{w}}}) = \frac{1}{m}\sum_{i=1}^{m} \Phi\Big(- \frac{y_i \langle \mathbf{w}, x_i\rangle}{\|x_i\|}\Big), \tag{7.5}$$

where Φ is the standard Gaussian cumulative distribution function: the 0-1 loss is replaced by a smooth *probit* loss of the normalized margin. The right panel of Fig. 48 explains where it comes from: a classifier $\mathbf{v}$ drawn around $\mathbf{w}$ errs on x when it falls on the wrong side of the hyperplane orthogonal to x, which happens with probability $\Phi(-y\langle \mathbf{w}, x\rangle / \|x\|)$. The left panel compares this loss with the 0-1 loss and with the hinge loss of the SVM: it is a smoothed 0-1 loss, which stays bounded for badly misclassified points.

Finally, the complexity is $\mathrm{KL}(Q_{\mathbf{w}} \| P) = \frac{1}{2}\|\mathbf{w}\|^2$. Figure 49 shows the resulting trade-off. Multiplying $\mathbf{w}$ by a constant does not change the vote, but it makes the sampled classifiers agree more with it: the Gibbs risk decreases towards the risk of the vote, while the KL grows quadratically.

Plugging these quantities into Catoni's bound (3.4) and removing the monotone transformation gives the objective of the algorithm PBGD (Germain et al., 2009):

$$\min_{\mathbf{w} \in \mathbb{R}^d} \; F(\mathbf{w}) = C \sum_{i=1}^{m} \Phi\Big(- \frac{y_i \langle \mathbf{w}, x_i\rangle}{\|x_i\|}\Big) + \frac{1}{2}\|\mathbf{w}\|^2. \tag{7.6}$$

It is instructive to compare it with the soft-margin SVM, $\min_{\mathbf{w}} C \sum_i \max(0, 1 - y_i\langle \mathbf{w}, x_i\rangle) + \frac{1}{2}\|\mathbf{w}\|^2$. PAC-Bayes *explains* the SVM objective: the regularization is the KL to the prior, the loss is a surrogate of the Gibbs risk, and the norm of $\mathbf{w}$ plays the role of the inverse of a margin (Langford and Shawe-Taylor, 2002; McAllester, 2003b). The two objectives differ mainly in the loss, and this difference has a practical consequence. The objective (7.6) is not convex; its gradient is

$$\nabla F(\mathbf{w}) = -C \sum_{i=1}^{m} \varphi\Big(\frac{y_i \langle \mathbf{w}, x_i\rangle}{\|x_i\|}\Big) \frac{y_i x_i}{\|x_i\|} + \mathbf{w},$$

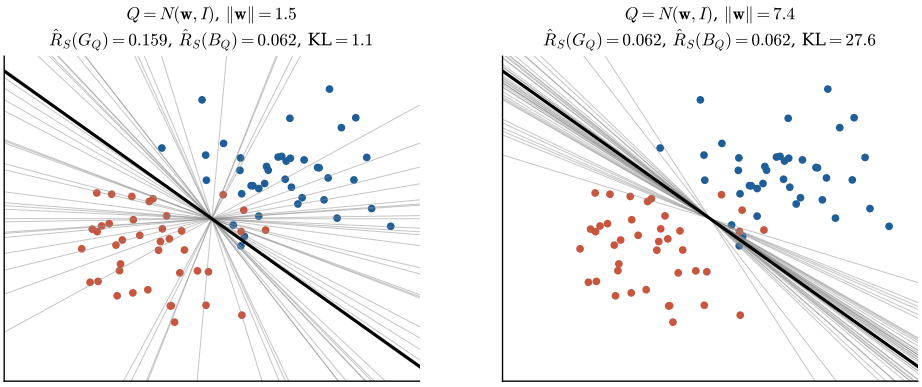


Figure 49: Gaussian posterior $\mathcal{N}(\mathbf{w}, I)$ over linear classifiers: 40 sampled hyperplanes (grey) and the majority vote $\operatorname{sign}\langle \mathbf{w}, x \rangle$ (black). Multiplying $\mathbf{w}$ by 5 does not change the vote but makes the sampled voters agree with it: the Gibbs risk decreases while the KL grows quadratically. [code: `make_figures.py: fig_linear_gaussian`]

where φ is the standard Gaussian density. Each example pushes $\mathbf{w}$ in its direction, with a weight that is maximal for points close to the decision boundary and vanishes for points with a large margin, positive or negative: unlike the hinge loss, badly misclassified points are eventually ignored, which makes PBGD robust to label noise. It is optimized by gradient descent with several restarts, and has a kernelized version (Germain et al., 2009).

Example 7.1 (PBGD selects its own regularization). As a self-bounding algorithm, PBGD has an important advantage: the parameter C only indexes posteriors, so it can be chosen by minimizing the certificate itself, without cross-validation. Figure 50 runs PBGD on a two-dimensional problem with $m = 200$ examples and 8% of flipped labels, for 25 values of C between 10^{-2} and 10^{2}, and computes Seeger's bound for each solution. For small C, $\mathbf{w}$ is short ($\|\mathbf{w}\| = 0.41$ for $C = 0.01$), the sampled classifiers are almost random and the certificate is 0.54. For large C, the Gibbs risk approaches the training error of the vote but the KL explodes ($\|\mathbf{w}\| = 13.6$ for $C = 100$), and the certificate climbs back to 0.58. The best certificate, 0.28, is reached for $C = 1$, with $\|\mathbf{w}\| = 3.5$, an empirical Gibbs risk of 0.133 and a training error of the vote of 0.110. The right panel shows the hyperplanes learned with a small C, with the C selected by the certificate, and with the largest C; no validation sample was used to choose among them.

Data-dependent Gaussian priors, centred on an SVM learned on part of the data, give some of the tightest bounds for SVMs (Ambroladze et al., 2006; Parrado-Hernández et al., 2012). The prior need not be Gaussian, however: a kernel can itself be read as a prior over voters.

7.4.2 Kernels as priors: random Fourier features

For a shift-invariant kernel $k(x, x') = k(x - x')$, Bochner's theorem gives $k(x - x') = \mathbb{E}_{\omega \sim p} \cos\left(\omega^\top (x - x')\right)$, where p is the Fourier transform of k (Rahimi and Recht, 2007). Letarte et al. (2019b) interpret this identity as a *majority vote*: the voters are the functions $h_\omega(x, x') = \cos(\omega^\top (x - x'))$ and the kernel is their vote under the *prior* p. Learning

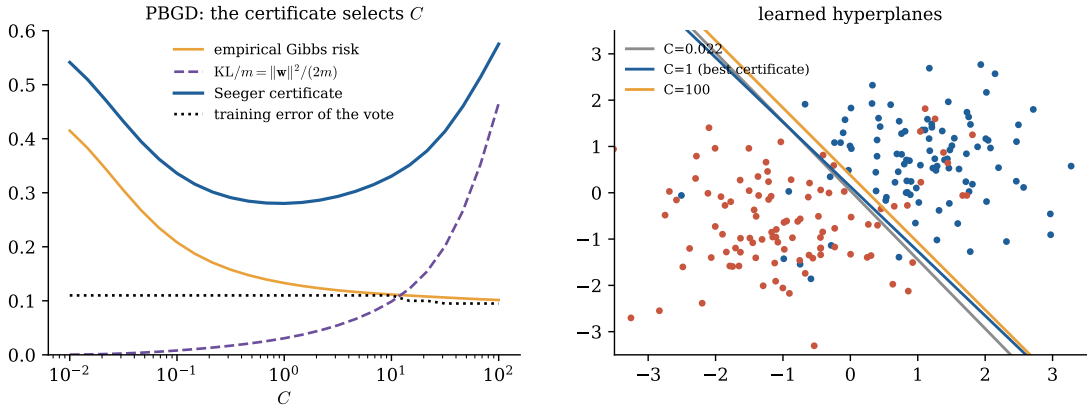


Figure 50: PBGD on a two-dimensional problem ($m = 200$, 8% label noise). Left: empirical Gibbs risk, complexity $\|\mathbf{w}\|^2/(2m)$ and Seeger certificate as functions of C. Right: hyperplanes learned for three values of C. [code: `make_figures.py: fig_pbgd`]

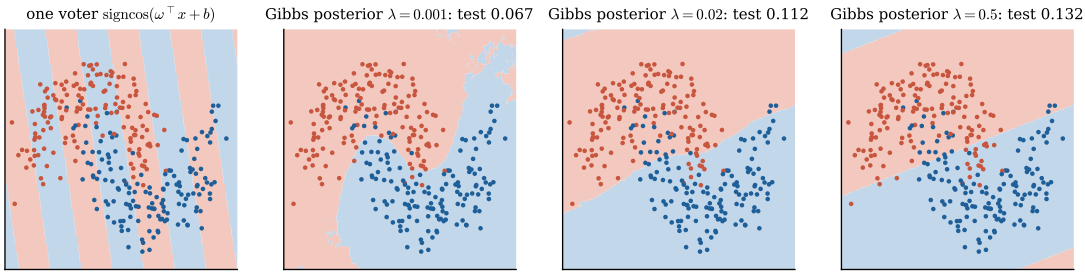


Figure 51: Random Fourier voters on the two moons. Left: one voter. Right: the majority vote under Gibbs posteriors at three temperatures (test risks in the titles). [code: `make_figures.py: fig_rff`]

a kernel adapted to the task then amounts to learning a *posterior* over frequencies. With N frequencies $\omega_1, \ldots, \omega_N$ drawn from p, the pseudo-posterior $Q(\omega_j) \propto e^{-\beta \widehat{L}_S(h_{\omega_j})}$ is exactly a Gibbs posterior, and a PAC-Bayesian bound certifies the learned kernel. This is closely related to the boosting-with-random-Fourier-features algorithm studied in the first project of the course.

Example 7.2 (Random Fourier voters on the two moons). A simpler variant uses the random features directly as voters, $h_{\omega,b}(x) = \operatorname{sign}\cos(\omega^\top x + b)$, with $\omega \sim \mathcal{N}(0, 2\gamma I)$ and $b \sim \mathcal{U}[0, 2\pi]$, together with their opposites. Figure 51 shows one such voter on the two moons dataset, and the vote of 4000 of them under Gibbs posteriors at three temperatures, learned on 900 examples. Under the uniform prior the vote is undecided, since each voter comes with its opposite. A very small tilt of the prior ($\lambda = 0.001$, KL $= 0.005$ nats) already gives a smooth decision boundary with a test risk of 0.067: the vote then behaves like a kernel classifier. Larger values of λ (lower temperatures) concentrate the posterior on fewer frequencies and give more irregular boundaries, with test risks of 0.11 ($\lambda = 0.02$) and 0.13 ($\lambda = 0.5$). Note that the best vote corresponds to a posterior whose empirical Gibbs risk is 0.49: the Gibbs certificate is vacuous, while the vote is excellent. This is an extreme instance of the gap studied in Section 5, and a reminder that votes must be certified with second-order bounds.

7.5 Beyond ensembles: stochastic neural networks and Gaussian processes

The Gaussian posteriors used for linear classifiers carry over to models with many more continuous parameters, and the same recipe applies. For a neural network with weights $\mathbf{w} \in \mathbb{R}^D$, consider a Gaussian posterior $Q = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$ and a Gaussian prior $P = \mathcal{N}(\boldsymbol{\mu}_0, \sigma_0^2 I)$; drawing a network from Q at each prediction defines a *stochastic neural network*. The centre $\boldsymbol{\mu}_0$ of the prior is the random initialization of the network or, much better, a network trained on a part S_1 of the data (Proposition 6.5); the scale σ_0 is chosen on a grid with a union bound. As for linear classifiers, the KL divergence has the closed form

$$\mathrm{KL}(Q\|P) = \frac{1}{2}\sum_{j=1}^{D}\Big(\frac{\sigma_j^2 + (\mu_j - \mu_{0,j})^2}{\sigma_0^2} - 1 + \ln\frac{\sigma_0^2}{\sigma_j^2}\Big).$$

The objective is one of the relaxations of Seeger's bound of Section 6.6, in which the 0-1 loss is replaced by a bounded surrogate, for instance the cross-entropy clipped to $[0, \ell_{\max}]$ and rescaled to $[0,1]$, and it is minimized by stochastic gradient descent on $(\boldsymbol{\mu}, \ln \boldsymbol{\sigma})$ with the reparametrization trick (see McAllester, 2013 for the related case of dropout, and Letarte et al., 2019a for networks with binary activations). The certificate, finally, uses the 0-1 loss: the empirical Gibbs risk is estimated with N networks sampled from Q, bounded by a first kl inversion, and plugged into Seeger's bound, as in Fig. 29.

Dziugaite and Roy (2017) obtained in this way the first *non-vacuous* bounds for over-parametrized networks on MNIST, and Pérez-Ortiz et al. (2021) showed that such PAC-Bayesian training yields networks with competitive accuracy together with tight certificates. Reeb et al. (2018) applied the same principle to learn the hyperparameters of Gaussian processes by minimizing a PAC-Bayesian bound, and Lotfi et al. (2022) obtained non-vacuous bounds for large models with compression-based priors. This line of work now reaches large language models (Lotfi et al., 2024a,b), see the Recent work box of Section 2.4. For networks with binary activations, Fortier-Dubois et al. (2023) compute exactly the output of an *aggregation* of stochastic networks, that is a majority vote over their weights, by propagating probabilities over the hidden representations, and minimize a PAC-Bayesian bound on it. Finally, *deep ensembles* (Lakshminarayanan et al., 2017) are majority votes of networks, to which the second-order bounds of Section 5 and the algorithms of this section apply.

7.6 The running example, one last time

Before summarizing the algorithms, let us step back and look at the whole journey of our running example.

> **Running example** Let us gather what we have learned about our forest of 100 trees on `digits`. Its vote errs on 4.4% of the test digits, while a tree drawn at random errs on 19% of them; Section 5 explained the gap by the diversity of the trees, whose disagreement (0.27) exceeds their Gibbs risk. Setting aside the 540 test digits certified the vote at 0.069 (Section 2). Without any test set, the out-of-bag tandem bound certified the uniform vote at 0.387 (Section 5), and learning the weights by minimizing this bound lowered the certificate to 0.361 while keeping the vote as accurate (0.043). Minimizing the first-order bound

instead gave the best certificate on the Gibbs risk, but destroyed the vote (0.150). Three lessons summarize the course: the diversity of the voters is what makes a vote work; a certificate for a vote must see this diversity, which requires second-order quantities; and a certificate that sees it can safely be used as a learning objective.

7.7 Summary

The following table summarizes the algorithms of this section with the five ingredients of the self-bounding recipe.

Algorithm	Posterior	Objective	Optimizer	Certifies
Gibbs posterior	finite, any	linear bound	closed form	Gibbs, vote
PB-λ (Thiemann et al., 2017)	finite (OOB)	PAC-Bayes-λ	alternating	Gibbs
TND (Masegosa et al., 2020)	finite (OOB)	tandem-λ	L-BFGS + λ	vote
MinCq (Roy et al., 2011)	quasi-uniform	empirical C-bound	QP	vote
CB-Boost (Bauvin et al., 2020)	greedy	empirical C-bound	closed-form steps	vote
Self-bounding C-bound (Viallard et al., 2021)	finite	PAC-Bayesian C-bound	gradient (kl inv.)	vote
StocMV (Zantedeschi et al., 2021)	Dirichlet	bound on stochastic vote	gradient	stochastic vote
PBGD (Germain et al., 2009)	Gaussian, linear	Catoni	gradient	Gibbs, vote
PBB (Pérez-Ortiz et al., 2021)	Gaussian, NN	Seeger relaxations	SGD, reparam.	Gibbs

To remember from Section 7

To certify an ensemble, each voter must be evaluated on examples it has not seen: an independent sample, or its out-of-bag examples (Theorem 7.1). Minimizing a first-order bound gives an excellent bound on the Gibbs risk but concentrates the weights on a few voters and destroys the diversity of the vote; second-order objectives such as the tandem bound keep it, and give the best certificates for votes. The C-bound leads to a family of algorithms (MinCq, CB-Boost, direct minimization) which explicitly trade the strength of the voters against their disagreement. The same recipe gives PAC-Bayesian versions of SVMs (PBGD, which selects its own regularization), of kernel learning and of stochastic neural networks, where data-dependent priors are the key ingredient.

Check your understanding

Short questions to answer without computation; answers in Appendix A.5.

Q7.1. Why can each tree of a bagged forest be evaluated on its out-of-bag examples without breaking the bound?

Q7.2. Why does minimizing the first-order bound degrade the vote?

Q7.3. What does the parameter μ of MinCq control?

Q7.4. In PBGD, which quantity plays the role of the regularization of the SVM?

Exercises

Exercise 7.1. Derive the gradient (7.3) of the tandem objective, and check it numerically with finite differences (notebook 4).

Exercise 7.2. Explain why the uniform prior over the trees of a random forest would *not* be a valid prior if the empirical losses were computed on the whole training sample.

Exercise 7.3. In PBGD, show that when all the training points have a large normalized margin, the gradient of the loss term vanishes, and interpret the resulting value of $\|\mathbf{w}\|$.

Exercise 7.4. With notebook 4, compute the test disagreement of the C-bound posterior of Table 4 and check whether the condition $d \geq R$ of Proposition 5.7 holds. Why is its certificate nevertheless larger than the tandem certificate?

Exercise 7.5. ($\star$) For a stochastic network with $D = 10^5$ weights, $\sigma_j = \sigma_0 = 0.03$ and $\|\boldsymbol{\mu} - \boldsymbol{\mu}_0\| = 3$, compute $\mathrm{KL}(Q\|P)$. How many examples are needed for the complexity term $\mathrm{KL}\,/m$ to be below 0.01? What does this suggest about data-dependent priors?

Exercise 7.6. **(Comprehension)** True or false? Justify briefly. (a) The OOB bound is invalid because every training example has been used to build some of the trees. (b) Adding trees to a forest with the uniform posterior increases $\mathrm{KL}(\rho\|\pi)$. (c) In PBGD, the parameter C must be selected by cross-validation, since the bound cannot be used to choose it. (d) The first-order posterior of Table 4 has a small Gibbs risk, hence necessarily a good vote.

Exercise 7.7. **(Application)** For the `digits` forest ($m = 1257$), compute the expected fraction and number of examples that are out-of-bag for one tree, and for two given trees, when the bootstrap samples have size m and when they have size $\lfloor m/2 \rfloor = 628$. Compare with the values $n_1 = 747$ and $n_2 = 423$ used in the notes, and explain the difference.

Exercise 7.8. **(Application)** Run Algorithm 3 by hand (with a calculator or `pbtools`) on $n = 4$ trees with OOB losses $\widehat{L} = (0.15, 0.17, 0.20, 0.25)$, uniform prior, $n_0 = 700$ and $\delta = 0.05$: compute ρ for $\lambda = 1$, the updated λ, and ρ again. Compare the final first-order certificate with the one of the uniform posterior, and comment on the weights.

Exercise 7.9. **(Application)** For a Gaussian posterior $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$ with $\|\mathbf{w}\| = 2$, three training points have normalized margins $y_i\langle \mathbf{w}, x_i\rangle / \|x_i\|$ equal to 2, 0.5 and -1. Compute the empirical Gibbs risk (7.5), the training error of the vote and $\mathrm{KL}(Q_{\mathbf{w}}\|P)$. Redo the computation for $3\mathbf{w}$, and relate the result to Fig. 49.

Exercise 7.10. **(Proof)** Let $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I_d)$ over linear voters $h_{\mathbf{v}}(x) = \mathrm{sign}\langle \mathbf{v}, x\rangle$, and $x \neq 0$. Show that $\mathbb{P}_{\mathbf{v}\sim Q_{\mathbf{w}}}\big(y\langle \mathbf{v}, x\rangle \leq 0\big) = \Phi\big(-y\langle \mathbf{w}, x\rangle / \|x\|\big)$, and deduce that the majority vote is $B_{Q_{\mathbf{w}}}(x) = \mathrm{sign}\langle \mathbf{w}, x\rangle$ (up to the tie $\langle \mathbf{w}, x\rangle = 0$) and the formula (7.5).

Part IV

Perspectives

This last part is a reading guide towards current research on PAC-Bayesian learning; it can be used as a starting point for the research part of the project.

8 Further topics and recent advances

About this section

Guiding question. *Where does PAC-Bayes go beyond this course?*

This last section is a reading guide. It briefly presents some active research directions — unbounded losses and dependent data, other divergences, information-theoretic bounds, disintegrated bounds, domain adaptation — with pointers to the literature. It is a good starting point for the research part of the project.

Figure 52 gives a map of the directions discussed below.

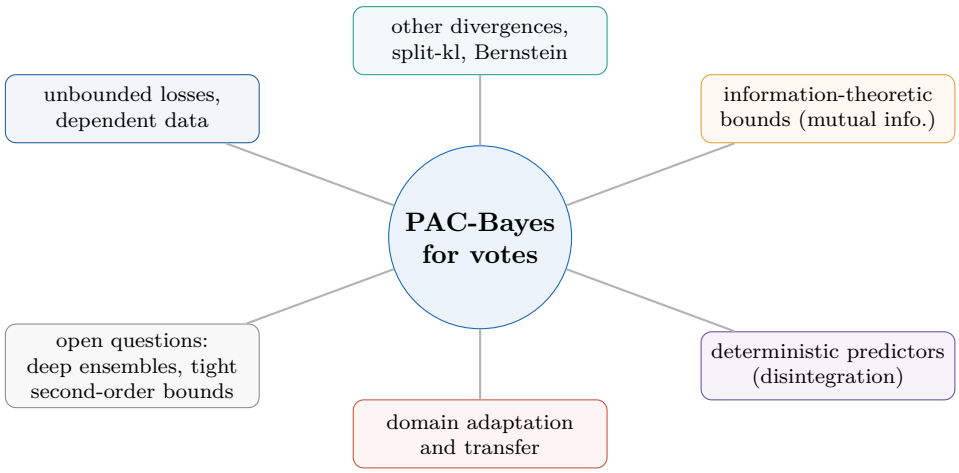


Figure 52: Some research directions around PAC-Bayesian bounds for majority votes, presented in this section. [code: `tikz/research_map.tex`]

8.1 Beyond bounded losses and i.i.d. data

All the bounds of this course assume a loss with values in $[0,1]$ and i.i.d. data. The boundedness enters the proofs at a single place: the control of the exponential moment $\mathcal{I}_\Delta(m)$ in the third step of Theorem 3.1. For unbounded losses, such as the squared loss or the log-loss, this term must be controlled through assumptions on the tails of the loss (sub-Gaussian, sub-exponential, or bounded moments). This is the route followed by Alquier et al. (2016) and Germain et al. (2016), whose analysis of Bayesian inference we met in Section 4, and by Haddouche et al. (2021) for losses whose range depends on the hypothesis; Alquier and Guedj (2018) relax both the boundedness and the independence assumptions, at the price of weaker rates. More recently, Rodríguez-Gálvez et al. (2024) and Chugg et al. (2023) gave unified treatments, the latter based on nonnegative supermartingales, which also yield *anytime-valid* bounds, holding simultaneously for all sample sizes. This property matters in online learning and sequential decision-making, where the number of observations is not fixed in advance. Martingale techniques similarly extend PAC-Bayes to dependent data and to reinforcement learning (Seldin et al., 2012; Chugg et al., 2023). Among the most recent results, Haddouche and Guedj (2023) only assume bounded second moments of the loss, and the PAC-Bayes-Chernoff bounds of Casado et al. (2024) optimize the free parameter of Catoni-type bounds exactly for unbounded losses (Section 3).

8.2 Other divergences and tighter inequalities

The KL divergence is not an intrinsic feature of PAC-Bayes: it comes from the change of measure (2.2). Other changes of measure, based for instance on Hölder's inequality, yield bounds involving other divergences, such as Rényi divergences (Bégin et al., 2016), which offer different trade-offs between the complexity term and the confidence term. Recent works go further: f-divergences (Ohnishi and Honorio, 2021), any strongly convex regularizer (Lugosi and Neu, 2022), divergences derived from coin-betting that can be much smaller than the KL (Kuzborskij et al., 2024), or the Wasserstein distance, which gives bounds for deterministic predictors (Viallard et al., 2023). Hellström and Guedj (2024) show that, on the concentration side, the kl comparator is optimal for bounded losses, and Jang et al. (2023) obtain time-uniform versions of the kl bound at no cost. On the concentration side, the kl inequality is optimal for Bernoulli variables but not for losses with a small variance, and Bernstein-type inequalities exploit this variance (Tolstikhin and Seldin, 2013; Mhammedi et al., 2019). For majority votes, the relevant excess losses take only three values, and the split-kl inequality of Wu and Seldin (2022) exploits this structure. On the prior side, data-dependent priors remain one of the most effective levers (Parrado-Hernández et al., 2012; Dziugaite and Roy, 2018; Pérez-Ortiz et al., 2021), as we saw in Fig. 28.

8.3 Information-theoretic generalization bounds

PAC-Bayesian bounds control, with high probability, the average risk under a posterior that depends on the sample. A closely related family of results bounds the *expected* generalization gap of a randomized algorithm by the *mutual information* $I(S;h)$ between the sample and the output of the algorithm (Xu and Raginsky, 2017). The connection is direct: averaging a PAC-Bayesian bound over S and choosing as prior the average posterior $P = \mathbb{E}_S Q_S$, which is the best data-independent prior in expectation, turns $\mathbb{E}_S \operatorname{KL}(Q_S \| P)$ into $I(S;h)$. The monograph of Hellström et al. (2025) presents both points of view in a unified way.

8.4 Deterministic predictors: disintegration

A recurring limitation of this course is that PAC-Bayes certifies a *randomized* predictor, and we had to work to transfer the guarantee to the majority vote. Disintegrated bounds (Blanchard and Fleuret, 2007; Rivasplata et al., 2020) take another route: they hold with high probability over the joint draw of the sample and of a *single* hypothesis $h \sim Q_S$, so that the one model that is actually deployed is certified. Viallard et al. (2024b) give a general framework and practical algorithms for such bounds. Applied to majority votes whose weights are drawn once from a distribution, as in the stochastic votes of Section 7, this approach certifies the vote that is actually used. Viallard et al. (2024a) use this framework to turn any complexity measure into a generalization bound, by sampling from a Gibbs distribution built on it, and Leblanc and Germain (2025) bound the risk of a deterministic majority vote through an exact identity (Section 5.9).

8.5 Domain adaptation and transfer

When the training distribution (the source) differs from the test distribution (the target), the risk of a majority vote on the target can be related to its risk on the source and

to a *divergence between the two domains*. PAC-Bayes is particularly well suited to this problem because the disagreement $d_{\mathcal{D}}(Q)$ can be computed on *unlabeled* target data. Germain et al. (2013, 2020) define a domain divergence as the difference between the disagreements of the voters on the source and on the target, and derive PAC-Bayesian domain adaptation bounds together with algorithms that minimize them. Similar ideas are used for multi-view learning and for fairness constraints. Sicilia et al. (2022) extended these bounds to multiclass learners.

8.6 Beyond classification: compression, meta-learning, generative models

PAC-Bayes keeps being extended to new settings. *Sample compression*, the other classical route to non-vacuous bounds, is close to the Occam bound of Section 2.4: a predictor that can be reconstructed from a few training examples is certified on the others. Bazinet et al. (2025) extend sample compression bounds to real-valued losses, with the kl comparator as the tightest choice for losses in $[0, 1]$, and certify random forests and neural networks trained with a compression procedure. In *meta-learning*, where the prior itself is learned from many related tasks, Riou et al. (2025) show that a Bernstein-type condition always holds at the level of tasks, so that learning the prior costs $O(1/T)$ instead of $O(1/\sqrt{T})$ with T tasks. Finally, PAC-Bayesian bounds now certify *generative* models: Mbacke et al. (2023) bound the distance between the data distribution and the distribution produced by adversarial generative networks.

8.7 Open questions

Several questions raised in this course remain open. The first concerns the relation between tightness and learning. The tightest certificates, such as the kl bounds or the CCPBB bound, are not always the best learning objectives: we saw that minimizing the first-order bound gives an excellent bound on the Gibbs risk and a poor vote. Understanding which bound should be minimized to obtain the best *predictor*, and not only the best certificate, is largely open. The second concerns scale. How should priors be designed for boosted ensembles with thousands of trees, or for ensembles of large neural networks, so that the bounds remain non-vacuous? Compression-based priors (Lotfi et al., 2022) give a partial answer. The third concerns diversity. Second-order bounds formalize the role of diversity in a vote, and one may hope to design ensemble algorithms that optimize the strength–diversity trade-off explicitly, with guarantees, beyond the implicit randomization of random forests. The diversity of deep ensembles is studied from this point of view by Masegosa (2020) and Ortega et al. (2022); see also the surveys on diversity in ensemble learning (Kuncheva and Whitaker, 2003; Brown et al., 2005).

Exercises

Exercise 8.1. **(Comprehension)** A practitioner collects examples one at a time, recomputes Seeger's bound with $\delta = 0.05$ after each new example, and stops as soon as the certificate is below a target. Explain why the reported certificate is not guaranteed to hold with probability 0.95. Show that applying the bound for sample size m with confidence $\delta_m = \frac{\delta}{m(m+1)}$ gives a bound valid simultaneously for all $m \geq 8$, and compute the extra cost in nats and the resulting Seeger bound for $m = 1000$, $\widehat{R}_S(G_Q) = 0.1$, $\mathrm{KL}(Q\|P) = 5$, $\delta = 0.05$.

Exercise 8.2. **(Proof)** Let $\mathcal{H}$ be finite, $S \mapsto Q_S$ a learning rule, and write $\bar{Q} = \mathbb{E}_S Q_S$ for the average posterior. Show that, for any prior P, $\mathbb{E}_S \operatorname{KL}(Q_S \| P) = \mathbb{E}_S \operatorname{KL}(Q_S \| \bar{Q}) + \operatorname{KL}(\bar{Q} \| P)$. Deduce that $\bar{Q}$ minimizes $\mathbb{E}_S \operatorname{KL}(Q_S \| P)$ over data-independent priors, and that the minimum equals the mutual information $I(S; h)$ between the sample and a hypothesis $h \sim Q_S$.

Exercise 8.3. **(Comprehension)** Explain why the disagreement $d_{\mathcal{D}}(Q)$ of a vote can be estimated on unlabeled target examples, whereas its Gibbs risk and its joint error cannot. Suppose that the disagreement measured on unlabeled target data is 0.30 and that the target joint error is (somehow) known to be 0.05: compute the target Gibbs risk and the oracle C-bound on the target risk of the vote.

Exercise 8.4. **(Comprehension)** A certificate $\mathcal{B}$ holds for the Gibbs risk $R_{\mathcal{D}}(G_Q)$ with probability at least $1-\delta$. A single network $h \sim Q$ is drawn once and deployed. Is $R_{\mathcal{D}}(h) \leq \mathcal{B}$ guaranteed with high probability? Using Markov's inequality over $h \sim Q$, show that $R_{\mathcal{D}}(h) \leq \mathcal{B}/\eta$ with probability at least $1 - \delta - \eta$, and compute this guarantee for $\mathcal{B} = 0.05$ and $\eta = 0.1$. What do disintegrated bounds improve?

Further reading

For a first contact with PAC-Bayes, the tutorial of Langford (2005) remains an excellent entry point, and the surveys of Guedj (2019) and Alquier (2024) cover the more recent developments, the latter from a statistical perspective. For majority votes, the main references are Germain et al. (2015) for the C-bound, Masegosa et al. (2020) for the tandem bound, Wu et al. (2021) for the Chebyshev–Cantelli family and Viallard et al. (2021) for self-bounding algorithms. The monograph of Catoni (2007) and the recent monograph of Hellström et al. (2025) are more advanced.

A Cheat sheet and code

A.1 Main bounds at a glance

All statements hold w.p. $\geq 1-\delta$ over $S \sim \mathcal{D}^m$, simultaneously for all posteriors Q, for a prior P fixed before seeing S. We write $\varepsilon_k = \dfrac{k\,\mathrm{KL}(Q\|P) + \ln \frac{2\sqrt{m}}{\delta}}{m}$.

Seeger (Gibbs)	$R_{\mathcal{D}}(G_Q) \leq \overline{\mathrm{kl}}^{-1}\big(\widehat{R}_S(G_Q), \varepsilon_1\big)$
McAllester (Gibbs)	$R_{\mathcal{D}}(G_Q) \leq \widehat{R}_S(G_Q) + \sqrt{\varepsilon_1/2}$
Catoni (Gibbs, C fixed)	$R_{\mathcal{D}}(G_Q) \leq \dfrac{1-\exp\!\Big(-C\widehat{R}_S(G_Q) - \frac{\mathrm{KL}(Q\|P)+\ln(1/\delta)}{m}\Big)}{1-e^{-C}}$
PAC-Bayes-λ (Gibbs)	$R_{\mathcal{D}}(G_Q) \leq \dfrac{\widehat{R}_S(G_Q)}{1-\lambda/2} + \dfrac{\varepsilon_1}{\lambda(1-\lambda/2)}$, all $\lambda \in (0,2)$
Gibbs posterior	$Q_\lambda(h) \propto P(h)\, e^{-\lambda m \widehat{R}_S(h)}$
First order (vote)	$R_{\mathcal{D}}(B_Q) \leq 2\,\overline{\mathrm{kl}}^{-1}\big(\widehat{R}_S(G_Q), \varepsilon_1\big)$
Tandem (vote)	$R_{\mathcal{D}}(B_Q) \leq 4\,\overline{\mathrm{kl}}^{-1}\big(\widehat{e}_S(Q), \varepsilon_2\big)$
C-bound (vote, oracle)	$R_{\mathcal{D}}(B_Q) \leq 1 - \dfrac{(1-2R_{\mathcal{D}}(G_Q))^2}{1-2d_{\mathcal{D}}(Q)}$ if $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$
CCTND (vote, oracle)	$R_{\mathcal{D}}(B_Q) \leq \dfrac{e_{\mathcal{D}}(Q) - 2\mu R_{\mathcal{D}}(G_Q) + \mu^2}{(1/2-\mu)^2}$, $\mu < \frac{1}{2}$
Identity	$R_{\mathcal{D}}(G_Q) = e_{\mathcal{D}}(Q) + \frac{1}{2} d_{\mathcal{D}}(Q)$
Linear classifiers	$Q = \mathcal{N}(\mathbf{w}, I)$: $\mathrm{KL} = \frac{\|\mathbf{w}\|^2}{2}$, $\widehat{R}_S(G_Q) = \frac{1}{m}\sum_i \Phi\big(-\frac{y_i\langle \mathbf{w}, x_i\rangle}{\|x_i\|}\big)$

A.2 Python snippets

The following functions (from `pbtools.py`, distributed with the notebooks) are all that is needed to compute the certificates of this course.

```
import numpy as np

def kl_bin(q, p):
    """Binary kl divergence kl(q || p)."""
    p = np.clip(p, 1e-12, 1 - 1e-12)
    t1 = q * np.log(q / p) if q > 0 else 0.0
    t2 = (1 - q) * np.log((1 - q) / (1 - p)) if q < 1 else 0.0
    return t1 + t2

def kl_inv_upper(q, eps, tol=1e-10):
    """sup { p >= q : kl(q || p) <= eps } by bisection."""
    lo, hi = q, 1.0
    while hi - lo > tol:
        mid = (lo + hi) / 2
        if kl_bin(q, mid) > eps: hi = mid
        else: lo = mid
    return hi

def kl_div(rho, pi):
    """KL(rho || pi) for discrete distributions."""
    mask = rho > 0
    return np.sum(rho[mask] * np.log(rho[mask] / pi[mask]))

def seeger_bound(emp_risk, KL, m, delta=0.05):
    eps = (KL + np.log(2 * np.sqrt(m) / delta)) / m
    return kl_inv_upper(emp_risk, eps)

def tandem_bound(rho, pi, L_tandem, m, delta=0.05):
    """4 * kl^{-1}( rho^T L rho , (2 KL + ln(2 sqrt(m)/delta)) / m )."""
    eps = (2 * kl_div(rho, pi) + np.log(2 * np.sqrt(m) / delta)) / m
    return 4 * kl_inv_upper(rho @ L_tandem @ rho, eps)
```

A.3 Code of the figures

All the figures of these notes can be reproduced, modified and reused. The Python figures are produced by the functions of the script `make_figures.py` (run `python make_figures.py` to regenerate all of them, or `python make_figures.py fig_condorcet` for a single one), which relies on the module `pbtools.py`; all random seeds are fixed, so that the numbers quoted in the text are recovered exactly. The TikZ figures are stored in the folder `tikz/`: each file contains a single `tikzpicture`, to be included with `\input{tikz/name}`, and the script `build_tikz.sh` compiles each of them alone. The style of the notes (boxes, colours, the annotated derivations with `\cause`) is defined in `polystyle.sty` and `usercommands.tex`. Everything is gathered in an archive on the course webpage; the table below gives the source of each figure.

Figure	Type	Source
Figure 1	TikZ	`tikz/roadmap.tex`
Figure 2	Python	`make_figures.py: fig_averaging_variance`
Figure 3	Python	`make_figures.py: fig_vote_correlation`
Figure 4	TikZ	`tikz/course_bridge.tex`
Figure 5	Python	`make_figures.py: fig_forest_growth`
Figure 6	Python	`make_figures.py: fig_adaboost`

Figure	Type	Source
Figure 7	TikZ	`tikz/ensembles_as_q.tex`
Figure 8	Python	`make_figures.py: fig_gibbs_vs_mv`
Figure 9	TikZ	`tikz/principle.tex`
Figure 10	TikZ	`tikz/timeline.tex`
Figure 11	TikZ	`tikz/markov.tex`
Figure 12	Python	`make_figures.py: fig_kl_inverse`
Figure 13	Python	`make_figures.py: fig_kl_pinsker`
Figure 14	Python	`make_figures.py: fig_concentration`
Figure 15	Python	`make_figures.py: fig_kl_examples`
Figure 16	Python	`make_figures.py: fig_change_of_measure`
Figure 17	Python	`make_figures.py: fig_occam`
Figure 18	Python	`make_figures.py: fig_uniformity`
Figure 19	TikZ	`tikz/proof_steps.tex`
Figure 20	Python	`make_figures.py: fig_anytime`
Figure 21	Python	`make_figures.py: fig_bound_anatomy`
Figure 22	Python	`make_figures.py: fig_seeger_landscape`
Figure 23	Python	`make_figures.py: fig_bounds_comparison`
Figure 24	Python	`make_figures.py: fig_prior_posterior`
Figure 25	Python	`make_figures.py: fig_temperature`
Figure 26	Python	`make_figures.py: fig_prior_choice`
Figure 27	TikZ	`tikz/recursive_pb.tex`
Figure 28	Python	`make_figures.py: fig_ddprior`
Figure 29	Python	`make_figures.py: fig_montecarlo`
Figure 30	TikZ	`tikz/wq_line.tex`
Figure 31	Python	`make_figures.py: fig_wq_distribution`
Figure 32	Python	`make_figures.py: fig_wq_scenarios`
Figure 33	Python	`make_figures.py: fig_condorcet`
Figure 34	TikZ	`tikz/tandem_table.tex`
Figure 35	Python	`make_figures.py: fig_cbound_landscape`
Figure 36	Python	`make_figures.py: fig_bound_regions`
Figure 37	Python	`make_figures.py: fig_cctnd`
Figure 38	Python	`make_figures.py: fig_split_kl`
Figure 39	Python	`make_figures.py: fig_mv_bounds_diversity`
Figure 40	Python	`make_figures.py: fig_selfbounding_selection`
Figure 41	TikZ	`tikz/data_usage.tex`
Figure 42	Python	`make_figures.py: fig_selfbounding_vs_holdout`
Figure 43	TikZ	`tikz/sb_pipeline.tex`
Figure 44	Python	`make_figures.py: fig_selfbounding_dynamics`
Figure 45	TikZ	`tikz/oob_sets.tex`
Figure 46	Python	`make_figures.py: fig_fo_iterations_and_weights`
Figure 47	Python	`make_figures.py: fig_rf_optimisation`
Figure 48	Python	`make_figures.py: fig_probit_loss`
Figure 49	Python	`make_figures.py: fig_linear_gaussian`
Figure 50	Python	`make_figures.py: fig_pbgd`
Figure 51	Python	`make_figures.py: fig_rff`
Figure 52	TikZ	`tikz/research_map.tex`

A.4 Proof of the refined Pinsker inequality

We prove that for $0 \le q < p \le 1$, $\mathrm{kl}(q\|p) \ge \frac{(p-q)^2}{2p}$. Fix p and let $g(q) = \mathrm{kl}(q\|p) - \frac{(p-q)^2}{2p}$ on $[0, p]$. We have $g(p) = 0$ and

$$g'(q) = \ln \frac{q}{p} - \ln \frac{1-q}{1-p} + \frac{p-q}{p}.$$

Using $\ln u \le u - 1$ for $u = \frac{q}{p} \cdot \frac{1-p}{1-q}$ we get $\ln \frac{q(1-p)}{p(1-q)} \le \frac{q-p}{p(1-q)} \le \frac{q-p}{p}$ (since $q - p < 0$ and $1 - q \le 1$), so $g'(q) \le 0$ on $(0, p)$: g is non-increasing and $g(q) \ge g(p) = 0$.

A.5 Answers to the quick checks

Section 1. **Q1.1** Its weights α_t are non-negative (weak learners are better than random), and dividing them by their sum does not change the sign of the vote. **Q1.2** About 19% for the Gibbs classifier (a tree drawn at random) and 4.4% for the vote. **Q1.3** The posterior may depend on the sample in any way; the prior must be fixed before seeing the sample used in the bound. **Q1.4** For a fixed sign vector, the correlation with the signs is linear in the predictor, so its supremum over weighted averages is reached at a single voter.

Section 2. **Q2.1** Because the variance of a Bernoulli variable is small when its mean is small; the kl adapts to it (width of order ε instead of $\sqrt{\varepsilon}$). **Q2.2** $\ln \frac{n}{k}$. **Q2.3** For the Gibbs distribution $Q_\phi \propto Pe^{\phi}$; the gap is $\mathrm{KL}(Q\|Q_\phi)$. **Q2.4** The description length $\ln \frac{1}{P(h)}$ of the chosen hypothesis.

Section 3. **Q3.1** The third step (Markov), where the expectations over the sample and over the prior are exchanged. **Q3.2** Because the posterior only appears in the deterministic steps; the probabilistic event does not depend on it. **Q3.3** Seeger's bound. **Q3.4** Because the PAC-Bayes-λ bound is a deterministic consequence of Seeger's bound, true for all λ on the same event, whereas each value of λ in the Hoeffding-based bound defines a different event.

Section 4. **Q4.1** When $\lambda \to 0$ it tends to the prior; when $\lambda \to \infty$ it concentrates on the empirical risk minimizers. **Q4.2** Because, conditionally on the part used to learn it, the prior is fixed; the bound must then be computed on the other part only. **Q4.3** Because the Monte Carlo estimate is itself random: its deviation from the exact Gibbs risk must be controlled, at the price of a second confidence level δ'. **Q4.4** Bounding the majority vote through the Gibbs risk and second-order quantities, using margins, and disintegrated bounds.

Section 5. **Q5.1** $R_{\mathcal{D}}(B_Q) = \mathbb{P}(W_Q \ge \frac{1}{2})$ and $R_{\mathcal{D}}(G_Q) = \mathbb{E}W_Q$. **Q5.2** It only uses the mean of W_Q, which does not tell whether the errors of the voters fall on the same examples or on different ones. **Q5.3** The disagreement $d_{\mathcal{D}}(Q)$. **Q5.4** When the disagreement is smaller than the Gibbs risk, i.e. when the voters are strongly correlated.

Section 6. **Q6.1** Because the bound holds on an event that does not depend on the posterior (Theorem 6.2): it holds in particular for the minimizer. **Q6.2** The temperature is free, since it only indexes posteriors; C in Catoni's bound is not, since it changes the event, and must be chosen on a grid with a union bound. **Q6.3** Replacing δ by δ/K, i.e. adding $\ln K$ to the complexity term. **Q6.4** The hold-out certificate is tighter and applies to any predictor; the self-bounding certificate uses all the data, guides learning and makes model selection almost free.

Section 7. **Q7.1** Conditionally on its bootstrap sample, a tree is fixed and its out-of-bag examples form an independent sample; the prior is put on the indices of the trees, which are fixed in advance. **Q7.2** Because the bound ignores the correlations of the errors: it concentrates the weights on the few trees with the smallest loss, which are similar, and the vote loses its diversity. **Q7.3** The targeted first moment of the margin; it acts as a regularization parameter. **Q7.4** The KL divergence $\frac{1}{2}\|\mathbf{w}\|^2$ between the posterior and the prior.

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